Bases and the City: Spatial Economics of US Military Bases in Okinawa *

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Abstract

Rising geopolitical tensions underscore the strategic importance of military bases and the consumption demand and employment they generate are central to policy debates about their economic benefits for host communities. We assess their spatial economic impacts using newly digitized data from postwar Okinawa, a region heavily exposed to US bases. We document substantial urbanization around US bases, with service-sector expansion but little manufacturing growth. We develop a quantitative spatial model calibrated to Okinawa in 1970 to quantify the role of base-sourced consumption demand and employment. Removing these channels reduces population in base-exposed municipalities and shifts activity from services toward agriculture and manufacturing. Okinawa-wide welfare falls by 7.25% and GNI by 11.99%, although these channels accounted for roughly 30% of GNI. Sectoral reallocation mitigates these losses alongside sizable migration. Suggestive evidence also points to an additional role for past investment in infrastructure and durable capital around US bases.

Keywords: Military bases, Service sector, Quantitative spatial model, Geoeconomics, Urbanization

JEL classification: O14, O18, N40, R11, R12

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1 Introduction

As military spending rises amid geopolitical tensions, its economic effects are attracting growing attention in policy discussions. In particular, military bases are expected to support local economies through consumption demand generated by military personnel and direct employment of local residents.[1] For instance, preparations for a German brigade in Lithuania have raised expectations that "new jobs will emerge, businesses will expand, demand for housing and services will increase."[2] Conversely, reporting on possible US troop withdrawals from Germany, Reuters describes a base town's dependence on "thousands of well-paying jobs for local people and steady stream of customers for local businesses."[3] Yet the impact of the local spending and base employment on the host economy is not straightforward to evaluate in the spatial economy. Workers can move across locations and sectors, while commuting and shopping transmit effects beyond host municipalities. How do base-sourced consumption demand and employment reshape the spatial and sectoral distribution of economic activity, and what are their aggregate effects on income and workers' welfare?

This paper addresses these questions using the distinctive case of Okinawa. The U.S. military constructed many bases for strategic reasons during and immediately after World War II, generating substantial spatial variation in base exposure. The bases brought large and spatially uneven income inflows through purchases of goods and services, employment of Okinawan residents, and rental payments to local landowners. Okinawa's GNI statistics separately record these three components and indicate that around 40% of aggregate income came from the US bases in 1955. These separate accounts allow us to discipline the scale and composition of base-related flows in a quantitative spatial model and assess their effects on local development and aggregate outcomes.

We collect and digitize a comprehensive dataset of the Okinawan economy, allowing us to observe various economic variables within the Okinawa island at the municipality level. Using these data, we first document two stylized facts on the relationship between US bases and local economic outcomes. First, a 10 percentage-point higher base-area share is associated with a roughly 36% higher ratio of 1970 to 1938 population density, implying substantial urbanization around the US bases. Second, areas that were highly exposed to the US bases experienced a large increase in the service employment share. In stark contrast, changes in the manufacturing employment share show little association with base exposure. Taken together, our stylized facts suggest that areas around the US bases were characterized by "urbanization without industrialization."

[1] In reviewing the US base-closure process, the U.S. General Accounting Office considers local purchases by military personnel, trainees, and civilian employees when assessing employment effects (U.S. General Accounting Office, 1995). A congressional hearing on Fort Ord likewise highlights the loss of civilian base jobs and secondary employment following closure (U.S. House of Representatives, 2002).

[2] Alexander Welscher, https://www.ahk-balt.org/en/publikationen/baltic-business-quarterly/bbq-autumn-2025/getting-ready-for-the-german-brigade, last accessed on October 2, 2026.

[3] https://www.reuters.com/world/small-town-germany-braces-end-decades-life-with-us-troops-2026-05-07/, last accessed on September 23, 2026.

Motivated by these stylized facts, we construct a quantitative spatial model that explicitly incorporates base-sourced consumption demand and base employment. Combined with the GNI statistics recording the bases' separate contribution to Okinawa's aggregate economy, the model allows us to quantify the effects of their consumption demand and employment while holding other local fundamentals fixed, including those that may reflect infrastructure and durable capital, noise, or pollution. Moreover, our quantitative spatial model is a general-equilibrium one, allowing us to investigate how consumption and base employment, which propagate through within-island migration, commuting, and cross-border shopping, translate into the aggregate impact on income and workers' welfare.

Our model has many locations, embedded in a larger economy, and workers choose their residence, workplace, and sector to maximize utility, paying commuting costs when residence and workplace differ. Each location has four sectors: agriculture, manufacturing, services, and the US military base sector. We assume that agriculture and manufacturing goods are tradable across locations while non-tradable services are not, although we allow workers to consume differentiated non-tradable services at different locations through cross-border shopping. Okinawan workers at the US military base do not produce any good in the Okinawan economy as their work has a militaristic value that is not traded in a market. The US base takes up a certain fraction of land in a location and pays rents for it, employs Okinawan residents, and demands tradable and non-tradable goods. In addition, developers provide floor space using tradable goods and land, implying that the presence of US bases may increase floor space prices by reducing available land for the private sector.

We calibrate the model to Okinawa in 1970 using GNI statistics that separately record US personnel's purchases, Okinawan base workers' labor income, and base land rents. Matching their observed GNI shares sets the base-sector wage scale, base rent payments, and personnel spending, disciplining the scale and composition of base-related flows. We combine these accounts with municipal residence and workplace employment by sector, commuting and shopping data, and parameters from existing studies to recover local amenities and productivity. The model exactly matches the observed population and sectoral employment distributions. Model-implied commuting patterns and floor-space rents are also closely correlated with untargeted commuting and housing-price data.

Our counterfactual isolates the effects of ongoing base spending and direct employment of local residents, the channels emphasized in policy discussions of the bases' economic benefits. To understand the effects of US-base-sourced expenditure and employment, we use our calibrated model to perform a counterfactual analysis that jointly removes expenditure and land-rent income associated with the bases and employment of Okinawan workers at the bases, while keeping base land unavailable for private use and Okinawa's aggregate worker population fixed. The counterfactual produces a sharp spatial reversal. A 10 percentage-point increase in the municipal base-area share is associated with a 4.77 percentage-point larger population decline in the counterfactual scenario. The corresponding decline in the tertiary sector is 2.16 percentage points larger, while the agricultural and manufacturing shares increase. Thus, US-base-sourced

expenditure and employment contributed substantially to urbanization without industrialization around the bases. The different tradability of goods by sector drives these results. While consumption and base employment increase demand for both tradable and non-tradable goods, non-tradable demand must be satisfied locally, whereas tradable goods can be supplied from outside Okinawa at fixed prices in the benchmark. As a result, areas around the US bases have higher service employment shares to satisfy the larger service demand. Separating the channels, we find that consumption has larger effects on civilian service employment, whereas base employment has larger effects on population concentration.

In the aggregate, the benchmark counterfactual reduces model-implied workers' welfare by 7.25% and gross national income (GNI) by 11.99%, accompanied by the growth of manufacturing and agricultural employment shares and the shrinkage of the non-tradable sector. Although the losses are large in absolute terms, they are modest relative to the scale of the initial shock: base-related consumption and employment accounted for roughly 30% of Okinawa's GNI in 1970. Once base-related demand and employment disappear, sectoral and spatial mobility can substantially cushion the economic cost. We find that sectoral reallocation is the primary adjustment margin, despite a sizable within-island migration response. Finally, returning base land to private use dampens population outflows from base-exposed municipalities and reduces the benchmark welfare loss from 7.25% to 1.95%, while leaving the economy's sectoral composition largely unchanged.

To understand how local employment gains relate to adjustment across Okinawa, we quantify the direct and spillover effects of base-related consumption and hiring using local and aggregate elasticities. Service employment gains from consumption are relatively concentrated near the source municipality, whereas manufacturing losses are more dispersed; additional base hiring likewise displaces civilian employment mainly elsewhere. These patterns show why employment gains in base-hosting municipalities alone do not capture the economy-wide adjustment to base-related inflows. Finally, by modifying our structural model, we assess the external validity of our results under alternative assumptions about external migration and manufacturing import supply. Allowing external migration leaves workers' welfare and sectoral reallocation close to the benchmark but increases the GNI loss through population outflows. Less elastic manufacturing import supply weakens, and can reverse, the aggregate manufacturing crowding-out effect because base-related demand then raises manufacturing prices and supports local production.

We close our paper by highlighting the potential importance of investment in infrastructure and durable capital accompanying the base development. Although our counterfactual analysis predicts that consumption demand and base employment induce "urbanization without industrialization," which we indeed observe in the data, the predicted tertiary-share response is smaller than the observed association. This comparison helps assess how much of the historical association can be accounted for by the ongoing demand and employment channels emphasized in the policy discussions above. We highlight one possible factor that could fill in this gap: investment in infrastructure and durable capital around the bases. Indeed, we find that infrastructure quality, proxied by water supply and road quality, is positively correlated with exposure to the

US bases. We also find that early designation of road corridors for authorized markets serving US military personnel is correlated with base exposure, suggesting that early access to military customers could have encouraged investment in infrastructure by the government or in durable capital by the private sector, yielding persistent advantages around the US bases. Conditional on base exposure and geographic controls, water supply is also positively associated with calibrated amenities, while road quality and early commercial access are positively associated with productivity. Overall, although the effects of US-sourced consumption demand and employment qualitatively match the observed pattern of spatial economic development in Okinawa, our structural analysis suggests that attributing all observed development to these ongoing flows, as a naive observation might suggest, could overstate their effects.

This paper contributes to several strands of literature. Most directly related is research on the local economic effects of military activity. Existing studies estimate how military spending and employment affect local output and employment (e.g., Hooker and Knetter 1997, 2001; Paloyo, Vance and Vorell 2010; Nakamura and Steinsson 2014; Lee 2018; Zou 2018; Biolsi 2019; Dahlberg, Martén and Öckert 2024; Brunet 2026).[4] We contribute to this literature by linking local employment responses to aggregate income and workers' welfare through sectoral reallocation and spatial spillovers. The separately recorded base-related GNI components allow us to quantify the distinct effects of spending and direct hiring on these outcomes. We also distinguish the effects of ongoing flows from the broader historical association between bases and local development, providing suggestive evidence on an additional role for past investment in infrastructure and durable capital.

Our paper also relates to research on urbanization without industrialization (Jedwab and Vollrath 2015; Henderson and Turner 2020). Research on the mechanisms underlying service-led growth and premature deindustrialization emphasizes technological change and trade (Rodrik 2016), productivity growth in consumer services (Fan, Peters and Zilibotti 2023), and differences in sectoral productivity and technology adoption (Huneeus and Rogerson 2024; Fujiwara and Matsuyama 2024). Most closely related, Gollin, Jedwab and Vollrath (2016) show how resource income supports consumption cities dominated by non-tradable services. Building on this mechanism, our quantitative spatial analysis of Okinawa shows how base-sourced consumption demand and direct employment can promote population concentration and service-oriented development around military bases while shifting aggregate employment away from manufacturing. By quantifying the distinct roles of these two channels in local and aggregate adjustment, we contribute to understanding the mechanisms behind urbanization without industrialization.

Methodologically, our framework builds on quantitative spatial models with commuting (e.g., Ahlfeldt, Redding, Sturm and Wolf 2015; Monte, Redding and Rossi-Hansberg 2018; Heblich, Redding and Sturm 2020; Miyauchi, Nakajima and Redding 2022; Dingel and Tintelnot 2023; Tsivanidis 2023; Takeda and Yamagishi 2026; see Redding 2025 for a survey). We extend this framework to incorporate a facility that purchases goods and services, employs local workers, and

[4]Existing quantitative assessments of US bases in Okinawa have largely relied on input–output analysis to estimate demand-induced output and employment effects, including studies of base employment and land redevelopment (e.g., Okinawa Prefecture Government 2007; Kusuyama 2012; Tomikawa 2018).

occupies land. Commuting, residential mobility, and shopping across municipalities transmit its effects beyond the host location, allowing us to distinguish local gains from aggregate effects. The framework can also be applied to other externally financed facilities, such as public offices and universities, that combine local spending, employment, and land use.

The rest of this paper is organized as follows. Section 2 introduces the institutional background of Okinawa and our data. Section 3 presents our stylized facts on the relationship between the US bases and urbanization without industrialization. Section 4 introduces our quantitative spatial model that incorporates consumption and base employment at the US bases. Section 5 calibrates our model using the data from Okinawa in 1970. Section 6 presents the counterfactual effects of consumption and base employment, reports direct and spillover elasticities, and examines how the benchmark results depend on external mobility and manufacturing import supply. Section 7 examines past investment in infrastructure and durable capital as a possible additional contributor to economic development. Section 8 concludes.

2 Institutional Background and Data

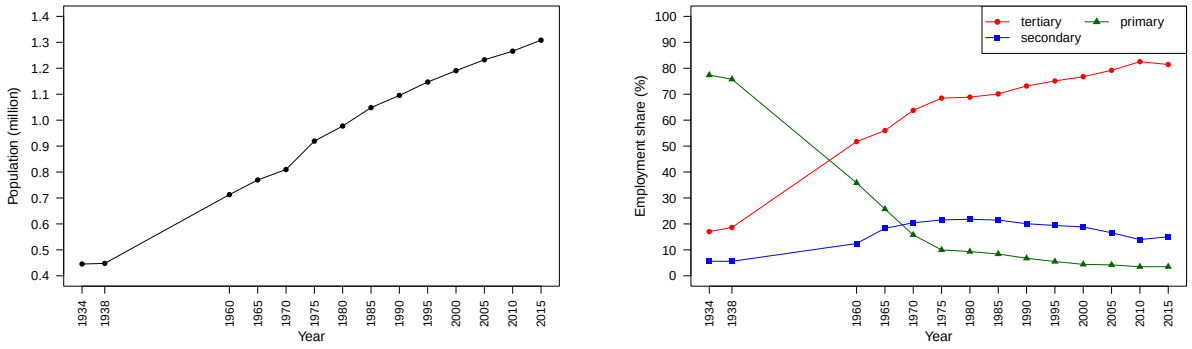
Section 2.1 introduces the institutional background of this study and Section 2.2 introduces our data.

2.1 Institutional background: US Bases in Okinawa

To facilitate the understanding of our study context, we briefly explain the institutional background of Okinawa and its relationship to the US bases.[5] The Okinawa island is located to the south-west of the main island of Japan.[6] Although Okinawa Prefecture of Japan contains islands other than the main island, we refer to the Okinawa main island by saying "Okinawa" unless explicitly noted. As of 2020, the population is around 1.3 million and its size is around 1,200 square kilometers. Figure 1a plots the long-run trend of the total population of Okinawa island, showing the increasing trend throughout the post-WW2 period. The growth in the total population accompanied substantial urbanization, especially in the central and southern parts of the island, such as Naha city and Koza city. Figure 1b documents that such an urbanization process entailed substantial growth in the service employment share. Notably, the manufacturing employment share did not see substantial growth and hit a plateau as early as in the 1960's. This development process is quite different from that of other Japanese regions. The post-war economy of Mainland Japan experienced a major shift of manufacturing sectors from light to heavy industries between the mid 1950's and the early 1970's, and then shifted to a service economy after the mid 1970's (Fujita and Tabuchi 1997), which is in line with Petty's law and is

[5]See, for instance, Okinawa Prefecture Government (2000), Taira (2012), Toriyama (2013), Franks (2023), and Oshiro (2023) for a more detailed description.

[6]Okinawa was also historically called "Ryukyus." In this paper, for simplicity, we use the name "Okinawa" throughout the paper.



(a) Total population

(b) Employment share by sector

Figure 1: Long-run dynamics of the total population and employment share in Okinawa

Note: Figures 1a and 1b plot the long-run trends of the total population and the sectoral employment share from 1934 to 2015 in the Okinawa main island, respectively. In Figure 1b, the lines with red circles, blue squares, and green triangles represent the employment shares of the tertiary, secondary, and primary sectors, respectively.

a widely observed pattern of economic development. Relative to this, Figure 1b highlights the uniqueness of Okinawa's development process.

The substantial presence of US military bases is a distinctive feature of Okinawa's postwar economy. Their spatial distribution reflects, at least in part, the sequence of occupation and military construction during the Battle of Okinawa in 1945. Figure 2 places the geography of the landings and early occupation alongside the distribution of US bases in 1970. Panel (a) marks the actual landing point of the US forces (Toguchi Beach) on April 1, 1945. The figure also presents two coastal sectors in an alternative plan for landing: a Marine landing between Minatogawa and Chinen Point, followed by an Army landing between Kuba and Yonabaru. The selection criteria for the landing point were driven primarily by military and logistical requirements, rather than by the local economy's development potential (see Appendix B.3 for more details). After the landing, US forces advanced rapidly through central Okinawa and into the north during the first two weeks, occupying the colored areas in Panel (a). In contrast, progress southward was much slower because Japanese forces concentrated their defenses in the southern part of the island, including the positions around Shuri. Resistance there delayed the occupation of southern locations even when they were relatively close to the landing beaches, and fighting in the south continued for several months. The early availability of land for base construction therefore depended on both proximity to the landing area and the geography of Japanese resistance.

Importantly, the timing of the occupation, at least in part, predicts the local presence of the US bases because the US military started constructing the bases even during the battle of Okinawa. Indeed, they started constructing the bases in an early-occupation area as they needed facilities to support the ongoing Okinawa campaign and to prepare for further operations against the Japanese mainland. Panel (b) shows the resulting concentration of US bases in central and northern Okinawa in 1970, which visually exhibits a strong correlation with the early-occupation

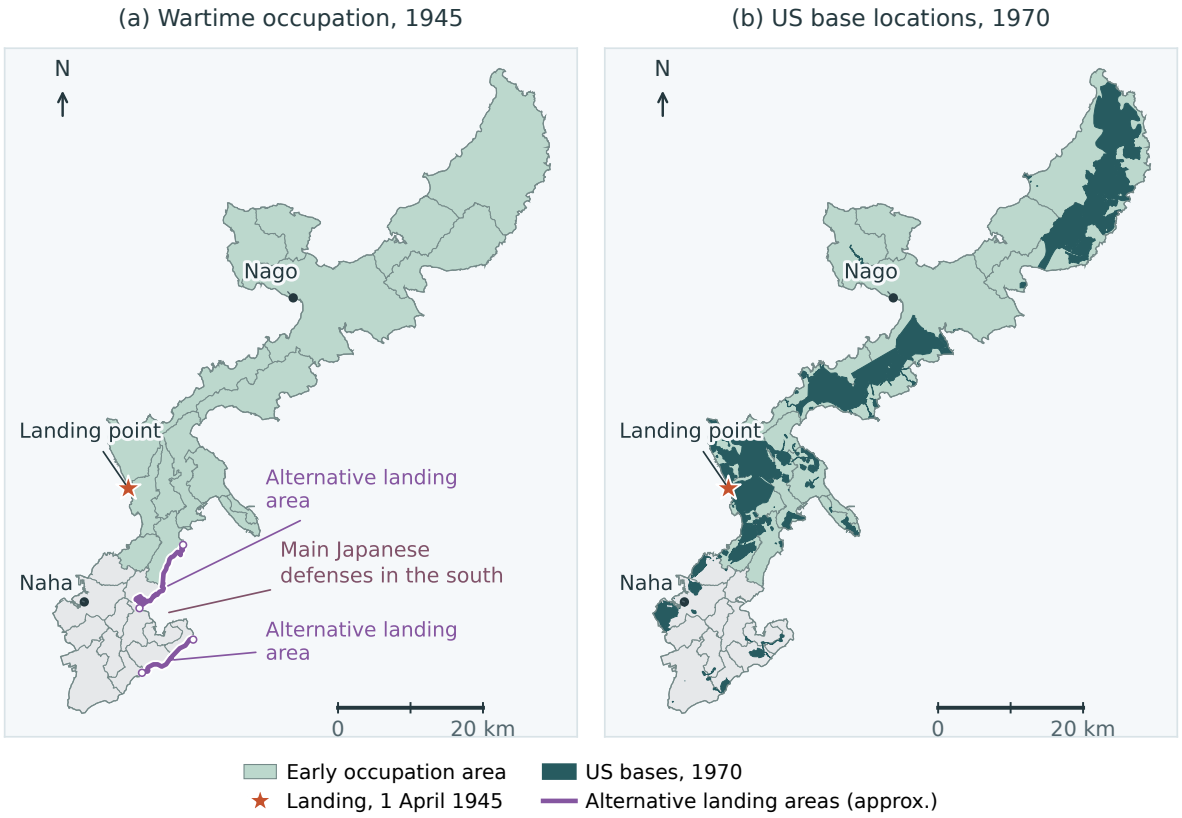


Figure 2: Wartime occupation and US base locations in Okinawa

Note: The stars labelled "Landing point" mark the reference point at Toguchi Beach in both panels. In panel (a), the northern and southern purple sectors labelled "Alternative landing area" correspond to Kuba–Yonabaru and Minatogawa–Chinen Point, respectively. Panel (a) also indicates the broad concentration of Japanese defenses in the south. Panel (b) shows US bases as of 1970. Both panels use the same scale and shade the early-occupation municipalities described in Appendix B.3; municipal boundaries are not precise wartime fronts or defensive lines. Only the main island is displayed. Sources: project GIS data and Okinawa Prefecture base-boundary data; the Cabinet Office account; the US Army, Marine Corps, and Navy histories cited in the text. Candidate-sector locators use the CODH historical place-name dataset and the Okinawa Prefectural Archives.

areas in Panel (a).[7] Overall, these historical accounts suggest that the US base locations were determined largely for non-economic reasons.[8]

Even after the war, the US bases have exhibited their substantial presence in the Okinawa island. The US bases constitute a large share of the geographical area of Okinawa Island, reaching up to about 20% in 1970, and it stabilizes at 15% today. In addition to their large geographical size, the US bases are an important economic actor that has greatly affected the Okinawan economy. Broadly speaking, the US bases are important as consumers, employers, and tenants. As consumers, the US military personnel consume goods and services provided

[7]Post-war land acquisition and construction, including during the 1950s, also contributed to the base construction. During the 1950s, the US administration used forcible land acquisition without seriously considering its economic consequences to expand military facilities for the Korean War and subsequent strategic needs (Taira 2012; Toriyama 2013; Franks 2023).

[8]Appendix B.1 examines prewar trends. Consistent with this, we do not find evidence of the pre-war trend that could explain a large part of post-war economic development.

by the Okinawan people, such as foods, restaurants, and daily commodities.[9] As employers, the US bases employ Okinawan people for their daily operations, such as cleaning and babysitting. As tenants, the US bases pay the land rents to landowners. In all these three dimensions, the US bases created large consumption demand and employment around the US bases.

Figure 3 shows the contribution of the US bases as the share of Okinawa's Gross National Income (GNI), separately for goods and service consumption, labor income payments for employment at US bases, and land rent payments for the US base area to local landlords. Note that these data present a distinctive feature of Okinawa for measuring the bases' economic impacts: In other contexts such as the US and mainland Japan, their GNI statistics do not separately record the contribution of the US bases. The US bases constituted around 40% of Okinawa's GNI in 1955. The share declined sharply in the 1960's and 1970's.[10] The share has been around 5% for the past 40 years. Unpacking the US-base sourced income before the return of Okinawa to Japan in 1972, the income from consumption of goods and services took up the highest share, the labor income of Okinawan residents at the US bases came next, and the land rent payment had the lowest share. After the return, however, the GNI shares of these three sources of income are close to each other due to the substantial drop in the first two. Given the very large share of the US-base sourced income in the GNI, reaching up to 40% in the 1950's, we may reasonably conjecture that the US bases have influenced the post-war economic development of Okinawa. As a first step for understanding such an impact, the next section documents stylized facts on how areas that were highly exposed to the US bases are characterized by their urbanization and industrial structure.

2.2 Data

We collect, and newly digitize when needed, the municipality-level data on various outcomes. Below we briefly describe the data used for our baseline analysis. Appendix A provides more details on the data description and Table A.1 summarizes our data sources.

Spatial unit and scope of the analysis. We focus on the data of Okinawa Prefecture. Okinawa Prefecture consists of the main island, which accounts for about 50% of the total area and about 90% of the population as of today, and the other smaller islands. Since most US bases are located in the Okinawa main island and a very large part of economic activity takes place there, we focus on the Okinawa main island in our main analysis unless explicitly noted.

Throughout the paper, our analysis is conducted at the municipality level. There are 35 municipalities on Okinawa Island as of 1970, although the number of municipalities changes over time due to mergers and divisions (see Figure 5 for an illustration of municipal boundaries

[9] Military contractors often involve large spending on weapons. However, the available data on the spending category of the US indicates that the US sourced virtually no weapons from Okinawa (Reversion Issues Study Group n.d.). The only primary contract for military activities in Okinawa is construction, which we classify as manufacturing following the Japan Standard Industrial Classification.

[10] This is due to the growth of the Okinawan economy and the shrinkage of the US-base sourced income due to reversion of Okinawa and the US's withdrawal from the Vietnam war.

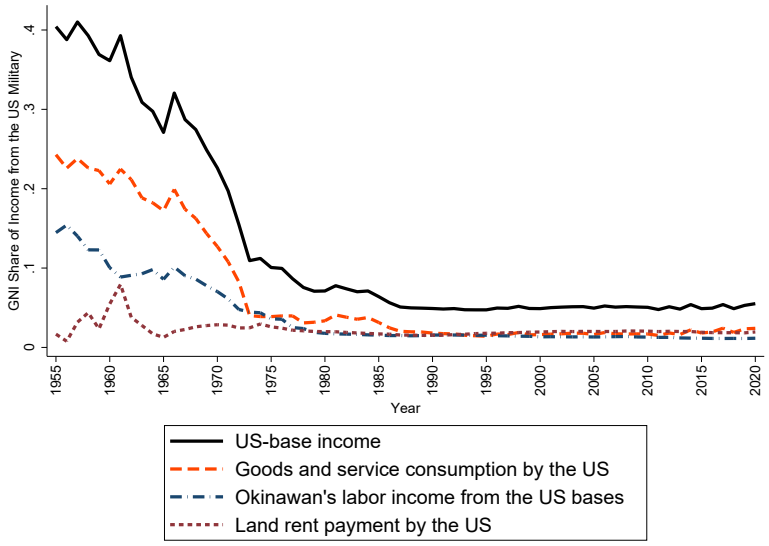


Figure 3: GNI share of the US bases in Okinawa island

Note: This figure plots the income sourced from the US military as the share of Okinawa's Gross National Income from 1955 to 2020. We used statistics from the Ryukyus Statistical Yearbook (*Ryukyus Toukei Nenkan*) and Okinawa Statistical Yearbook (*Okinawa Toukei Nenkan*). The black solid line is the sum of the three sources of US-base income: goods and service consumption by the US, labor income of Okinawans who are employed at the US bases, and land rent the US pays to Okinawan landowners for their bases.

in 1970). To address the mergers and divisions of municipalities during our sample period, we aggregate municipalities to the largest definitions of municipalities when our analysis involves different years (see Appendix A for details).

Population. For population at the municipality level after 1945, we use the Population Census data, which have been available every five years since 1950. The US-occupation-period data are newly digitized. The population includes only the Japanese population, implying that the US citizens are not covered by the Census. The population census also reports the number of workers by industry category living in each municipality. The census also tells us the number of people working for the US bases. In addition, we obtain population data by industry category in 1934 and 1938 from the Okinawa Prefectural Statistical Yearbook (*Okinawa-ken toukei sho*).

Employment. At the workplace level, we use the 1970 Population Census to obtain the number of workers by industry category in each municipality. The census also reports workplace counts of Okinawan workers employed at the US bases, which we treat as a separate sector in the model. Although sourced from the same dataset, workplace employment by sector differs from the number of resident workers in that sector because of commuting across municipalities.

Size of the US bases. In our reduced-form analysis, we use the area share of the US bases in a municipality taken from official statistics, which we use as an exposure index for base-sourced consumption and base employment. This measure is motivated by the assumption that the

amount of demand of the US bases is proportional to the size of each base, and each base spends its money locally. We also observe in Komeito (1969) the number of US military personnel and Okinawan workers at each base. We use these data to validate our assumption that the US base area share indeed captures the base's economic importance (see Section B.1).

GNI and US-base-sourced income. Gross National Income (GNI) is reported annually from 1955 in the Ryukyus (Okinawa) Statistical Yearbook. We use the prefecture-wide series as an approximation for the main island. The statistics distinguish base-related purchases of goods and services, salaries of Okinawan base workers, and land rents.[11] Figure 3 shows these income shares over time, and Appendix A provides further details.

Road network and trip data. We use the travel-level microdata of the Okinawa Main Island Central and Southern Urban Area Person Trip Survey in 1977.[12] For each respondent, the survey asks for details of the trips they make in a day. The survey asks about the purpose of each trip and the worker's industry, allowing us to identify commuting and shopping trips and to classify commuters by sector. Based on these data, we construct separate bilateral commuting matrices for workers in agriculture, manufacturing, and non-tradable services, as well as a bilateral shopping matrix, and use them to estimate gravity models. The bilateral travel distance is calculated based on the newly-digitized road network data in 1973, which are taken from the Geospatial Information Authority of Japan Map.[13]

3 Stylized Facts

We first document the cross-sectional relationship between the economic outcomes and the area share of the US bases in 1970, when the US-base sourced income accounted for around 30% of the Gross National Income of Okinawa (Figure 3). Figure 4 presents the scatterplot of each municipality, where the horizontal axis is the share of land occupied by the US bases and the vertical axis shows the changes in various economic outcomes from 1938 to 1970. Here, we take the difference from 1938, the pre-WW2 period with no US base, to investigate economic development relative to the period without the US bases. That said, Appendix B.2 shows that our stylized facts also hold without taking the difference. We also show a corresponding fitted line and report its slope, along with the standard error.

We measure urban concentration as population density per unit of land outside US bases. Figure 4a relates base exposure to the log change in population density, while Figure 4b reports the corresponding change in population as a complementary measure of the number of residents.

[11] US-base-sourced income includes receipts from base-related transactions, including payments financed by the Japanese government, but excludes economic aid and intergovernmental transfers, including base-related grants to local governments.

[12] The data do not cover the northern part of the Okinawa island. We have also used the same survey conducted in 1989 and 2006 and found that our results are hardly affected.

[13] Digitizing the road network data in 1930 and 2005, we find that using these alternative road network data hardly affected the bilateral travel distance, suggesting that the road network did not experience a large change.

Population density adjusts for the amount of land available for civilian use: its postwar change reflects both population change and the reduction in available land caused by base occupation.[14] Both measures show that municipalities with greater base exposure experienced stronger urbanization. The relationship is also economically strong: the estimated β_t for population density in Figure 4a is 3.045, implying that a 10 percentage-point increase in the US-base area share is associated with a roughly 36% higher ratio of 1970 to 1938 population density. Overall, we establish the following Fact 1:

Fact 1 (Urbanization around the US bases). *Municipalities with a higher share of land occupied by US military bases tend to have higher urbanization rates, whether urbanization is measured by population size or population density.*

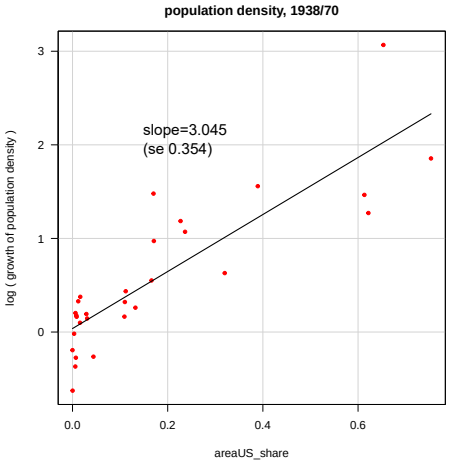
Such rapid urbanization around the US bases accompanied significant growth of the service sector, but not manufacturing. Figure 4c shows that municipalities with higher US base shares experienced rapid growth in the service sector. The association is strong: an approximately 10 percentage-point increase in the US base share is associated with a 5 percentage-point increase in the service employment share. In stark contrast, Figure 4d shows no association between manufacturing employment share and the US base area share.[15] As a result, almost the entire decline in the agricultural employment share around the US bases, shown in Figure 4e, was absorbed by the increase in the service employment share. These patterns can be summarized by the following Fact 2:

Fact 2 (Service-sector growth without industrialization around the US bases). *Municipalities with a higher area share of the US bases were characterized by the rapid growth in the employment share of the service sector and the decline of the agriculture employment share. In contrast, the change in the manufacturing employment share had little association with the US base area share.*

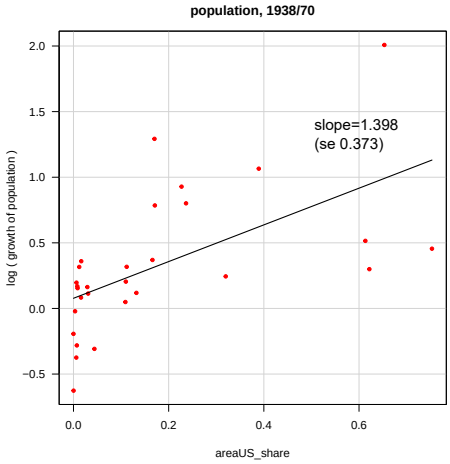
Appendix B reports additional analyses of these stylized facts and their robustness. First, we find that the stylized facts also hold when the vertical axis is replaced with the corresponding levels in 1970, rather than the 1970–1938 difference. Second, we consider the potential endogeneity of the US base locations by (i) controlling for topographic conditions and (ii) using the landing point and the occupation timing shown in Figure 2(a) as an instrumental variable, both of which do not change the main conclusion. Third, we do not find statistically-significant evidence that pre-war trends in population density and sectoral employment shares drive our results. Population density exhibits some positive pre-trend but its magnitude is far from explaining the post-war growth in population density, Sectoral employment shares also show no statistically significant pre-trends. Fourth, using the predetermined 1956 base share to address

[14] Prewar population density divides population by municipal area; postwar population density divides population by municipal area minus US-base area.

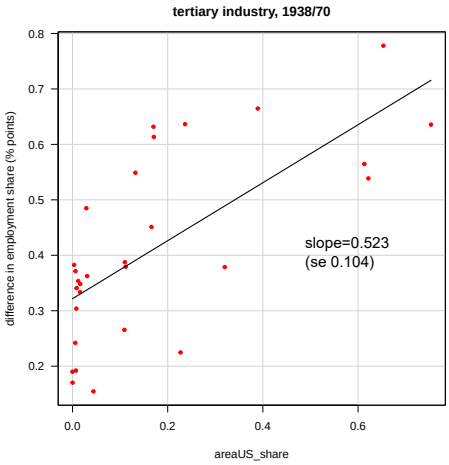
[15] Following the Japan Standard Industry Classification, we include the construction sector as manufacturing. We expect this tends to create a positive association between manufacturing and US base area share because the US military demands construction for the base operation. Despite this, we find little evidence of a positive association between these two.



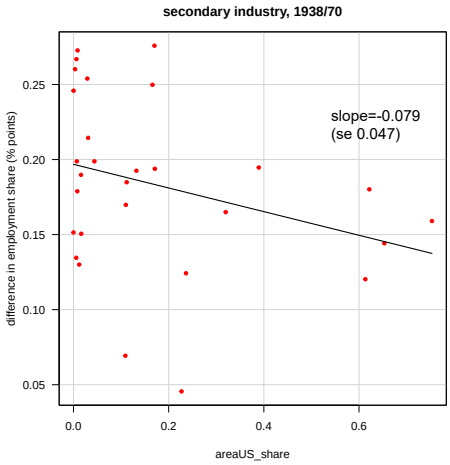
(a) Population density



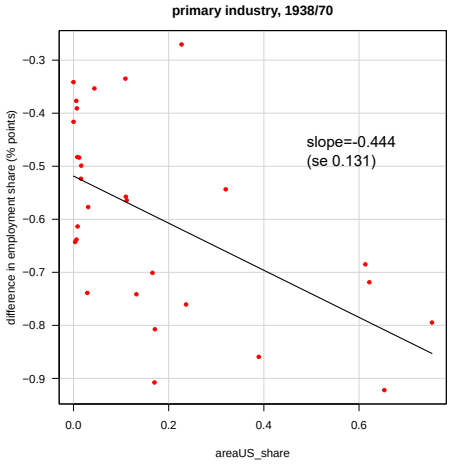
(b) Population level



(c) Service share



(d) Manufacturing share



(e) Agriculture share

Figure 4: Relationship between US bases, Urbanization, and Industry Structure (1970)

Note: We present a scatterplot of each municipality using the 1970 data. The horizontal axis is the US base area share. Each dot represents a municipality. As the vertical axis, we plot the values relative to those of 1938, a pre-war period without any US base. Figure 4a takes the change in the log population density, Figure 4b takes the change in the log population, Figure 4c takes the change in the service employment share, Figure 4d takes the change in the manufacturing employment share, and Figure 4e takes the change in the agriculture employment share. The linear fitted lines are also shown.

the potential endogeneity arising from base returns yields the same qualitative conclusions. Fifth, our results are robust to alternative measures of base importance based on the densities of Okinawan base workers or U.S. military personnel. Sixth, the association between U.S. bases and local economic outcomes remains qualitatively similar across different years, although the magnitude generally declines over time as the aggregate importance of the bases in the Okinawan economy diminishes. Finally, supplementary sectoral evidence suggests that the positive employment effects are particularly pronounced in consumer-oriented service activities such as wholesale and retail and intangible services.

These stylized facts document a strong relationship between US-base exposure and urbanization without industrialization. They do not, however, establish how much ongoing base-related expenditure and employment contribute to this pattern or reveal their aggregate income and welfare implications. Municipal comparisons may reflect both current base-related flows and the legacy of past investment, while migration, commuting, and shopping transmit effects across municipal boundaries. To address these questions, we develop a quantitative spatial model disciplined by GNI data on base-related purchases, labor income, and land rents. We evaluate a counterfactual that removes base-related expenditure, land-rent income, and employment while holding other local fundamentals fixed at their calibrated 1970 levels. This exercise quantifies how ongoing base-related consumption demand and employment shape urbanization and sectoral composition and affect aggregate income and welfare. The next section introduces the model.

4 Model

We consider an economy (the Okinawa main island) consisting of several locations (municipalities). The total number of locations in this economy is I, and each location is indexed by $i = 1, 2, \cdots, I$. The economy as a whole is inhabited by a measure L of workers, which is exogenously given as we assume a closed economy in the baseline model.[16] Workers inelastically supply a unit of labor and earn the wage. Each worker is allowed to reside in a location different from their workplace, implying that the mass of residents in location i (R_i) may differ from that of workers working in location i (L_i).

The present model has four sectors: agriculture (A), manufacturing (M), non-tradable services (N), and the US military base sector (B). The agricultural and manufacturing sectors are perfectly competitive and produce homogeneous goods. We assume that Okinawa is a small open economy and both goods are freely traded, indicating that the prices of agricultural and manufacturing goods are common across all locations in Okinawa. The non-tradable service sector is monopolistically competitive, implying that firms in this sector produce horizontally differentiated goods. These goods cannot be traded across locations and are consumed only locally, although we allow workers to make a trip to consume non-tradable services outside their residence. The US military base sector does not produce any good, but some labor force is

[16] Section 6.3 examines the sensitivity of the results to relaxing this fixed-population assumption.

required to support the operation of the US army stationed in Okinawa.

Each location is endowed with a fixed amount of land, S_i. However, some parts of the land may be occupied by the US military bases. Let $m_i \in [0, 1]$ represent the share of land used as US military bases in location i. We posit that m_i is determined for a military reason and is exogenously given to the model. Then, a fraction $1 - m_i$ of the land is available for developing floor space that can be used for private purposes, including housing and the production of both tradable goods and non-tradable services. The floor space is supplied by perfectly competitive developers, and we let H_i represent the amount of floor space in location i.

4.1 Consumption

Workers. Workers derive utility from consuming agricultural and manufacturing goods, non-tradable services and housing. We assume that the preference of a worker ν residing in location i and working in sector $s \in \{A, M, N, B\}$ in location j is represented by a Cobb-Douglas function.[17] The indirect utility is given by

$$U_{ijs}(\nu) = \frac{\varepsilon_{ijs}(\nu) w_{js}}{\tau_{ij} \mathbb{P}_i^{\alpha} Q_i^{1-\alpha}}, \tag{1}$$

where $\alpha \in (0, 1)$ is the expenditure share on goods and services; $\mathbb{P}_i$ is the price index of goods and services in location i; Q_i is the floor-space rent in location i; w_{js} is the wage earned in sector s in workplace j; and τ_{ij} is the utility cost of commuting from residence i to workplace j. The term $\varepsilon_{ijs}(\nu)$ is an idiosyncratic preference shock for the residence–workplace–sector choice, which captures idiosyncratic reasons for living and working in particular locations and sectors. We assume that these shocks are independently drawn from a Fréchet distribution, $\Pr(\varepsilon_{ijs} \leq x) = \exp(-A_{is} x^{-\theta})$, where $A_{is} > 0$ governs the amenity of residing in location i and working in sector s, and $\theta > 1$ is the (inverse) measure of the dispersion of idiosyncratic preferences. Workers observe the shocks before choosing where to live, where to work, and in which sector to work.

The price index, $\mathbb{P}_i$, is defined as

$$\mathbb{P}_i \equiv \left[\left(P_i^T \right)^{1-\kappa} + \left(\mathbb{P}_i^N \right)^{1-\kappa} \right]^{\frac{1}{1-\kappa}}, \tag{2}$$

where P_i^T is the price index of tradable goods (i.e., agricultural and manufacturing goods); $\mathbb{P}_i^N$ is the price index of non-tradable services faced by residents in location i; and $\kappa > 0$ is the

[17] Preferences are homothetic: at given prices, expenditure shares do not vary with income, as in Faber and Gaubert (2019) and Fajgelbaum and Redding (2022). To gauge the omitted income effect, we use an illustrative Engel-curve calculation similar in spirit to Faber and Gaubert (2019). Using Comin, Lashkari and Mestieri's (2021) estimated service expenditure elasticity of 1.20, including housing, a 12% decline in household expenditure, comparable in magnitude to our benchmark GNI decline, would lower an initial service expenditure share of 40% by about one percentage point at fixed prices. This modest implied response suggests that the direct income channel is of limited quantitative importance in our setting, motivating our use of homothetic preferences for tractability.

elasticity of substitution between tradable goods and non-tradable services.[18]

The price index of tradable goods, P_i^T, is defined over the prices of agricultural goods in location i (P_i^A) and manufacturing goods in location i (P_i^M) as follows: $P_i^T \equiv \Upsilon(P_i^A, P_i^M)$, where $\Upsilon(\cdot,\cdot)$ is a homothetic aggregator. Since the prices of agricultural and manufacturing goods are common across locations because of perfect competition and free trade, we can suppress the location subscript and write $P_i^A = P^A$ and $P_i^M = P^M$. It then follows that the price index of tradable goods is also common across locations: $P_i^T = P^T = \Upsilon(P^A, P^M)$. In our quantitative analysis, we normalize the price index of tradable goods to one, i.e., $P^T = 1$.[19]

We assume that workers can consume non-tradable services not only at their residence but also in other locations. That is, residents in location i can make trips for shopping and enjoy non-tradable services provided in locations other than their residence. For simplicity, it is assumed that the set of destinations and the itinerary of shopping trips are identical across all residents in location i. The non-tradable price index $\mathbb{P}_i^N$ takes the following form:

$$\mathbb{P}_i^N \equiv \left[\sum_j \xi_{ij} (P_j^N)^{1-\sigma}\right]^{\frac{1}{1-\sigma}} \tag{3}$$

$$\text{with} \quad P_j^N \equiv \left[\int_0^{n_j} (p_j^N(\iota))^{1-\sigma} \mathrm{d}\iota\right]^{\frac{1}{1-\sigma}}, \tag{4}$$

where P_j^N is the price index of non-tradable services in location j; $p_j^N(\iota)$ is the price of a non-tradable service indexed by ι and produced in location j; $\sigma > 1$ is the elasticity of substitution among non-tradable varieties; $\xi_{ij} \in [0,1]$ satisfies $\sum_j \xi_{ij} = 1$ and measures the extent to which residents in location i spend on consumption activities in location j; and n_j is the mass of varieties produced in location j. It is straightforward to assume that the farther away the locations i and j, the smaller ξ_{ij}. Note that this formulation nests the standard case of purely local consumption of non-tradables by letting $\xi_{ii} = 1$ and $\xi_{ij} = 0$ for $j \neq i$.

By Roy's identity, the shares of expenditure on agricultural goods (χ_i^A), manufacturing goods (χ_i^M) and non-tradable services (χ_i^N) are derived as follows:

$$\chi_i^A = \alpha \left(\frac{P^T}{\mathbb{P}_i}\right)^{1-\kappa} \frac{\partial \ln \Upsilon(P^A, P^M)}{\partial \ln P^A},$$

$$\chi_i^M = \alpha \left(\frac{P^T}{\mathbb{P}_i}\right)^{1-\kappa} \frac{\partial \ln \Upsilon(P^A, P^M)}{\partial \ln P^M},$$

$$\chi_i^N = \alpha \left(\frac{\mathbb{P}_i^N}{\mathbb{P}_i}\right)^{1-\kappa}. \tag{5}$$

[18]Demand weights are normalized to one because they are not separately identified from sectoral productivity; calibrated productivity therefore incorporates these weights. Appendix C.1 provides details.

[19]In Section 6.3, we endogenize the manufacturing goods prices, which requires us to specify $\Upsilon(P^A, P^M)$ to model the endogenous response of P^T. We assume that $\Upsilon(P^A, P^M)$ takes a Cobb-Douglas form in this extension.

The homotheticity of Υ implies that $\partial \ln \Upsilon / \partial \ln P^A + \partial \ln \Upsilon / \partial \ln P^M = 1$ holds and that the share of expenditure on tradable goods is $\chi_i^A + \chi_i^M = \alpha \left(P^T / \mathbb{P}_i\right)^{1-\kappa}$. Note also that the share of expenditure on non-tradable services provided in location j and spent by residents in location i is $\xi_{ij} \chi_i^N \left(P_j^N / \mathbb{P}_i^N\right)^{1-\sigma}$.

Landowners. We assume that the land is owned by local landowners. Unlike workers, local landowners neither supply labor to any sector nor commute to other locations. The total income of local landowners in location i is the aggregate land rents from the land owned by them, which we denote by Ω_i. Let $\mathbb{Q}_i^{US}$ represent the land rent for the US military bases in location i, which is exogenously given to the model. Note that $\mathbb{Q}_i^{US}$ can differ from the land rent in the private market, which we denote by $\mathbb{Q}_i$, because landowners were forced to lend the land even though $\mathbb{Q}_i^{US} < \mathbb{Q}_i$ holds, which is the empirically relevant case. Since location i's total land endowment is S_i and a fraction m_i of it is occupied by the US bases, Ω_i is determined as follows:

$$\Omega_i = [(1 - m_i)\mathbb{Q}_i + m_i \mathbb{Q}_i^{US}] S_i. \tag{6}$$

We assume that local landowners consume the same set of goods and services as workers except for housing and that the expenditure shares are also the same, implying that they spend a fraction χ_i^A / α of income on agricultural goods, a fraction χ_i^M / α on manufacturing goods, and a fraction χ_i^N / α on non-tradable services.

The US military personnel. The US military bases are extensively distributed on Okinawa Island. Each base hosts some American military personnel for operation, and the mass of those working in the base in location i is denoted by L_i^{US}, which is determined by military considerations and exogenously given to the model.

We assume that they are endowed with a fixed amount of housing in the base area where they work.[20] This implies that they do not account for housing when they make a consumption decision and that their residence and workplace are in the same location. We also assume that they consume both tradable goods and non-tradable services and that their expenditure shares are the same as those of workers. Therefore, they spend a fraction χ_i^A / α of income on agricultural goods, a fraction χ_i^M / α on manufacturing goods, and a fraction χ_i^N / α on non-tradable services. We denote by e_{US} the individual expenditure of the US military personnel on Okinawa Island.[21]

4.2 Production

In location i, sector $s \in \{A, M, N\}$ hires a measure L_{is} of workers and uses a measure H_{is} of floor space as production factors.

[20]The US military personnel are required to live in housing within the US base area. As an exception, some are eligible to live outside bases, but they are still required to live in housing authorized by the military and such housing tends to be located around the bases (Tomochi 2009).

[21]The US personnel may also receive income spent elsewhere (e.g., remittance to the US mainland). We do not consider such income to focus on the Okinawan economy.

Agricultural sector. We assume that the agricultural sector's production technology is represented by a Cobb-Douglas function and exhibits constant returns to scale. Then, profit maximization and free entry imply that the unit cost of production is equal to the price of agricultural goods:

$$P^A = \frac{w_{iA}^{1-\gamma^A} Q_i^{\gamma^A}}{b_i^A}, \tag{7}$$

where b_i^A is productivity of agricultural goods in location i and $\gamma^A \in [0, 1]$ is the agricultural sector's cost share of floor space. We do not consider agglomeration economies in the agricultural sector, indicating that b_i^A is an exogenous parameter that reflects location i's geographic conditions such as soil and terrain.

Manufacturing sector. As in the case of the agricultural sector, we assume that the manufacturing sector's production technology is given by a Cobb-Douglas function and exhibits constant returns to scale. Then, the unit cost of production is equal to the price of the manufacturing goods:

$$P^M = \frac{w_{iM}^{1-\gamma^M} Q_i^{\gamma^M}}{B_i^M}, \tag{8}$$

where B_i^M is productivity of manufacturing goods in location i and $\gamma^M \in [0, 1]$ is the manufacturing sector's cost share of floor space.

To incorporate the agglomeration economies in the manufacturing sector, we assume that B_i^M takes the following form:

$$B_i^M = b_i^M L_{iM}^{\eta}, \tag{9}$$

where b_i^M is the exogenous production fundamental that captures the geographic features such as good access to rivers or the sea, and $\eta > 0$ measures the strength of agglomeration forces.[22]

Non-tradable service sector. The non-tradable service sector is characterized by monopolistic competition, and each variety is produced by a single firm under increasing returns to scale. To start production, f_i^N units of a Cobb-Douglas composite of labor and floor space are needed as a fixed input. Then, producing a unit of output requires $1/b_i^N$ units of the same Cobb-Douglas composite of labor and floor space, where b_i^N is an exogenous parameter that measures productivity of non-tradable services in location i. We let $\gamma^N \in [0, 1]$ represent the non-tradable service sector's cost share of floor space, implying that the total costs of producing $x_i^N(\iota)$ units of output of variety ι are given by

$$c_i^N(\iota) = \left(f_i^N + \frac{x_i^N(\iota)}{b_i^N} \right) w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}. \tag{10}$$

[22]We assume that there is no spillover from the US military personnel and Okinawans working for the US bases. We believe that no spillover is natural given that the military activities are largely unrelated to private economic activities. Indeed, Okinawan residents were not even permitted to enter the US bases except in special circumstances.

Since consumers' preferences are characterized by the constant elasticity of substitution among varieties, the profit-maximizing price set by a firm in location i exhibits the constant-markup property:

$$p_i^N = \frac{\sigma}{\sigma - 1} \frac{w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}}{b_i^N}, \tag{11}$$

where we omit the firm index ι because there is no longer heterogeneity across firms in the same location. Thus, the zero profit condition implies that the equilibrium output for each firm in location i is $x_i^N = (\sigma - 1) f_i^N b_i^N$. Plugging this into (10), the total costs incurred by a firm in location i can be calculated as $\sigma f_i^N w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}$. Because each variety is produced by a single firm, the total mass of firms producing non-tradable services in location i is represented by n_i. The aggregate payments to labor and floor space are $w_{iN} L_{iN} = \sigma(1 - \gamma^N) f_i^N n_i w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}$ and $Q_i H_{iN} = \sigma \gamma^N f_i^N n_i w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}$, respectively. From these two equations, the equilibrium mass of firms in the non-tradable service sector in location i can be expressed as follows:

$$n_i = \frac{1}{\sigma f_i^N} \left(\frac{L_{iN}}{1 - \gamma^N} \right)^{1-\gamma^N} \left(\frac{H_{iN}}{\gamma^N} \right)^{\gamma^N}. \tag{12}$$

Substituting (11) and (12) into (4), the price index of non-tradable services in location i can be written as

$$P_i^N = \frac{w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}}{B_i^N}, \tag{13}$$

where B_i^N is given by

$$B_i^N = \widetilde{b}_i^N L_{iN}^{\frac{1-\gamma^N}{\sigma-1}} H_{iN}^{\frac{\gamma^N}{\sigma-1}}, \tag{14}$$

and $\widetilde{b}_i^N$ is defined as $\widetilde{b}_i^N \equiv (\sigma - 1) \left[\sigma^\sigma (1 - \gamma^N)^{1-\gamma^N} (\gamma^N)^{\gamma^N} f_i^N \right]^{1/(1-\sigma)} b_i^N$. Note that B_i^N increases in L_{iN} because $(1 - \gamma^N)/(\sigma - 1) > 0$ holds, implying that the non-tradable service sector also exhibits agglomeration economies. Finally, substituting equation (13) into equation (3) gives the price index for workers living in i as follows:

$$\mathbb{P}_i^N = \left[\sum_j \xi_{ij} \left(\frac{w_{jN}^{1-\gamma^N} Q_j^{\gamma^N}}{B_j^N} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \tag{15}$$

The US military base sector. Although the US military base sector does not produce any good transacted in the market, it needs to employ some Okinawan people to support the operation of the US army.[23] Note the distinction between the Okinawan people working in the US bases and the US military personnel, who are Americans. The mass of Okinawan workers demanded by the base in location i is represented by L_{iB}, which is exogenously given to the model because it

[23] We can consider that as in Becker, Heblich and Sturm (2021), US bases produce global public goods (i.e., military forces), which is irrelevant for location choices within the Okinawa island.

is determined based on military considerations.

We denote the set of locations hosting the US bases by $\mathcal{I}_B$ and let the number of such locations be I_B (i.e., $|\mathcal{I}_B| = I_B$). For locations included in $\mathcal{I}_B$, the wage in the base in location i, w_{iB}, is endogenously determined such that the labor supply matches the demand in each location. For locations without bases, we let $w_{iB} = 0$ hold because there is no demand for labor in the base sector in such locations.

4.3 Floor space supply

Floor space is supplied by a perfectly competitive construction sector that uses land and tradable goods as inputs. Following Epple, Gordon and Sieg (2010) and Combes, Duranton and Gobillon (2021), we assume that the production function for the floor space takes a Cobb-Douglas form and the corresponding cost function is given by $C_i(H_i) = \varkappa \left(P^T\right)^{\mu} \mathbb{Q}_i^{1-\mu} H_i$, where $\mu \in [0, 1]$ is the cost share of tradable inputs and $\varkappa$ is a positive constant that governs the level of marginal cost.[24]

Although the land endowment of location i is S_i, a fraction m_i is occupied by the US bases, indicating that the amount of land available for development is $(1 - m_i)S_i$. Thus, Shephard's lemma yields

$$(1 - m_i)\mathbb{Q}_i S_i = (1 - \mu) Q_i H_i. \tag{16}$$

This equation allows us to compute the land rents, given the floor space rents and the total floor space supply.

Note also that perfect competition implies $Q_i = \varkappa \left(P^T\right)^{\mu} \mathbb{Q}_i^{1-\mu}$. Substituting (16) into this equation and arranging the terms, we obtain the following floor space supply function:

$$H_i = (1 - m_i) S_i \left(\frac{Q_i}{P^T}\right)^{\phi},$$

where $\phi \equiv \mu/(1 - \mu)$ represents the floor space supply elasticity.[25]

4.4 Market equilibrium

Labor market. Each worker chooses the residence–workplace–sector combination that maximizes utility in (1). Since $\varepsilon_{ijs}(\nu)$ follows a Fréchet distribution, the probability that a worker chooses to live in location i and work in sector s in location j is

$$\lambda_{ijs} = \frac{A_{is} \left(w_{js} \mathbb{P}_i^{-\alpha} Q_i^{-(1-\alpha)}\right)^{\theta} \tau_{ij}^{-\theta}}{\sum_k \sum_l \sum_r A_{kr} \left(w_{lr} \mathbb{P}_k^{-\alpha} Q_k^{-(1-\alpha)}\right)^{\theta} \tau_{kl}^{-\theta}}. \tag{17}$$

[24] Yoshida (2018) and Kii, Tamaki, Kajitani and Suzuki (2022) show evidence that the Cobb-Douglas housing production function is also a good approximation of reality in Japan.

[25] Here, without loss of generality, we simplify notations by setting the unit of floor space such that $\varkappa = (1 - \mu)^{-(1-\mu)}$ holds.

Summing these probabilities across all workplaces for a given pair of residence i and sector s, we obtain the probability that a worker chooses to live in location i and work in sector s:

$$\lambda_{is}^{R} \equiv \sum_{j} \lambda_{ijs}. \tag{18}$$

Therefore, the probability that a worker commutes to location j conditional on living in location i and working in sector s is

$$\lambda_{ij|is} \equiv \frac{\lambda_{ijs}}{\lambda_{is}^{R}} = \frac{w_{js}^{\theta}\tau_{ij}^{-\theta}}{\sum_{l} w_{ls}^{\theta}\tau_{il}^{-\theta}}. \tag{19}$$

Let R_{is} represent the mass of workers who live in location i and work in sector s. By equation (18), R_{is} is determined as follows:

$$R_{is} = \lambda_{is}^{R} L. \tag{20}$$

Using equation (19), we obtain the mass of workers in location i for each sector:

$$\begin{aligned} &L_{is} = \sum_{k} \lambda_{ki|ks} R_{ks} \quad \text{for all } i \text{ and } s \in \{A, M, N\}, \\ &\text{and} \quad L_{iB} = \sum_{k} \lambda_{ki|kB} R_{kB} \quad \text{for } i \in \mathcal{I}_B. \end{aligned} \tag{21}$$

Note that the mass of US-base workers in location i, L_{iB}, is exogenously given and locations without bases exhibit $L_{iB} = 0$. The masses of workers living and employed in location i are, respectively, expressed as

$$R_i = \sum_{s} R_{is} \quad \text{and} \quad L_i = \sum_{s} L_{is}. \tag{22}$$

We now turn to describing wages. We can express private-sector wages by using (7), (8), (9), (13), (14) as follows:

$$w_{iA} = (\widetilde{b}_i^A)^{\frac{1}{1-\gamma^A}} Q_i^{-\frac{\gamma^A}{1-\gamma^A}}, \tag{23}$$

$$w_{iM} = (\widetilde{b}_i^M)^{\frac{1}{1-\gamma^M}} Q_i^{-\frac{\gamma^M}{1-\gamma^M}} L_{iM}^{\frac{\eta}{1-\gamma^M}}, \tag{24}$$

$$w_{iN} = (\widetilde{b}_i^N)^{\frac{1}{1-\gamma^N}} (P_i^N)^{\frac{1}{1-\gamma^N}} Q_i^{-\frac{\gamma^N}{1-\gamma^N}} L_{iN}^{\frac{1}{\sigma-1}} H_{iN}^{\frac{\gamma^N}{(\sigma-1)(1-\gamma^N)}}, \tag{25}$$

where $\widetilde{b}_i^A \equiv P^A b_i^A$ and $\widetilde{b}_i^M \equiv P^M b_i^M$ are the price-adjusted productivities of agricultural and manufacturing goods in location i, respectively.[26] In locations with US bases, the wage rate of US-base workers is determined so that it solves the second equation of (21).

Floor space market. Since equation (1) implies that the share of expenditure on housing is $1-\alpha$, the total amount of floor space demanded by residents in location i, denoted by H_{iR}, is

[26] We cannot identify P^A and b_i^A (resp. P^M and b_i^M) separately in our calibration process. Instead, we recover the value of $\widetilde{b}_i^A$ (resp. $\widetilde{b}_i^M$) such that it rationalizes the observed equilibrium.

given by

$$H_{iR} = (1-\alpha)\sum_j \sum_s \frac{\lambda_{ij|is} w_{js} R_{is}}{Q_i}. \tag{26}$$

Private uses of floor space include not only housing but also the production of both tradable goods and non-tradable services. Because each sector exhibits a Cobb-Douglas production technology, the amount of floor space used for production in sector $s \in \{A, M, N\}$ can be written as follows:

$$H_{is} = \frac{\gamma^s w_{is} L_{is}}{(1-\gamma^s) Q_i}. \tag{27}$$

In location i, the equilibrium of the floor space market is characterized by $H_i = H_{iR} + \sum_{s\in\{A,M,N\}} H_{is}$. Combining this equation with (26) and (27), we obtain the following condition for floor space market clearing:

$$Q_i H_i = (1-\alpha)\sum_j \sum_{s\in\{A,M,N,B\}} \lambda_{ij|is} w_{js} R_{is} + \sum_{s\in\{A,M,N\}} \frac{\gamma^s w_{is} L_{is}}{1-\gamma^s}, \tag{28}$$

where the first and second terms on the right-hand side represent the aggregate rents from housing and production, respectively.

Non-tradable service market. In each location, the total revenue of firms producing non-tradable services must be equal to the total expenditure on non-tradable services spent by workers living or working there, local landowners, and the US military personnel in that location.

Workers living in location i spend a fraction $\xi_{ij}\chi_i^N (P_j^N/\mathbb{P}_i^N)^{1-\sigma}$ of their income on non-tradable services served in location j. Multiplying this expenditure share by $1/\alpha$, we obtain the shares of expenditure spent by local landowners and the US military personnel in location i on non-tradable services produced in location j. Thus, the market clearing condition for non-tradable services in location i can be expressed as follows:

$$\begin{aligned}\left(\frac{w_{iN}L_{iN}}{1-\gamma^N}\right)^{1-\gamma^N}\left(\frac{Q_i H_{iN}}{\gamma^N}\right)^{\gamma^N} = &\sum_k\sum_j\sum_s \xi_{ki}\chi_k^N\left(\frac{P_i^N}{\mathbb{P}_k^N}\right)^{1-\sigma}\lambda_{kj|ks} w_{js} R_{ks} \\ &+ \sum_k \frac{\xi_{ki}\chi_k^N}{\alpha}\left(\frac{P_i^N}{\mathbb{P}_k^N}\right)^{1-\sigma}\left(\Omega_k + e_{US} L_k^{US}\right),\end{aligned} \tag{29}$$

where the left-hand-side captures the total revenue of non-tradable firms and the first and second terms on the right-hand side represent the total expenditure on non-tradable services in location i from all workers and that from local landowners and the US military personnel, respectively.[27]

[27]Note that the total costs incurred by all non-tradable firms in location i are expressed as $\sigma f_i^N n_i w_{iN}^{1-\gamma^N} Q_i^{\gamma^N}$. As the present model features free entry, the total revenue equals the total costs. Then, using (12), the total revenue of firms in location i can be written as $\left[w_{iN}L_{iN}/(1-\gamma^N)\right]^{1-\gamma^N}(Q_i H_{iN}/\gamma^N)^{\gamma^N}$.

From (6), (16), (27) and (28), the market clearing condition (29) can be transformed into

$$\begin{aligned}\frac{w_{iN}L_{iN}}{1-\gamma^N} =& \frac{1-\mu+\alpha\mu}{\alpha}\sum_k\sum_j\sum_{s\in\{A,M,N,B\}}\xi_{ki}\chi_k^N\left(\frac{P_i^N}{\mathbb{P}_k^N}\right)^{1-\sigma}\lambda_{kj|ks}w_{js}R_{ks}\\ &+\frac{1-\mu}{\alpha}\sum_k\sum_{s\in\{A,M,N\}}\xi_{ki}\chi_k^N\left(\frac{P_i^N}{\mathbb{P}_k^N}\right)^{1-\sigma}\frac{\gamma^s w_{ks}L_{ks}}{1-\gamma^s}+\frac{1}{\alpha}\sum_k\xi_{ki}\chi_k^N\left(\frac{P_i^N}{\mathbb{P}_k^N}\right)^{1-\sigma}\Phi_k,\end{aligned} \tag{30}$$

where $\Phi_k \equiv m_k\mathbb{Q}_k^{US}S_k + e_{US}L_k^{US}$ denotes base-sourced consumption in municipality k: US personnel's purchases plus consumption financed by rents on base land. Local landowners receiving these rents are assumed to have the same spending structure as US military personnel. This measure excludes Okinawan base workers' payroll, whose consumption effects enter through the base-employment channel. We refer to Φ_k as consumption below, retaining the full term when needed to distinguish it from total consumer expenditure.

Workers' welfare. Prior to the realization of the idiosyncratic preference shocks, expected utility from living in Okinawa is

$$\mathbb{W} = \Gamma\left(\frac{\theta-1}{\theta}\right)\left[\sum_i\sum_j\sum_s A_{is}\left(w_{js}\mathbb{P}_i^{-\alpha}Q_i^{-(1-\alpha)}\right)^{\theta}\tau_{ij}^{-\theta}\right]^{\frac{1}{\theta}}, \tag{31}$$

where $\Gamma(z) = \int_0^\infty t^{z-1}\mathrm{e}^{-t}\mathrm{d}t$ is the Gamma function. In our baseline model, we assume a closed economy so that the total population L is exogenously determined. In Section 6.3, we consider a small-open economy situation in which workers choose whether to live in Okinawa, which gives the expected utility $\mathbb{W}$, or live outside Okinawa.

Equilibrium conditions. The present model has $19\times I + I_B$ endogenous variables: the spatial distribution of residents and employment, $\{R_i, L_i\}_{i=1}^I$; the masses of residents and employment by sector, $\{R_{is}\}_{i=1}^I$ for $s\in\{A,M,N,B\}$ and $\{L_{is}\}_{i=1}^I$ for $s\in\{A,M,N\}$; the total amount of floor space used for housing, $\{H_{iR}\}_{i=1}^I$; the total amount of floor space used for production in each sector, $\{H_{is}\}_{i=1}^I$ for $s\in\{A,M,N\}$; the wages in private sectors, $\{w_{is}\}_{i=1}^I$ for $s\in\{A,M,N\}$; the wages in the US base sector, $\{w_{iB}\}_{i\in\mathcal{I}_B}$; the floor space rent, $\{Q_i\}_{i=1}^I$; the land rent, $\{\mathbb{Q}_i\}_{i=1}^I$; and the price index of non-tradable services in location i, $\{P_i^N\}_{i=1}^I$. These variables are determined in equations (16), (20), (21), (22), (23), (24), (25), (26), (27), (28), and (30).

5 Calibration

In this section, we calibrate the model parameters to match the Okinawan economy around 1970. We choose 1970 as our target of calibration in light of the substantial fraction of Okinawa's GNI in 1970 (see Figure 3) and rich data availability. The calibration is done in steps, which we

Parameter	Value	Description	Source
$1-\alpha$	0.25	Housing expenditure share	Davis and Ortalo-Magné (2011)
σ	4.6	Elasticity of substitution among non-tradable varieties	Miyauchi et al. (2022)
κ	1.87	Elasticity of substitution between tradable and non-tradable goods	Own estimation
θ	2.19	Fréchet shape parameter	Hayakawa, Koster, Tabuchi and Thisse (2021)
ζ	0.845	Commuting cost elasticity	Own estimation
ψ_S	2.3	Travel cost elasticity for shopping	Own estimation
γ^A	0.3	Cost share of floor space in the agricultural sector	Hayashi and Prescott (2008)
γ^M	0.2	Cost share of floor space in the manufacturing sector	Ahlfeldt et al. (2015)
γ^N	0.2	Cost share of floor space in the non-tradable service sector	Ahlfeldt et al. (2015)
η	0.05	Agglomeration elasticity in the manufacturing sector	Rosenthal and Strange (2004)
ϕ	2	Floor space supply elasticity	Heblich et al. (2020)

Table 1: Values of structural parameters in the model calibration

describe below. Appendix C.1 provides supplementary details on calibration procedure. For convenience, Table 1 summarizes calibrated parameter values and their sources.

Step 1: Calibrating parameters from external information. We first determine some parameter values based on external studies. The housing expenditure share $(1-\alpha)$ is set at 0.25 based on Davis and Ortalo-Magné (2011). The elasticity of substitution among non-tradable varieties (σ) is set at 4.6 based on Miyauchi et al. (2022), which is also consistent with Hobijn and Nechio (2019). We set the Fréchet shape parameter $\theta = 2.19$, following Hayakawa et al.'s (2021) estimate using across-municipality Japanese data.[28] The cost share of floor space in the agricultural sector (γ^A) is set at 0.3 following Hayashi and Prescott (2008). The cost share of floor space in the manufacturing sector and the non-tradable service sector (γ^M, γ^N) is set at 0.2 following Ahlfeldt et al. (2015). The agglomeration elasticity in manufacturing productivity (η) in each municipality is set at 0.05, which is consistent with Rosenthal and Strange (2004) and Ahlfeldt and Pietrostefani (2019). Finally, the floor-space supply elasticity is set at 2 following Heblich et al. (2020), which is consistent with Japanese floor-space supply technology in land-scarce environments like Okinawa (Yoshida 2018).

Step 2: Gravity equation estimation. We estimate the commuting cost parameter using commuting flows observed separately for workers in agriculture, manufacturing, and non-tradable services. We assume that the commuting cost is written as $\tau_{ij} = d_{ij}^{\zeta}$, where d_{ij} is the distance between residence i and workplace j and ζ is the commuting cost elasticity with respect to commuting distance. Then, taking the log of equation (19) gives the sector-conditioned gravity equation

$$\ln \lambda_{ij|is} = -\vartheta \ln d_{ij} + o_{is}^R + o_{js}^L, \tag{32}$$

where ϑ is defined as $\vartheta \equiv \zeta\theta$, and o_{is}^R and o_{js}^L are residence-by-sector and workplace-by-sector fixed effects, respectively.

Using trip survey data, we observe $\lambda_{ij|is}$ and estimate the equation (32) by PPML and

[28] We also experimented with estimating θ to match the observed dispersion of residential income across municipalities, analogously to Ahlfeldt et al. (2015). While this procedure has limitations in our context due to the relatively small number of municipalities, it yields a value close to $\theta = 2.19$.

OLS. The estimated distance coefficient identifies $\vartheta = \zeta\theta$; given the externally calibrated value of θ, we recover the commuting cost elasticity as $\zeta = \vartheta/\theta$. Table C.1 presents estimation results, which yield our preferred estimate of 1.85. Since we have set $\theta = 2.19$, this implies $\zeta = 1.85/2.19 \simeq 0.845$.

We next turn to estimating the travel cost for shopping using a gravity specification. The distance-based shopping-trip shares used in the model are given by

$$\xi_{ij}^{\text{trip}} = \frac{d_{ij}^{-\psi_S}}{\sum_k d_{ik}^{-\psi_S}}, \tag{33}$$

where d_{ij} is the travel distance between municipalities i and j, and ψ_S is the parameter governing travel costs. Appendix C.2 reports shopping gravity estimates using destination shares, and we set $\psi_S = 2.3$ based on Column 1 of Table C.2. We use this estimate as a calibration benchmark and construct the model's shopping weights ξ_{ij} from the distance-based shares in equation (33).

Step 3: Wages and rents. By combining (19) and (21), we obtain the following labor market clearing condition for each sector:

$$L_{is} = \sum_k \frac{\omega_{is} d_{ki}^{-\vartheta}}{\sum_l \omega_{ls} d_{kl}^{-\vartheta}} R_{ks}, \tag{34}$$

where $\omega_{is} \equiv w_{is}^{\theta}$ is the transformed wage of sector s in location i. Note also that the probability that a worker resides in location i conditional on working in sector s in location j is given by $\lambda_{ijs}/\sum_k \lambda_{kjs} = \widetilde{A}_{is} d_{ij}^{-\vartheta}/\sum_k \widetilde{A}_{ks} d_{kj}^{-\vartheta}$, where $\widetilde{A}_{is} \equiv A_{is}\left(\mathbb{P}_i^{\alpha} Q_i^{1-\alpha}\right)^{-\theta}$ captures the composite attractiveness of living in i and working in sector s, inclusive of amenities, goods prices, and housing rents. Then, $\{R_{is}\}_{i=1}^{I}$ and $\{L_{is}\}_{i=1}^{I}$ satisfy the following system of equations:

$$R_{is} = \sum_j \frac{\widetilde{A}_{is} d_{ij}^{-\vartheta}}{\sum_l \widetilde{A}_{ls} d_{lj}^{-\vartheta}} L_{js}. \tag{35}$$

Given the estimate of ϑ obtained in the previous step and the values of L_{is} and R_{is} observed in the data, we solve equation (34) for ω_{is} and equation (35) for $\widetilde{A}_{is}$. Note that equations (34) and (35) are homogeneous of degree zero in ω_{is} and $\widetilde{A}_{is}$, respectively, implying that we need to impose normalization conditions to determine the absolute levels of both variables. For $\widetilde{A}_{is}$ ($s \in \{A, M, N\}$), we normalize the geometric means of $\widetilde{A}_{is}$ across all locations to one, which implicitly assumes that the scale of the amenities A_{is} is common across all private sectors $s \in \{A, M, N\}$. For the US-base sector, we allow the geometric mean of $\widetilde{A}_{iB}$ to differ from one and denote it by a_B, reflecting that the US-base sector may have specific job (dis)amenities compared with the private sectors.[29] The introduction of the US-base job amenity a_B allows us

[29]For example, military-base workers were subject to a distinctive and restrictive labor-law regime, such as the restrictions on union activities (Toya 2019). Fukuyama (2023) also documents the language and cultural barriers faced by the Okinawan maids employed by US military personnel.

to match the observed GNI share of the US-base sourced labor income. For the values of ω_{is}, we normalize the geometric mean of ω_{iA} to one by suitably choosing the unit of labor. Then, the scales of ω_{iM}, ω_{iN}, and ω_{iB} are chosen to match the employment share of each sector (see Appendix C.1).

Based on the normalization described above, we first back out $\widetilde{A}_{is}$ for $s \in \{A, M, N\}$ and then ω_{is} for $s \in \{A, M, N\}$. Given the value of θ, we can readily recover the wages in sectors A, M, and N by using $w_{is} = \omega_{is}^{1/\theta}$. For the US-base sector, we first back out $\widetilde{A}'_{iB} \equiv \widetilde{A}_{iB}/a_B$, which is the amenity level renormalized to have the geometric mean of one by netting out the US-base job amenity. We then calibrate a_B so that the model-implied GNI share of wages earned by workers in the US-base sector matches the observed value, which we describe in more detail in Appendix C.3. The knowledge of a_B, together with the observed employment share of the US bases, also pins down the scale of ω_{iB} and hence w_{iB}.

Finally, given the recovered wages w_{is}, we back out the floor space rents Q_i from the floor space market clearing condition (28). We also back out the land rents $\mathbb{Q}_i$ from equation (16).

Step 4: Price index of non-tradable services. Given the values of w_{is} for any pair of i and s and ξ_{ij} for any pair of i and j, we can back out the equilibrium values of P_i^N that rationalize the observed patterns of shopping and commuting trips. Specifically, we solve the goods market clearing condition (30) with respect to P_i^N. In solving equations (30), we need the value of consumption Φ_i, which is defined as $\Phi_i \equiv m_i \mathbb{Q}_i^{US} S_i + e_{US} L_i^{US}$. We allocate US military personnel across municipalities in proportion to US-base land area and calibrate $\mathbb{Q}_i^{US}$ and e_{US} so that US-base land rents and US personnel's purchases of Okinawan goods and services separately match their observed shares of GNI. More details are provided in Appendix C.3. Together with the base-sector wage level calibrated in the previous step, this procedure matches the three separately reported base-related GNI shares: labor income from base employment, base land rents, and US personnel's purchases of goods and services.

In solving the goods market clearing condition (30) with respect to P_i^N, we also need to determine the value of κ, governing the substitutability between tradable and non-tradable goods. After solving the equation (30) under many values of κ, we set κ to match the relative spending share for restaurants in the largest urban place (Naha) and the entire Okinawan economy, which is 1.16 according to the Okinawa Household Expenditure Survey in 1970. We find $\kappa = 1.87$, which is well within the 95% confidence interval reported by Hobijn and Nechio (2019).[30]

Step 5: Locational fundamentals. We back out the exogenous amenities and productivity fundamentals. First, equation (7) enables us to identify the price-adjusted productivity of agricultural goods in location i, defined as $\widetilde{b}_i^A \equiv P^A b_i^A$, because the values of w_{iA} and Q_i are

[30]Note that in our model, housing consumption is excluded from the non-tradable sector in defining the substitutability between tradable and non-tradable sectors. Hobijn and Nechio's (2019) estimate is conceptually in line with ours in distinguishing housing consumption from other non-tradable goods. Studies that include housing as a non-tradable good (e.g., Duarte and Restuccia 2010; Fajgelbaum and Redding 2022; Buera, Kaboski, Rogerson and Vizcaino 2022) tend to find a smaller elasticity of substitution between tradable and non-tradable goods.

already known. Similarly, we define the price-adjusted productivity of manufacturing goods in location i as $\widetilde{b}_i^M \equiv P^M b_i^M$, and combine equations (8) and (9) to obtain the following expression: $\widetilde{b}_i^M = w_{iM}^{1-\gamma^M} Q_i^{\gamma^M} L_{iM}^{-\eta}$. Substituting the values of w_{iM}, Q_i, and L_{iM}, we can recover the value of $\widetilde{b}_i^M$. For the non-tradable service sector, we can solve equation (15) with respect to B_i^N, given the values of w_{iN}, Q_i, and $\mathbb{P}_i^N$. After obtaining the value of B_i^N, we can readily back out $\widetilde{b}_i^N$ from equation (14). Finally, we back out the amenity associated with living in i and working in sector s. Now that $\widetilde{A}_{is}$, $\mathbb{P}_i$ and Q_i are known, A_{is} can be obtained by using $A_{is} = \widetilde{A}_{is} \left(\mathbb{P}_i^{\alpha} Q_i^{1-\alpha}\right)^{\theta}$.

Model fit. As we have used information on the population and employment of each sector in each municipality in our calibration, our model prediction perfectly matches the observed values. To assess predictions for variables not used in the calibration, we compare model-predicted bilateral commuting probabilities with observed probabilities in 1967, and model-implied floor-space rents with observed municipal house prices. Figure C.1 reports correlations of 0.958 for commuting probabilities and 0.873 between model-implied floor-space rents and observed house prices. Overall, the model successfully predicts these variables not used for our calibration.

6 Counterfactual Analysis

We quantify the effects of US-base-sourced expenditure and employment by comparing the calibrated 1970 equilibrium with a series of counterfactual equilibria. Section 6.1.1 jointly removes consumption and base employment while keeping base land unavailable for private use, holding other exogenous fundamentals at their calibrated levels, and fixing Okinawa's aggregate worker population. This benchmark case allows us to isolate the effect of consumption demand and base employment at the US bases. Section 6.1.2 then examines the role of sectoral and spatial mobility. Section 6.1.3 adds the return of base land to private use, while Section 6.1.4 summarizes the Okinawa-wide outcomes across counterfactuals. Section 6.2 then summarizes the direct and spillover effects in elasticity form under the benchmark specification. Finally, Section 6.3 assesses external validity of our results by examining how the benchmark counterfactual results depend on external mobility and manufacturing import supply. Appendix D describes further details.

6.1 Counterfactual effects of US-base-related flows and land use

6.1.1 Benchmark: Removing consumption and base employment

To isolate the effects of base-sourced consumption and employment, we first ask how jointly removing consumption and base employment changes the spatial distribution of economic activities. More specifically, we set to zero the sum of the consumption of the US military personnel and land-rent income (i.e., $\Phi_i = 0$) and employment of Okinawans at the bases (i.e., $L_{iB} = 0$ for every location i), while keeping base area unavailable for private use. Therefore, the counterfactual considered in this section is different from the "US-base closure," in which the

base acreage is returned to Okinawans for private use in addition to eliminating consumption and base employment, which we analyze in Section 6.1.3.

Geographical and sectoral reallocation. We begin with the population maps in Figure 5, which show both the total number of residents and the numbers of residents working in agriculture, manufacturing, and non-tradable services.

The top-left map shows considerable geographical mobility away from the US bases in response to the elimination of consumption and base employment. More specifically, the number of worker residents falls by 26.57% in Chatan, 18.51% in Ishikawa, 20.45% in Kin, 16.88% in Koza, and 14.11% in Ginowan, all of which have relatively large US-base area shares. In contrast, Naha, Nago, and Itoman, which have relatively low US-base area shares, gain worker residents by 5.62%, 11.01%, and 10.16%.

In addition to such large-scale geographical mobility, the sector maps in the rest of Figure 5 reveal considerable sectoral mobility. The number working in civilian non-tradable services falls in every municipality, with an aggregate-level decline of 9.63%. By contrast, the number of residents working in agriculture and manufacturing rises in every municipality, with the aggregate-level increases of 33.73% and 47.87%. Moreover, Figure 5 also reveals that these sectoral changes are stronger in areas with a higher share of the US bases, highlighting that these sectoral changes are driven by the evaporation of consumption and base employment.

Intuitively, the difference in the tradability of the outputs is the key factor in explaining these patterns. The loss of consumption and base employment induces the decrease in the local non-tradable service demand, which reduces the non-tradable goods prices in Okinawa and reduces its employment share in Okinawa. In contrast, although the loss of consumption and base employment also reduces the demand for agricultural and manufacturing goods, these goods are tradable so that it does not reduce the output prices. Therefore, relative to the non-tradable prices, the price of these tradable goods becomes higher, especiallly in areas around the US bases, and this makes the employment share in these sectors expand.

To quantify how this geographical and sectoral mobility varies with initial base exposure, Figure 6 summarizes the municipal responses in a scatterplot. Panel 6a plots the counterfactual-to-observed population ratio, while Panel 6b reports counterfactual-minus-observed changes in broad sector-of-employment shares among residents: primary (agriculture), secondary (manufacturing), and tertiary (civilian non-tradables plus base employment). The population ratio has a fitted slope of -0.477 (standard error 0.059). Thus, a municipality with a base-area share 10 percentage points higher is predicted to lose population by 4.77%. The broad sector composition also reverses most strongly in base-intensive municipalities: A municipality with a base-area share 10 percentage points higher is predicted to experience a 2.16-percentage-point larger decline in the tertiary employment share, while its manufacturing and agricultural employment shares are predicted to increase by an additional 1.44 and 0.72 percentage points, respectively.

Note that these numbers in Figure 6 are comparable to the stylized facts in Figure 4 (Section 3) because we report the association between the US bases and the change in the relevant

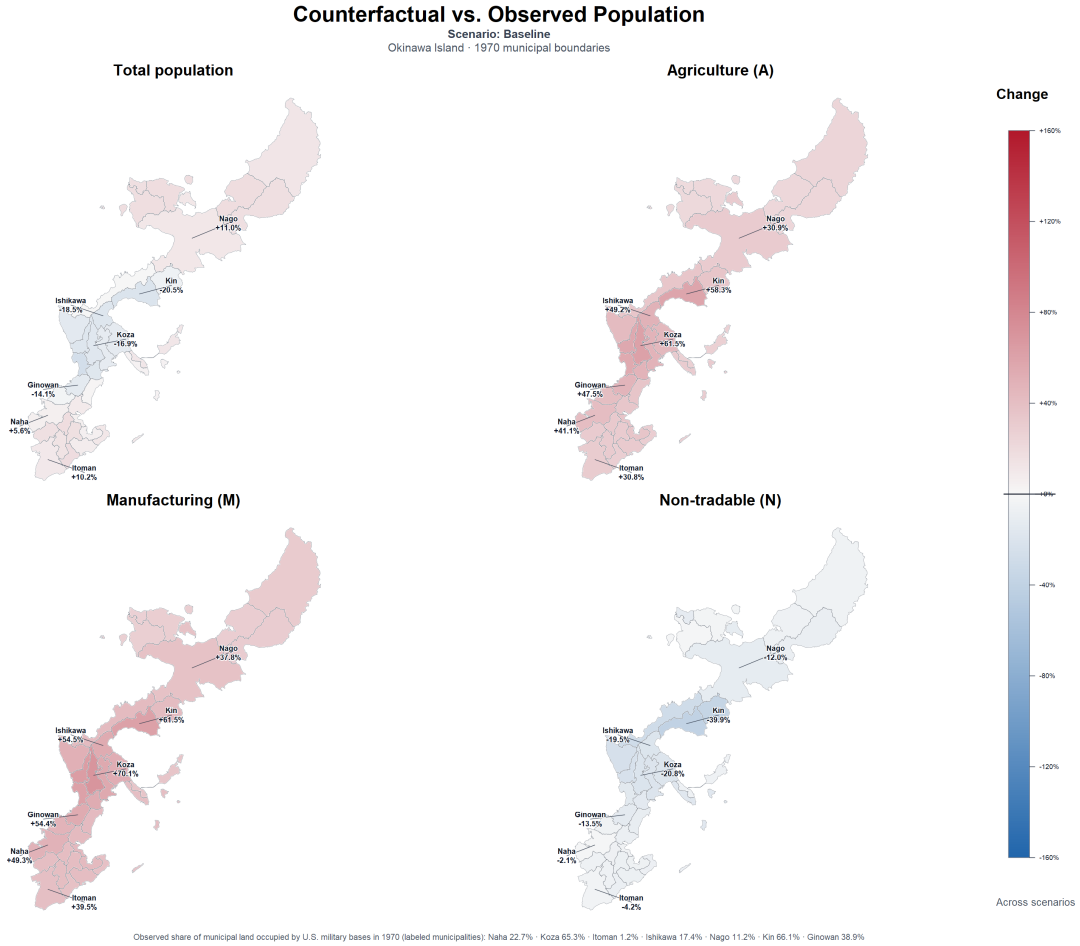
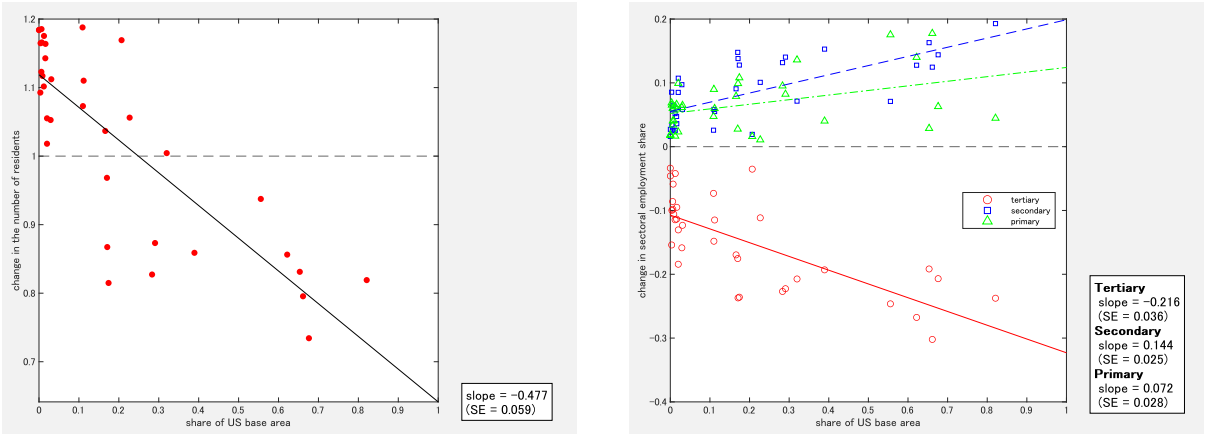


Figure 5: Municipal worker-resident responses, overall and by civilian sector, in the benchmark counterfactual

Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total residents and for residents working in agriculture (A), manufacturing (M), and non-tradable services (N). Red denotes an increase and blue a decrease. Base employment is included in total residents but is not included in the three sector panels. The benchmark sets $\Phi_i = L_{iB} = 0$ while leaving base acreage unavailable for private use. Municipal boundaries are those of 1970; labels report changes for selected municipalities. For the labeled municipalities, the shares of 1970 municipal area occupied by US bases are Nago 11.2%, Kin 66.1%, Ishikawa 17.4%, Koza 65.3%, Ginowan 38.9%, Naha 22.7%, and Itoman 1.2%.



(a) Residents

(b) Broad sector-of-employment shares

Figure 6: Exposure gradients in population and sector-of-employment shares

Note: Each point is a municipality, and the horizontal axis is the share of municipal area occupied by US bases in 1970. Panel 6a plots the counterfactual-to-observed ratio of residents. Panel 6b plots counterfactual-minus-observed changes in sector-of-employment shares among residents. Red circles denote the tertiary share, which combines civilian non-tradable services and base employment; blue squares and green triangles denote manufacturing and agriculture. The three share changes therefore sum to zero within each municipality. Lines are linear fits, and the boxes report slopes and standard errors. The benchmark sets $\Phi_i = L_{iB} = 0$ but does not return base land to private use.

outcomes in the same form.[31] Generally, although they qualitatively match, the association is stronger in the stylized facts than in Figure 6. A possible reason for this discrepancy is that unlike our counterfactual scenario, the stylized facts include the effect of not only the consumption and employment of the US bases but also other effects such as infrastructure investment around the bases. We return to this issue in Section 7.

The effects on wages, rents, and non-tradable prices and varieties. Non-tradable wages fall in every municipality, and service varieties decline in 32 of 35 municipalities. These outcomes reflect the loss of base-related spending and the departure of residents from more exposed municipalities, which reduce demand for local services and therefore local labor demand in the non-tradable sector. Floor-space rents tend to fall around the bases, which is natural given the population outflows. These rent changes affect wages because firms use floor space in production. With tradable output prices fixed, lower floor-space costs allow firms to pay higher wages in the tradable sector. In addition to this, manufacturing wages rise because of agglomeration economies: the expansion of manufacturing employment around the bases raises the manufacturing productivity. Regarding the service price index (equation 4), lower non-tradable wages and, in the most exposed municipalities, lower rents decrease marginal costs (equation 11). These cost reductions outweigh the loss of non-tradable varieties, so the non-tradable price index falls in every municipality despite the contraction in service provision.

[31] To ensure comparability, we include the US-base sector as tertiary in Figure 6, while Figure 5 does not include the US-base sector as tertiary for transparency.

Appendix D.2 reports the corresponding maps and exposure gradients in Figures D.1 and D.2.

Separating the consumption-side and employment channels. The benchmark removes consumption demand and base employment jointly. To distinguish these two operating channels, we compare a consumption-only counterfactual, which sets $\Phi_i = 0$ and therefore removes both US personnel's purchases and consumption coming from the US-base rents while retaining local base employment, with an employment-only counterfactual, which sets $L_{iB} = 0$ while retaining Φ_i. Both experiments leave base land occupied to focus on the role of consumption and base employment. Appendix D.3 reports the corresponding maps and fitted exposure gradients, including the side-by-side comparison in Figure D.5.

The two shocks have qualitative similarities: both shift residents away from more base-exposed municipalities, increase the numbers of residents working in agriculture and manufacturing in every municipality, and lower non-tradable wages and prices throughout Okinawa. Their magnitudes and service-sector quantity responses, however, differ sharply. The fitted resident gradient is -0.190 when Φ_i is set to zero but -0.275 when base employment is removed, indicating that base jobs play the larger role in attracting population around the US bases. The consumption-side shock reduces non-tradable employment in every municipality and by 14.20% islandwide, and reduces service varieties everywhere. Removing base employment instead raises the non-tradable employment in every municipality and by 5.64% islandwide, while varieties rise in 34 of 35 municipalities as former base workers move into the private non-tradable sector. Overall, the comparison highlights that US personnel's purchases and base-rent income play a larger role in sustaining non-tradable demand and market size, whereas base employment plays a larger role in attracting more population around the US bases. We return to this distinction in the elasticity analysis in Section 6.2.

6.1.2 Migration as a spatial adjustment margin

The benchmark allows residents to adjust their municipality of residence as well as their sector and workplace, implying that both the "sectoral reallocation" and "geographical reallocation" are feasible in response to the removal of consumption and base employment.

To separate the role of these two adjustment margins, we re-solve the benchmark while imposing $R_i^{cf} = R_i$ in every municipality. This restriction fixes municipal resident totals but leaves sector and workplace choices, commuting and shopping patterns, and local wages, rents, and prices endogenous.[32]

Figures D.8 and D.10 in Appendix D.4 report the corresponding municipal maps, while Figure D.9 summarizes the fitted gradients. Under fixed residence, the fitted gradients in the tertiary, manufacturing, and agricultural sector-of-employment shares are -0.191, $+0.141$, and $+0.050$, compared with -0.216, $+0.144$, and $+0.072$ in the benchmark case. Thus, the

[32]Therefore, precisely speaking, this exercise still accommodates changes in the commuting pattern as a geographical response.

structural reallocation does not arise because of workers' reallocation toward tradable-intensive municipalities, but sectoral reallocation *within* the same municipality plays the key role.

Residential reallocation matters in determining where and how the local adjustment occurs. Even though residence is fixed, the employment gradients are −0.238 for non-tradables, +0.882 for manufacturing, and +0.732 for agriculture, compared with −0.606, +0.345, and +0.371 in the benchmark. This implies that commuting responses could generate a substantial reallocation of employment even if workers' residences are fixed. At the same time, the floor-space-rent gradient flattens from −0.167 to −0.030, and the variety gradient falls in magnitude from −0.610 to −0.273. These attenuated gradients are consistent with residential sorting amplifying local housing and non-tradable demand responses.

6.1.3 Returning base land to private use

The benchmark and fixed-residence exercises keep base acreage unavailable for private use. We now ask how the results change when that land is released for housing and private production. By increasing the amount of available land for production or residence, this additional land-supply channel may offset some of the economic losses caused by removing US-base-sourced expenditure and employment. The value of returned base land is also central to policy discussions in assessing the economic impact of base returns: some studies argue that the land-occupying effect is large (e.g., Okinawa Prefecture Government 2007), while others find a smaller effect (e.g., Kusuyama 2012).

Specifically, we repeat the benchmark analysis in Section 6.1.1, but the only difference is that we additionally set $m_i = 0$ for every municipality: all base acreage becomes available for private use. Note that we assume that the returned land has the same municipality-specific technology as other private land in the same municipality. We therefore abstract from any additional productivity associated with releasing a large contiguous site (Hornbeck and Keniston 2017). Consistent with this abstraction, Yamasaki, Nakajima and Teshima (2023) documents no lot-size premium in Japan in 1970, when high-rise construction was not yet widely feasible, although large sites could become especially valuable if skyscraper technology were available.

Figure D.11 in Appendix D.5 shows that returning the land makes the residential reallocation much more muted. In Panel D.11a, the resident-ratio slope is −0.034 (standard error 0.061), compared with −0.477 in the benchmark, implying that the US base share and the change in population are no longer correlated. Intuitively, more availability of land around the US bases directly increases utility through inducing lower housing costs and indirectly through increasing labor productivity due to more land input in production, offsetting the negative effect due to the loss of the local non-tradable demand from the base and the US-base employment. By contrast, Panel D.11b shows a broad tertiary sector-of-employment-share slope of −0.220, close to the benchmark value of −0.216, while agriculture and manufacturing gain share. Overall, at least in our context, land availability therefore strongly dampens residential reallocation without undoing the sectoral reallocation.

Table 2: Okinawa-wide effects of counterfactual scenarios

	Aggregate effects (%)		Employment-share changes (pp)			
Scenario	Workers' Welfare	GNI	Agric.	Manuf.	Non-trad.	U.S. base
Benchmark and model variants						
Benchmark counterfactual	−7.25	−11.99	+5.58	+10.26	−4.92	−10.93
Fixed residence; sector mobility allowed	−7.10	−14.04	+4.85	+10.39	−4.31	−10.93
Base land also returned	−1.95	−11.23	+5.35	+10.19	−4.61	−10.93
Partial-channel counterfactuals						
Consumption removed	−2.42	−11.60	+2.57	+4.68	−7.25	0.00
Base employment removed	−5.22	−1.87	+2.92	+5.12	+2.88	−10.93

Notes: Welfare and GNI entries are percentage changes from the initial equilibrium, 100(counterfactual/initial−1). Employment entries are changes in Okinawa-wide sectoral employment shares in percentage points (pp). Initial employment shares are 16.55, 21.44, 51.08, and 10.93 percent for agriculture, manufacturing, non-tradable services, and US-base employment, respectively.

6.1.4 Aggregate outcomes across counterfactuals

The preceding subsections focused on the geography and sectoral composition of adjustment. We now turn to the aggregate economic consequences for Okinawa as a whole. Table 2 summarizes the effects on welfare, GNI, and economy-wide employment shares. The first three rows report the benchmark (Section 6.1), the fixed-residence variant (Section 6.1.2), and the counterfactual that also returns base land to private use (Section 6.1.3). As in the last part of Section 6.1, the final two rows isolate the consumption-side and employment channels: one sets $\Phi_i = 0$ while retaining local base employment, and the other sets $L_{iB} = 0$ while retaining Φ_i.

In the first row of Table 2, the benchmark counterfactual predicts both a substantial sectoral reallocation and sizable aggregate losses: Welfare falls by 7.25% and GNI by 11.99%. While this is large in absolute value, it is relatively small given that base-sourced consumption and income from base employment together constituted around 30% of the GNI in the observed Okinawan economy in 1970. The employment shares of agriculture and manufacturing rise by 5.58 and 10.26 percentage points, while the shares of civilian non-tradable services and base employment fall by 4.92 and 10.93 percentage points. Thus, the islandwide counterpart of the spatial patterns documented above is a shift from services and base employment toward tradable production, accompanied by lower income and welfare.

In this sense, this result is broadly consistent with the "Dutch disease" hypothesis that a large facility, which accompanies large consumption demand and employment, does not contribute to economic activities by preventing efficient allocation of resources (Gollin et al. 2016; Allcott and Keniston 2018). Our result on Okinawa suggests that even if the military base's demand and employment disappear, sectoral and geographical mobility can substantially mitigate the negative economic impact and restore a large part of the lost income and welfare. That said, the overall economic impact of the US bases in 1970 is still estimated to be positive.

In the second row of Table 2, restricting residential mobility leaves this sectoral realloca-

tion broadly unchanged compared with the benchmark. Under fixed municipal residence, the agricultural, manufacturing, and non-tradable employment shares change by +4.85, +10.39, and −4.31 percentage points. GNI also falls by 14.04%, compared with 11.99% in the benchmark. This comparison is consistent with internal migration somewhat cushioning the aggregate income loss by reallocating residents toward locations with higher productivity.[33] However, the aggregate impacts with or without residential reallocation are similar, implying that sectoral adjustments play the key role in mitigating the negative economic impact of the loss of the US-base consumption and employment.

In the third row of Table 2, returning base land affects a different aggregate margin: It reduces the welfare loss from 7.25% to 1.95%, a recovery of 5.30 percentage points, but reduces the GNI loss by only 0.76 percentage points, from 11.99% to 11.23%. Its civilian-sector employment-share changes—+5.35, +10.19, and −4.61 percentage points in agriculture, manufacturing, and non-tradable services—also remain close to the benchmark. Additional land therefore offsets much of the welfare loss without reversing either the income decline or the service-to-tradable reallocation.

In the fourth row, removing consumption lowers welfare by 2.42% and GNI by 11.60%; the non-tradable employment share falls by 7.25 percentage points, which is a larger response than in the benchmark case. In the fifth row, removing the US-base employment lowers welfare by 5.22% but GNI by only 1.87%; displaced base workers enter all three civilian sectors, and the non-tradable share rises by 2.88 percentage points. The consumption-side counterfactual therefore produces the larger GNI decline and service-sector contraction, whereas the employment-only counterfactual produces the larger welfare decline.

Overall, the aggregate effects are economically sizable, but the benchmark GNI loss remains modest relative to the scale of the initial base-related flows. The aggregate results broadly align with the local effects discussed in Sections 6.1–6.1.3, and neither the benchmark welfare decline nor its GNI decline approaches contemporary claims that an immediate US withdrawal would reduce Okinawan living standards by more than half (Toriyama, 2009).

6.2 Direct and Spillover Elasticities of Consumption and Base Employment

We next trace how employment gains in the municipality receiving base-related inflows are connected to adjustment elsewhere in Okinawa. Starting from the calibrated 1970 equilibrium, we examine separate 1% increases in consumption, Φ_j, and base employment, L_{jB}, in one municipality, holding other exogenous inputs fixed. Expressing the responses as local and aggregate elasticities provides normalized measures that facilitate comparisons of effect sizes across settings and with existing studies.

[33]The reported welfare in the fixed residence case hardly differs from the benchmark value. However, note that the reported entry evaluates the same free-residence inclusive-value index used in the benchmark at the fixed-residence equilibrium, which is computable but does not have the standard expected welfare interpretation because the location choice has been fixed.

Table 3 reports the average elasticity across all municipalities. The first column reports the local elasticity in the municipality receiving the shock; the second reports the elasticity of the Okinawa-wide aggregate with respect to consumption or base employment in that municipality; and the third reports the percentage of the Okinawa-wide impact accounted for by the local effect in the source municipality. Subtracting this percentage from 100% gives the share attributable to effects in all other municipalities, that is, the net spillover contribution. Appendix D.6 provides the calculation details.

Elasticities with respect to consumption. A 1% increase in consumption raises local non-tradable employment by 0.1074% and non-base employment by 0.0394%. Each additional local service job is accompanied by losses of 0.112 agricultural and 0.088 manufacturing jobs in the same municipality. Local tradable-sector contraction thus offsets only one fifth of the service employment gain. Intuitively, additional spending supports service production, and the labor market competition reallocates some workers from tradable sectors to the non-tradable sector, with the overall positive employment impact.

In terms of the magnitude of the spillovers, service expansion is more concentrated in the source municipality than manufacturing contraction. The source municipality accounts for 47.3% of the aggregate service employment gain but only 13.2% of the manufacturing loss. Shopping costs concentrate additional service demand near the income source, directly supporting nearby service providers. The resulting service expansion, however, draws workers from tradable sectors across municipalities through commuting and relocation, spreading manufacturing employment losses beyond the source municipality. Our results show that manufacturing accounts for a large part of the job losses, with a smaller contribution from agriculture (Appendix D.6).

Both commuting and relocation yield spillover effects onto other municipalities. The importance of commuting can be seen in the contrast between local employment and population responses: local workplace employment rises by 0.0310% and population increases by 0.0126%. While population increase implies the role of residential relocation, workers can also respond to the additional consumption demand by commuting to the source municipality from elsewhere, yielding a larger response of employment.

Local service wages, prices, and floor-space rents rise by 0.0537%, 0.0126%, and 0.0096%, respectively, with most of their aggregate adjustment occurring elsewhere. Service demand supports higher wages and prices, whereas higher rents lower agricultural and manufacturing wages at fixed tradable output prices. Manufacturing contraction also reduces manufacturing wages by weakening agglomeration economies.

The estimated impact of base-sourced consumption on employment and wages is broadly consistent with the existing literature in different settings. After normalization by the induced change in nominal consumer expenditure, our result implies a local non-tradable employment response of 0.412% per 1% increase in expenditure, close to the 0.48% inferred by Mian and Sufi (2014). The impact of the consumption shock on employment and wages is also broadly

Table 3: Local and aggregate responses to a 1% increase in one municipality

Panel A. Consumption: a 1% increase in one municipality

Outcome	Local response (%)	Okinawa-wide response (%)	Local share of aggregate impact (%)
Agricultural employment	−0.0187	−0.00482	11.7
Manufacturing employment	−0.0176	−0.00656	13.2
Non-tradable employment	+0.1074	+0.00432	47.3
Non-base employment	+0.0394	0.00000	NA
Resident workers	+0.0126	0.00000	NA
Agricultural wage	−0.0041	−0.00017	80.8
Manufacturing wage	−0.0035	−0.00070	21.2
Non-tradable wage	+0.0537	+0.00405	24.6
Floor-space rent	+0.0096	+0.00107	33.3
Local non-tradable price index	+0.0126	+0.00236	18.7

Panel B. Base employment: a 1% increase in one municipality

Outcome	Local response (%)	Okinawa-wide response (%)	Local share of aggregate impact (%)
Agricultural employment	−0.0233	−0.00713	6.3
Manufacturing employment	−0.0216	−0.00939	14.9
Non-tradable employment	+0.0023	−0.00112	−2.9
Non-base employment	−0.0091	−0.00423	9.5
Resident workers	+0.0409	0.00000	NA
Agricultural wage	−0.0053	+0.00009	−72.1
Manufacturing wage	−0.0044	−0.00065	27.3
Non-tradable wage	+0.0068	+0.00308	18.6
Floor-space rent	+0.0123	+0.00030	91.7
Local non-tradable price index	+0.0075	+0.00272	18.0

Note: Each experiment raises consumption or base employment by 1% in one municipality, holding the other channel and occupied base land fixed. Consumption comprises US personnel's purchases and consumption financed by base land rents, excluding Okinawan base workers' payroll. The first two columns average local and aggregate responses across the same 29 sources; all 35 municipalities enter the aggregate. The third column gives the local share of the net aggregate change; negative shares indicate offsetting local changes and NA denotes a zero aggregate response due to the assumption of fixed total population of Okinawa. Non-base employment comprises agriculture, manufacturing, and non-tradables. Appendix D.6 defines the weights and calculations.

consistent with the cash-transfer evidence of Gerard, Naritomi and Silva (2026) and Mendes, Miyamoto, Nguyen, Pennings and Feler (2026), although employment and wage adjustment differ across settings.[34] Gerard et al. (2026) report increased labor force participation and little wage adjustment holding worker composition fixed. Their estimates imply a local private formal employment response of about 0.15% per 1% increase in transfers, compared with our local non-base employment response of 0.0394% per 1% increase in base-sourced consumption with local service wages rising by 0.0537%. In our benchmark, Okinawa's workforce is fixed, so service expansion draws on existing workers through sectoral and spatial reallocation, which induces a wage increase rather than the employment adjustment. In contrast, Mendes et al. (2026) find higher formal-sector real wages without a statistically significant increase in total employment, which is likely if labor supply elasticity is smaller than in our setting. Taken together, our results are consistent with the available relevant estimates in the literature, although the impact seems to depend on labor supply elasticity that differs by context.

Elasticities with respect to base employment. A 1% increase in Okinawan employment at the bases, holding base-sourced consumption and occupied base land fixed, raises total local workplace employment by 0.1201% and reduces non-base employment by 0.0091%. Converting elasticities into numbers of jobs, we find that, on average, each additional base job displaces 0.044 non-base jobs locally, leaving a net increase of 0.956 local jobs. This suggests a limited crowding-out of local private jobs. However, limited local displacement coexists with substantial displacement elsewhere: Only 9.5% of the aggregate decline occurs in the source municipality. Commuting and relocation allow local hiring to draw on workers from other municipalities, leading to the spillover of the private-sector employment contraction effect. Indeed, resident workers increase by 0.0409%, or 0.473 workers per base job, with the remaining 0.483 local jobs accounted for by increased net inward commuting.

Note that the elasticity of the local service employment is substantially smaller than under the consumption shock. The base-employment channel resembles the consumption channel in that it also expands spending through the payroll of Okinawan base workers. Base hiring, however, also competes directly with civilian employers for Okinawa's fixed workforce, while the inflow of residents increases housing demand and floor-space rents. These demand and cost pressures leave local service employment only 0.0023% higher on average. It also induces increases in nominal service wages, service prices, and floor-space rents of 0.0068%, 0.0075%, and 0.0123%, respectively. The small local service employment gain is more than offset by service job losses elsewhere, yielding an Okinawa-wide decline of 0.00112%.

Our total local employment multiplier of 0.956 is broadly consistent with the mostly below-one base-closure estimates of Hooker and Knetter (2001) and the near-unit employment multiplier from prison openings in Chirakijja (2024). Our small non-base response of -0.044 is also comparable in magnitude to the statistically insignificant estimate of 0.085 private jobs

[34] Faber and Gaubert (2019) also find positive local employment effects of external demand. Appendix D.6 discusses differences in the shocks and outcomes across these studies.

per additional public-sector job in Faggio and Overman (2014). Zou (2018) further highlights population adjustment accompanying civilian job losses after military personnel reductions. While our results are in line with these results, the comparisons remain suggestive because estimands are different: Hooker and Knetter (2001) and Zou (2018) capture concurrent changes in personnel and spending, Chirakijja (2024) examines prison openings, and Faggio and Overman (2014) examine broader changes in public-sector employment. Our results shed new light on this discussion by distinguishing the consumption and base-employment channels.

6.3 External validity: international mobility and trade

Our baseline counterfactual analysis so far has assumed fixed population of Okinawa and no trade cost for tradable goods, which is justified given Okinawa's island geography with restrictions on movement to mainland Japan and active external trade. We relax these assumptions separately to assess how the benchmark counterfactual results on welfare, aggregate income, and sectoral reallocation depend on external mobility and manufacturing import supply.[35] Appendix D.7 describes the population adjustment, and Appendix D.8 reports the extended specifications and full results in Table D.1.

External migration. We accommodate migration in and out of Okinawa, which was prohibited in our baseline analysis. As described in Appendix D.7, the migration elasticity parameter θ_{out} governs how strongly workers' choice between Okinawa and the outside economy responds to relative expected welfare. We use $\theta_{\text{out}} = 1.1$ as an intermediate sensitivity case, informed by Ortega and Peri (2013), whose preferred income elasticity of international migration flows is 0.76 and whose estimate for migration within Europe is 1.90.

Table D.1 shows that our main findings on workers' welfare and sectoral reallocation are robust to relaxing the fixed-population assumption. Across external-migration elasticities of 0.5–2.19, welfare losses remain between 7.26% and 7.28%, almost identical to the benchmark's 7.25%. Sectoral employment-share changes also remain close to the benchmark: manufacturing gains 9.96–10.19 percentage points, compared with 10.26 points, and civilian non-tradable services lose 5.01–5.32 points, compared with 4.92 points. Thus, the welfare loss and the shift from services and base employment toward tradable production do not depend on ruling out migration between Okinawa and the outside economy. With the external migration elasticity set to $\theta_{\text{out}} = 1.1$, the aggregate population falls by 7.91% and GNI by 18.93%, compared with the benchmark GNI decline of 11.99%. These larger aggregate income losses coexist with stable welfare and sectoral-composition results. The main spatial pattern of population is also preserved: a municipality with 10 percentage-points increase in the base-area share is predicted to lose population by 4.45%, very close to 4.77% in our benchmark prediction.

[35] Note that altering these assumptions affect the counterfactuals, not the calibration of the observed equilibrium.

Manufacturing imports and the trade environment. We accommodate the possibility that manufacturing imports face frictions. To capture such frictions, we denote by ε_M the price elasticity of manufacturing import supply, which measures the percentage response of the quantity supplied to Okinawa by external producers to a 1% increase in the manufacturing-goods price. The benchmark assumes perfectly elastic import supply ($\varepsilon_M \to \infty$): imports can freely adjust at a fixed manufacturing price, so the loss of base-related demand lowers service prices relative to tradable prices and shifts workers toward tradable production. With finite ε_M, the decline in import demand due to eliminating consumption and base employment also lowers the manufacturing price. The resulting fall in manufacturing returns relative to agriculture weakens the expansion of manufacturing and redirects more workers toward agriculture. At the same time, cheaper manufactured goods reduce consumption costs and the cost of inputs used to construct floor space, cushioning the decline in workers' real welfare.

Table D.1 shows that these price adjustments affect welfare, aggregate income, and sectoral composition differently under different values of ε_M. A useful benchmark is $\varepsilon_M = 10$, guided by Nicita, Olarreaga and Silva (2018), who estimate that the median elasticity of export supply faced by World Trade Organization member importers is approximately 10. In this case, welfare falls by 6.25% and GNI by 18.64%, compared with 7.25% and 11.99% in the benchmark. Manufacturing's employment share rises by 7.13 percentage points rather than 10.26, while agriculture's gain increases from 5.58 to 9.41 points. In this case, eliminating consumption and base employment decreases the manufacturing prices, which dampens reallocation to manufacturing from the US-base sector. While this substantially lowers the nominal income, the welfare impact is cushioned because a lower manufacturing price also benefits workers as consumers. That said, qualitative implications remain similar to our benchmark case.[36] However, even under elastic import supply, removing base-related flows induces a stronger shift toward manufacturing from services or base employment in more base-exposed municipalities. In this sense, the elastic import elasticity does not change the spatial tendency for bases to favor services over manufacturing.

7 Role of Past Investment in Infrastructure and Durable Capital around the Bases

The counterfactuals in Section 6 quantify the effects of the ongoing consumption demand and direct employment emphasized in the policy discussions introduced at the outset of the paper. They remove base-related expenditure, land-rent income, and local base employment while holding other local fundamentals fixed at their calibrated 1970 levels. We now ask how far these ongoing flows can account for the broader historical association between the US bases and local

[36]However, the aggregate effect can even reverse in an extreme scenario: when import quantities are fixed ($\varepsilon_M = 0$), removing consumption and base employment reduces manufacturing's islandwide employment share by 1.77 percentage points (Table D.1), implying that their presence supports aggregate manufacturing employment in that case.

development.

Our counterfactual analysis suggests that these flows alone may not fully account for the observed association. To illustrate this point, we compare the changes in the tertiary employment share in the stylized facts and the benchmark counterfactual. In Figure 4c, a 10-percentage-point higher base-area share is associated with an approximately 5.2-percentage-point larger increase in the tertiary employment share in 1970, relative to the period without US bases. In contrast, in the benchmark counterfactual of eliminating consumption demand and base employment, the same difference in exposure is associated with a 2.16-percentage-point larger decline in this share (Figure 6b). Therefore, the effects of consumption demand and base employment qualitatively match the observed pattern of service-sector development, but the association might be weaker than in the observed data. Note that our counterfactual analysis holds local fundamentals fixed at their calibrated 1970 levels and therefore does not evaluate how postwar investment shaped these fundamentals.[37] By contrast, the observed data in Figure 4c cover changes from the pre-war period (1938), which may reflect both ongoing consumption demand and base employment and changes in local amenities and productivity during the postwar period.

This comparison motivates examining past investment in infrastructure and durable capital as a possible source of the local advantages inherited in 1970. Base development may have encouraged both public infrastructure provision and private investment in durable capital, such as buildings and local human capital. The US bases could thereby have improved subsequent living conditions and production opportunities, consistent with the role of historical agglomeration in shaping later amenities and productivity emphasized by Allen and Donaldson (2022). Related evidence documents lasting local effects of early land-use regulation (Twinam 2018), transport infrastructure (Jedwab and Moradi 2016), and publicly financed industrial investment (Garin and Rothbaum 2025). In our static model, its calibrated fundamentals, which are held constant in our counterfactual analysis, can embody the legacy of such investments made before 1970.

We examine three municipal measures that capture physical infrastructure and the early commercial opportunities that may have encouraged durable private investment. The first two directly measure infrastructure provision: the share of settlements with piped water around 1963 and a road quality index combining the z-scores of municipal-road improvement, paving, and road density per unit of non-base land. These measures capture infrastructure in place before 1970, the calibration year. The third measure identifies municipalities containing road corridors designated to host authorized markets serving US personnel in 1949.[38] By concentrating early access to military customers, designation may have encouraged business entry and commercial agglomeration, raising the returns to investment in shops, commercial premises, and related equipment. As in Allen and Donaldson (2022), such investment could have left a durable local advantage beyond the contemporaneous spending that initially attracted businesses. We therefore

[37] Likewise, the potential impact of other base-related externalities, such as noise and pollution, is also absent in our baseline counterfactual analysis.

[38] Military Government Directive No. 19, issued on September 6, 1949, restricted occupation personnel's purchases of Ryukyuan services and goods not designated as essential commodities to authorized markets along specified highways and at designated places.

use designation as an indicator of historical conditions conducive to private capital formation, not as a direct measure of infrastructure or capital stocks. Together, the three measures allow us to examine public infrastructure provision and a potential source of durable private investment within the same historical mechanism.

Table 4 examines the first link in this argument: whether these infrastructure and early commercial conditions were more prevalent in municipalities with greater US-base exposure. It reports separate cross-sectional regressions of the three measures on the 1970 base-area share for 35 municipalities defined by their 1970 boundaries. We include region fixed effects and standardized elevation and slope measures to control for topographic conditions. We find positive and statistically significant associations for all three measures. For a 10-percentage-point higher base-area share, the estimated associations correspond to 0.17, 0.25, and 0.16 outcome standard deviations for piped-water provision, road quality, and corridor designation, respectively. Thus, greater base exposure is associated with better infrastructure and earlier commercial access.

Table 4: U.S. Military-Base Land Share, Infrastructure, and Early Market Access

	Water-installed settlement share (1)	Road quality index (2)	1949 designated road corridor (3)
1970 base share (+10 pp)	5.686**	0.565**	8.063**
HC1 robust standard error	(2.210)	(0.209)	(3.502)
Outcome mean	62.075	0.000	54.286
Outcome standard deviation	33.013	2.287	50.543
Observations	35	35	35
R^2	0.423	0.500	0.548
Region fixed effects	Yes	Yes	Yes
Terrain controls	Yes	Yes	Yes

Notes: The unit of observation is the municipality as of 1970. Each column is estimated separately by OLS with North and South indicators (Central omitted), standardized log(1 + mean elevation), standardized mean slope, and HC1 standard errors. Coefficients report the change associated with a 10-percentage-point increase in 1970 U.S. military-base land share. Coefficients, standard errors, and outcome standard deviations are in percentage points in columns (1) and (3) and index points in column (2); outcome means are in percent in columns (1) and (3) and index points in column (2). The road quality index is the unweighted sum of full-35 sample z-scores for the municipal-road improvement rate, paving rate, and road density per historical non-base km^2. The 1949 outcome is a dummy variable for containing roads designated to host authorized markets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The next question is whether these observed conditions are also associated with the local fundamentals that we backed out in our calibration. For this purpose, we regress log calibrated amenities and productivity on water supply, road quality, and commercial-corridor designation, conditioning on the base-area share and the same region and terrain controls as in Table 4. The regressions ask whether municipalities with similar base exposure and observed geography, but different infrastructure or early commercial access, exhibit advantages in fundamentals in 1970.[39] If these are positively correlated with the calibrated fundamental amenities and productivity,

[39]Methodologically, this approach is closely related to Ang, Angel and Parkhomenko (2026), who relate amenities recovered from quantitative spatial models to observable local characteristics. We apply this logic to both amenities and productivity to investigate whether past infrastructure and durable investment improved both amenities and productivity.

combined with Table 4, we can establish that US bases are associated with better fundamentals through these channels.

Table 5: Infrastructure, Early Commercial Access, and Local Fundamentals in 1970

	Agriculture (1)	Manufacturing (2)	Non-tradables (3)	US-base (4)	Private mean (5)	All-four mean (6)
Panel A: Log amenity						
Base share (+10 pp)	-0.087	-0.026	0.067	0.276	-0.015	0.058
	(0.104)	(0.127)	(0.105)	(0.183)	(0.108)	(0.114)
Water share (+10 pp)	0.065	0.155**	0.143**	0.049	0.121**	0.103*
	(0.046)	(0.059)	(0.056)	(0.093)	(0.051)	(0.058)
Road quality index	-0.021	-0.059	-0.018	-0.235**	-0.033	-0.083
	(0.053)	(0.081)	(0.063)	(0.110)	(0.063)	(0.068)
Commercial road (1949)	-0.285	0.066	0.416	0.160	0.066	0.089
	(0.267)	(0.366)	(0.306)	(0.518)	(0.290)	(0.319)
Observations	35	35	35	35	35	35
R^2	0.562	0.505	0.551	0.388	0.472	0.413
Panel B: Log productivity						
Base share (+10 pp)	-0.067	-0.144**	-0.254***	—	-0.155***	—
	(0.040)	(0.055)	(0.079)		(0.053)	
Water share (+10 pp)	0.030	0.039	0.090*	—	0.053	—
	(0.021)	(0.035)	(0.051)		(0.034)	
Road quality index	0.024	0.107**	0.152**	—	0.095*	—
	(0.028)	(0.044)	(0.073)		(0.047)	
Commercial road (1949)	0.116	0.549**	0.822**	—	0.496**	—
	(0.134)	(0.258)	(0.369)		(0.240)	
Observations	35	35	35	—	35	—
R^2	0.232	0.378	0.377	—	0.339	—

Notes: All regressions include an intercept, North and South indicators (Central omitted), standardized log(1 + mean elevation), and standardized mean slope. Outcomes are natural logs of calibrated sectoral fundamentals. Private and all-four means are equal-weight geometric means over agriculture, manufacturing, and non-tradables, and over these sectors plus US-base employment, respectively. US-base productivity is not a calibrated outcome, so the corresponding productivity columns are unavailable. Base share and water share coefficients are per 10-percentage-point increase. Water share is the fraction of settlements with water installations in June 1963. Road quality is the sum of full-sample z-scores of municipal-road improvement, paving, and density per historical non-base km^2. Commercial road is a dummy variable of containing 1949 road centerlines designated to host authorized markets. HC1 robust standard errors appear in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5 shows that all three channels are positively and significantly associated with at least one dimension of calibrated fundamentals amenities and productivity. Water supply is positively associated with amenities for manufacturing and non-tradable service workers, as well as with the geometric mean across the three private sectors. This is consistent with Devoto, Duflo, Dupas, Parienté and Pons (2012), who show that piped-water connections improve well-being by freeing time and reducing conflicts over water. Road quality is positively associated with productivity in manufacturing and non-tradable services. Early commercial-corridor designation is also positively associated with productivity in both sectors and with the private-sector geometric mean. These patterns have intuitive interpretations: better roads can facilitate production and exchange, while early access to military customers may have encouraged investment in durable capital such as buildings and human capital. The persistence of such local advantages is consistent with the role of historical agglomeration emphasized by Allen and Donaldson

(2022). Together with Table 4, these results suggest that base development was accompanied by improvements in civilian living and production conditions.[40]

Taken together, our results suggest that attributing all observed urbanization and service-sector development around the US bases to their consumption demand and employment may overstate the effects of these channels. We provide suggestive evidence that the accompanying investment in infrastructure and durable capital, driven by public investment and early zoning regulation, may also have contributed to local development. This distinction is important for evaluating base expansion or closure in that the gains from additional consumption demand and employment from the bases may differ from what a simple observation of Okinawa in Section 3 suggests: The base expansion might not bring a large economic benefit if it does not accompany infrastructure investment, and the base closure might not have an immediate devastating impact because the associated infrastructure and durable capital do not immediately disappear. Indeed, we can find examples of recent policy discussions that recognize the potential benefits of infrastructure investment accompanying base development. For example, in March 2026, Canadian Prime Minister Mark Carney linked new Arctic military hubs and upgrades to existing facilities to investment in energy, telecommunications, and wastewater infrastructure, stating that "as we build these military operations, we can transform and benefit the surrounding communities."[41] In addition to the role of base-related consumption demand and employment, our findings also underscore the potential importance of such infrastructure investment when evaluating the local economic effects of base development.

8 Conclusion

To understand how military bases, which accompany large local consumption demand and employment opportunities, have spatial economic impacts, this paper examines how US bases in Okinawa affected its local and aggregate economy. Using newly digitized municipal data, we document "urbanization without industrialization" around the bases: greater base exposure is associated with growth in population density on non-base land and in the broad tertiary employment share, including base jobs, but has little association with changes in the manufacturing employment share. Combining base-related GNI components with a quantitative spatial model calibrated to 1970, we quantify the effects of consumption demand and employment.

Removing base-sourced consumption and employment, while holding local fundamentals and Okinawa's workforce fixed and leaving base land unavailable for private use, shifts residents away from more base-exposed municipalities and reallocates employment from tertiary activities toward agriculture and manufacturing. Consumption has larger effects on civilian service employment, whereas base employment has larger effects on population concentration. In the

[40] Conditional on these channels and geographic controls, however, base share is negatively and significantly associated with productivity in manufacturing, non-tradable services, and the geometric mean across the three private sectors. This residual association may reflect restrictions on civilian business expansion and investment near bases beyond the aggregate land-supply constraint captured by the model.

[41] Prime Minister of Canada, speech transcript, March 12, 2026, last accessed on October 1, 2026.

aggregate, the benchmark counterfactual reduces workers' welfare by 7.25% and GNI by 11.99%. These losses are large in absolute value but substantially smaller than the roughly 30% initial base-related GNI share, with sectoral reallocation cushioning the shock alongside sizable within-island migration. Returning base land dampens population outflows around the bases and reduces the welfare loss to 1.95%, with little change in sectoral composition. Base-related consumption generates relatively localized service employment gains, while manufacturing losses spread more widely across municipalities. Allowing external migration leaves welfare and sectoral responses close to the benchmark but increases GNI losses through population outflows. Less elastic manufacturing import supply weakens aggregate manufacturing crowding out and can reverse it when import quantities are fixed. Finally, the consumption and employment channels predict a smaller tertiary-share response than the observed postwar association. Evidence on infrastructure and early commercial access suggests that past investment in infrastructure and durable capital may also have contributed to development around the bases.

As rising geopolitical tensions renew attention to military basing decisions, our findings inform debates over hosting military bases by quantifying the contribution of ongoing spending and employment to host economies. Our analysis on Okinawa suggests that consumption spending and job creation around the bases, which many policymakers expect to stimulate the local economy, does induce economic development that accompany a large-scale in-migration and service-sector development. Despite the positive impact, there are two important qualifications. First, the sectoral reallocation away from manufacturing and toward the base employment or service-sector jobs may partially dampen its economic benefits. Second, the impact of consumption spending and employment could be smaller than a naive observation of Okinawa suggests: our analysis also highlights the potential importance of investment in infrastructure and durable capital. Incorporating the impact of base development on public and private investment in a dynamic spatial model, which requires more detailed data on infrastructure and durable capital, remains for future research.

References

Ahlfeldt, Gabriel M and Elisabetta Pietrostefani, "The economic effects of density: A synthesis," *Journal of Urban Economics*, 2019, *111*, 93–107.

_ , Stephen J Redding, Daniel M Sturm, and Nikolaus Wolf, "The economics of density: Evidence from the Berlin Wall," *Econometrica*, 2015, *83* (6), 2127–2189.

Allcott, Hunt and Daniel Keniston, "Dutch disease or agglomeration? The local economic effects of natural resource booms in modern America," *The Review of Economic Studies*, 2018, *85* (2), 695–731.

Allen, Treb and Dave Donaldson, "Persistence and Path Dependence in the Spatial Economy," NBER Working Paper 28059, National Bureau of Economic Research November 2022.

Ang, Amanda, Daniel M. Angel, and Andrii Parkhomenko, "Amenities in Quantitative Spatial Models," June 2026. Working paper, June 15, 2026.

Becker, Sascha O, Stephan Heblich, and Daniel M Sturm, "The impact of public employment: Evidence from Bonn," *Journal of Urban Economics*, 2021, *122*, 103291.

Biolsi, Christopher, "Local effects of a military spending shock: Evidence from shipbuilding in the 1930s," *Review of Economic Dynamics*, 2019, *32*, 227–248.

Brunet, Gillian, "Stimulus on the home front: The state-level effects of WWII spending," *Review of Economics and Statistics*, 2026, pp. 1–17.

Buera, Francisco J., Joseph P. Kaboski, Richard Rogerson, and Juan I. Vizcaino, "Skill-biased structural change," *Review of Economic Studies*, 2022, *89* (2), 592–625.

Chirakijja, Janjala, "The local economic impacts of prisons," *Review of Economics and Statistics*, 2024, *106* (6), 1442–1459.

Combes, Pierre-Philippe, Gilles Duranton, and Laurent Gobillon, "The production function for housing: Evidence from France," *Journal of Political Economy*, 2021, *129* (10), 2766–2816.

Comin, Diego, Danial Lashkari, and Martí Mestieri, "Structural Change with Long-Run Income and Price Effects," *Econometrica*, 2021, *89* (1), 311–374.

Dahlberg, Matz, Linna Martén, and Björn Öckert, "Coping with job loss: evidence from military base closures," *The Scandinavian Journal of Economics*, 2024, *126* (3), 440–464.

Davis, Morris A and François Ortalo-Magné, "Household expenditures, wages, rents," *Review of Economic Dynamics*, 2011, *14* (2), 248–261.

Devoto, Florencia, Esther Duflo, Pascaline Dupas, William Parienté, and Vincent Pons, "Happiness on Tap: Piped Water Adoption in Urban Morocco," *American Economic Journal: Economic Policy*, 2012, *4* (4), 68–99.

Dingel, Jonathan I and Felix Tintelnot, "Spatial economics for granular settings," *mimeograph*, 2023.

Duarte, Margarida and Diego Restuccia, "The role of the structural transformation in aggregate productivity," *Quarterly Journal of Economics*, 2010, *125* (1), 129–173.

Epple, Dennis, Brett Gordon, and Holger Sieg, "A new approach to estimating the production function for housing," *American Economic Review*, 2010, *100* (3), 905–924.

Faber, Benjamin and Cecile Gaubert, "Tourism and economic development: Evidence from Mexico's coastline," *American Economic Review*, 2019, *109* (6), 2245–2293.

Faggio, Giulia and Henry Overman, "The effect of public sector employment on local labour markets," *Journal of Urban Economics*, 2014, *79*, 91–107.

Fajgelbaum, Pablo and Stephen J. Redding, "Trade, structural transformation, and development: Evidence from Argentina 1869-1914," *Journal of Political Economy*, 2022, *130* (5), 1249–1318.

Fan, Tianyu, Michael Peters, and Fabrizio Zilibotti, "Growing like India — The unequal effects of service-led growth," *Econometrica*, 2023, *91* (4), 1457–1494.

Franks, Luke, "The endless postwar: Okinawa at the modern frontier," *Reconsidering Postwar Japanese History: A Handbook*, 2023, pp. 276–291.

Fujita, Masahisa and Takatoshi Tabuchi, "Regional growth in postwar Japan," *Regional Science and Urban Economics*, 1997, *27* (6), 643–670.

Fujiwara, Ippei and Kiminori Matsuyama, "A technology-gap model of 'premature' deindustrialization," *American Economic Review*, 2024, *114* (11), 3714–3745.

Fukuyama, Michelle Ann Brogdan, "A Study of Okinawan Maids' Experiences in American-Controlled Okinawa." Doctoral dissertation, University of the Ryukyus September 2023.

Garin, Andrew and Jonathan Rothbaum, "The Long-Run Impacts of Public Industrial Investment on Local Development and Economic Mobility: Evidence from World War II," *The Quarterly Journal of Economics*, 2025, *140* (1), 459–520.

Gerard, François, Joana Naritomi, and Joana Silva, "Cash Transfers and the Local Economy: Evidence from Brazil," *Econometrica*, 2026. Forthcoming.

Gollin, Douglas, Remi Jedwab, and Dietrich Vollrath, "Urbanization with and without industrialization," *Journal of Economic Growth*, 2016, *21* (1), 35–70.

Hayakawa, Kazunobu, Hans R.A. Koster, Takatoshi Tabuchi, and Jacques-François Thisse, "High-speed rail and the spatial distribution of economic activity: Evidence from Japan's Shinkansen," 2021. RIETI Discussion Paper Series 21-E-003.

Hayashi, Fumio and Edward Prescott, "The depressing effect of agricultural institutions on the prewar Japanese economy," *Journal of Political Economy*, 2008, *116* (4), 573–632.

Heblich, Stephan, Stephen J Redding, and Daniel M Sturm, "The making of the modern metropolis: evidence from London," *The Quarterly Journal of Economics*, 2020, *135* (4), 2059–2133.

Henderson, J Vernon and Matthew A Turner, "Urbanization in the developing world: too early or too slow?," *Journal of Economic Perspectives*, 2020, *34* (3), 150–173.

Hobijn, Bart and Fernanda Nechio, "Sticker shocks: Using VAT changes to estimate upper-level elasticities of substitution," *Journal of the European Economic Association*, 2019, *17* (3), 799–833.

Hooker, Mark A and Michael M Knetter, "The effects of military spending on economic activity: Evidence from state procurement spending," *Journal of Money, Credit, and Banking*, 1997, pp. 400–421.

_ and _ , "Measuring the economic effects of military base closures," *Economic Inquiry*, 2001, *39* (4), 583–598.

Hornbeck, Richard and Daniel Keniston, "Creative destruction: Barriers to urban growth and the Great Boston Fire of 1872," *American Economic Review*, 2017, *107* (6), 1365–98.

Huneeus, Federico and Richard Rogerson, "Heterogeneous paths of industrialization," *Review of Economic Studies*, 2024, *91* (3), 1746–1774.

Jedwab, Rémi and Alexander Moradi, "The Permanent Effects of Transportation Revolutions in Poor Countries: Evidence from Africa," *The Review of Economics and Statistics*, 2016, *98* (2), 268–284.

Jedwab, Remi and Dietrich Vollrath, "Urbanization without growth in historical perspective," *Explorations in Economic History*, 2015, *58*, 1–21.

Kii, Masanobu, Tetsuya Tamaki, Yoshio Kajitani, and Tatsuya Suzuki, "Estimation of floor production function for urban economic model," *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, 2022, *77* (5). (In Japanese).

Komeito, *Okinawa beigun kichi no jittai chosa (Survey on the actual conditions of U.S. military bases in Okinawa)* 1969. (In Japanese).

Kusuyama, Hiroaki, "Okinawa-ken ni okeru churyugun tou roudousha no keizai kouka ni tsuite (Economic Effect Brought by USFJ Employees in Okinawa)," *Journal of Public Policy Studies*, 2012, *12*, 128–140. (In Japanese).

Lee, Jim, "The regional economic effects of military base realignments and closures," *Defence and Peace Economics*, 2018, *29* (3), 294–311.

Mendes, Arthur, Wataru Miyamoto, Thuy Lan Nguyen, Steven Pennings, and Leo Feler, "The Macroeconomic Effects of Cash Transfers: Evidence from Brazil," Working Paper 2024-02, Federal Reserve Bank of San Francisco July 2026.

Mian, Atif and Amir Sufi, "What Explains the 2007–2009 Drop in Employment?," *Econometrica*, 2014, *82* (6), 2197–2223.

Miyauchi, Yuhei, Kentaro Nakajima, and Stephen J Redding, "The economics of spatial mobility: Theory and evidence using smartphone data," *mimeo*, 2022.

Monte, Ferdinando, Stephen J Redding, and Esteban Rossi-Hansberg, "Commuting, migration, and local employment elasticities," *American Economic Review*, 2018, *108* (12), 3855–90.

Nakamura, Emi and Jón Steinsson, "Fiscal stimulus in a monetary union: Evidence from US regions," *American Economic Review*, 2014, *104* (3), 753–792.

Nicita, Alessandro, Marcelo Olarreaga, and Peri Silva, "Cooperation in WTO's Tariff Waters?," *Journal of Political Economy*, 2018, *126* (3), 1302–1338.

Okinawa Prefecture Government, *The History and Culture of Okinawa* 2000.

_ , "Churyu-gun youchi atochi riyou ni tomonau keizai hakyu kouka tou kentou chosa houkokusho (Study Report on the Economic Ripple Effects of the Utilization of Former Military Land)," 2007. (In Japanese).

Ortega, Francesc and Giovanni Peri, "The Effect of Income and Immigration Policies on International Migration," *Migration Studies*, 2013, *1* (1), 47–74.

Oshiro, Jun, "Ryukyu Okinawa no keizai hatten shi (The Economic Development History of Ryukyus and Okinawa)," in 1950 Club, ed., *Ryukyu Okinawa no keizai hatten shi*, Kobundo Communications, 2023. (In Japanese).

Paloyo, Alfredo R, Colin Vance, and Matthias Vorell, "The regional economic effects of military base realignments and closures in Germany," *Defence and Peace Economics*, 2010, *21* (5-6), 567–579.

Redding, Stephen J., "Quantitative urban economics," *Handbook of Regional and Urban Economics*, 2025, *6*, 73–141.

Reversion Issues Study Group, "Documents concerning materials of the Economic Committee (Keizai Iinkai Shiryō ni Kansuru Shorui)." (In Japanese).

Rodrik, Dani, "Premature deindustrialization," *Journal of Economic Growth*, 2016, *21* (1), 1–33.

Rosenthal, Stuart S. and William C. Strange, "Evidence on the nature and sources of agglomeration economies," in J. Vernon Henderson and Jacques-François Thisse, eds., *Handbook of Regional and Urban Economics*, Vol. 4, Amsterdam: North-Holland, 2004, pp. 2119–2171.

Taira, Yoshitoshi, *Sengo Okinawa to beigun kichi: jyuyou to kyozetsu no hazama de 1945~1972 nen (Postwar Okinawa and U.S. military bases: Between acceptance and rejection, 1945–1972)*, Hosei University Press, 2012. (In Japanese).

Takeda, Kohei and Atsushi Yamagishi, "The economic dynamics of city structure: Evidence from Hiroshima's recovery," *Journal of Political Economy*, 2026. forthcoming.

Tomikawa, Moritake, "Okinawa no beigun kichi no keizai kouka (The U.S. military base economic effects in Okinawa," *Journal of General Industrial Research*, 2018, *26*, 35–58. (In Japanese).

Tomochi, Masaki, "A research note on issues of off-base residence of the United States Forces Personnel in Okinawa: A case study of Sunabe area in Chatan, Okinawa," *Journal of Economics and Environmental Policy*, 2009, *5*, 83–103. (In Japanese).

Toriyama, Atsushi, "Senryou to genjitsu shugi (Occupation and realism)," in Atsushi Toriyama, ed., *Imo to Hadashi: Senryou to genzai*, 2009. (In Japanese).

_ , *Okinawa: Kichi shakai no kigen to soukoku: 1945–1956 (Okinawa: The Origins and conflicts of a military base society (1945−1956))*, Keisou shobo, 2013. (In Japanese).

Toya, Yoshiharu, "The Authority and Function of the Labour Relations Commission in Okinawa/Ryukyu Islands under the American Military Government (1)," *Ryudai Law Review*, March 2019, (100), 31–41. Research note; in Japanese.

Tsivanidis, Nick, "Evaluating the impact of urban transit infrastructure: Evidence from Bogotá's TransMilenio," 2023. Conditionally accepted at *American Economic Review*.

Twinam, Tate, "The Long-Run Impact of Zoning: Institutional Hysteresis and Durable Capital in Seattle, 1920–2015," *Regional Science and Urban Economics*, 2018, *73*, 155–169.

U.S. General Accounting Office, "Military Bases: Analysis of DOD's 1995 Process and Recommendations for Closure and Realignment," Report GAO/NSIAD-95-133, U.S. General Accounting Office, Washington, DC April 1995.

U.S. House of Representatives, "Local Economy, Environment, and Intergovernmental Cooperation: What Can Be Learned from Fort Ord?," House Hearing Serial No. 107-76, Committee on Government Reform, Subcommittee on Government Efficiency, Financial Management and Intergovernmental Relations, Washington, DC 2002. 107th Congress, 1st Session; hearing held August 28, 2001.

Yamasaki, Junichi, Kentaro Nakajima, and Kensuke Teshima, "From samurai to skyscrapers: How transaction costs shape Tokyo," *mimeo*, 2023.

Yoshida, Jiro, "The real estate production function: Variation by property type, country, and building age," *mimeograph*, 2018.

Zou, Ben, "The local economic impacts of military personnel," *Journal of Labor Economics*, 2018, *36* (3), 589–621.

Appendix to "Bases and the City: Spatial Economics of US Military Bases in Okinawa" (Not for Publication)

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A Details on the data

Okinawan workers at the US bases. Municipal counts of Okinawan base workers by residence and workplace are available in a special 1970 Census tabulation preserved by the Okinawa Prefectural Archives. For 1980 onward, we obtain these counts through custom-made Census tabulations. Both sources are in Japanese (last accessed on May 1, 2025).

US base areas. The US base area in 1956 is obtained from *Okinawa Nenkan*, while that during the occupation period is obtained from the Ryukyus (Okinawa) Statistical Yearbook, and that after Okinawa's return to Japan is obtained from *Okinawa-no beigun kichi* and *Okinawa-no beigun oyobi jieitai kichi*. We also use the shape file of the US bases constructed by the Okinawa Prefecture government, as illustrated in Figure 2. The data are available online at http://gis.pref.okinawa.jp/OpenData/ (in Japanese, last accessed on May 2 2025).

Addressing municipal mergers and divisions. We construct municipal crosswalks between years t_1 and t_2 and aggregate municipalities to common boundaries. For postwar comparisons, these generally follow the later boundaries because of municipal mergers. For prewar–postwar comparisons, we also combine postwar municipalities that belonged to the same prewar municipality. Two prewar municipalities, *misato-son* and *oozato-son*, were divided around 1950 and subsequently merged into different municipalities. We allocate their 1938 variables, such as population, using the corresponding shares after the division, then aggregate to the boundaries used in each comparison.

Geographic scope and treatment of GNI. The GNI statistics cover Okinawa Prefecture. We use them as an approximation for the main island, which accounts for roughly 90% of the prefecture's population today. Table A.1 describes the component shares and their interpolation. Foreign aid from the United States and Japan was about one-fourth of US-base-sourced income around 1970. We exclude this aid from base-related demand and assume that it is spent uniformly across Okinawa.

Summary of the data sources For convenience, Table A.1 summarizes the dataset used in this paper.

Table A.1: Variables and Data Sources

Category	Variable or measure	Data source(s)	Coverage and construction
Population	Municipality-level resident population	Population Census	Available at five-year intervals from 1950 onward. Tabulations for the US occupation period were newly digitized by the authors. The data cover the Japanese population and exclude US citizens.
Employment (residence-based)	Residents employed by industry	Population Census; *Okinawa Prefectural Statistical Yearbook* (*Okinawa-ken toukei sho*) for 1934 and 1938	Used to construct residence-based sectoral employment shares. The reduced-form analysis groups industries into agriculture, manufacturing (including construction), and services (including base employment). The model treats base employment separately by utulizing the separate base-employment data below.
Employment (workplace-based)	Number of workers by industry and municipality of workplace	1970 Population Census	This is a workplace-based measure and may differ from residence-based employment because of commuting across municipalities. These data provide the sectoral employment distribution used in the 1970 calibration.
US-base employment	Okinawan workers employed at US military bases, by municipality of residence and workplace	Population Census; a special 1970 Census tabulation preserved by the Okinawa Prefectural Archives; custom-made Census tabulations provided by the Ministry of Internal Affairs and Communications from 1980 onward	These data identify US-base workers separately from other sectors, by both residence and workplace.

Continued on next page

Table A.1 — Continued from previous page

Category	Variable or measure	Data source(s)	Coverage and construction
US-base size	US-base area and the share of municipal land occupied by US bases	*Okinawa Nenkan* for 1956; *Ryukyus Statistical Yearbook* and *Okinawa Statistical Yearbook* for the occupation period; *Okinawa-no beigun kichi* and *Okinawa-no beigun oyobi jieitai kichi* for the post-reversion period	The municipal base-area share is the main measure of exposure to consumption and base employment.
US-base geography	Geographic boundaries of US military bases	GIS shapefiles provided by the Okinawa Prefectural Government	Used to identify and map the locations of bases. For Figure 2, the land boundary of Okinawa Island is taken from the Digital National Land Information, using the 1975 geography as an approximation for 1970.
Alternative measures of base importance	Number of Okinawan workers and US military personnel at each base	Komeito (1969), *Okinawa beigun kichi no jittai chosa* [Survey on the Actual Conditions of US Military Bases in Okinawa]	Used to validate the base-area measure and in supplementary robustness analyses. Base-level counts are aggregated to municipalities and divided by total municipal land area, including land occupied by US bases.
Aggregate income	Gross National Income of Okinawa	*Ryukyus Statistical Yearbook* (*Ryukyus Toukei Nenkan*) and *Okinawa Statistical Yearbook* (*Okinawa Toukei Nenkan*)	Available annually from 1955 onward.
US-base-sourced income	Share of Okinawa's GNI derived from US bases, separately for purchases of goods and services; salaries of Okinawan base workers; and rents paid for base land	*Ryukyus Statistical Yearbook* and *Okinawa Statistical Yearbook*	Includes base-related payments financed by the Japanese government. Component shares for 1966–1971 are linearly interpolated. Economic aid and intergovernmental transfers, including base-related grants to local governments, are excluded.

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Table A.1 — Continued from previous page

Category	Variable or measure	Data source(s)	Coverage and construction
Household expenditure	Restaurant expenditure share in Naha relative to Okinawa as a whole	1970 Okinawa Household Expenditure Survey (National Diet Library Digital Collections: https://dl.ndl.go.jp/pid/12101674)	The observed relative expenditure share, 1.16, is used as the calibration target for the elasticity of substitution between tradable and non-tradable goods.
Model validation	Bilateral commuting probabilities in 1967 and municipality-level house prices	1967 commuting origin–destination data; 1970 edition of *Shichoson koufuzei seido kaisetsu*	These variables are not calibration targets. Commuting probabilities are compared across aggregated origin–destination pairs. House prices and model-implied floor-space rents are standardized before comparison.
Agricultural trade	Okinawa's agricultural-goods trade balance around 1970	*Okinawa Statistical Yearbook*	Used in the external-trade extension to discipline the manufacturing share in the tradable-goods composite.
Travel behavior	Trip origins, destinations, purposes, and workers' industries at the individual-trip level	1977 Okinawa Main Island Central and Southern Urban Area Person Trip Survey, provided by the Okinawa Prefectural Government	Trip purposes allow commuting and shopping trips to be identified, and worker industries allow commuting flows to be constructed separately by sector. The 1977 survey does not cover northern Okinawa Island. The corresponding 1989 and 2006 surveys are used for robustness checks.
Commuting and shopping flows	Bilateral commuting matrices for agriculture, manufacturing, and non-tradable services, and a bilateral shopping matrix between municipalities	Authors' construction using the 1977 Person Trip Survey	The sectoral commuting matrices are pooled to estimate the sector-conditioned commuting gravity equation. Shopping and leisure trips are used in the calibration of shopping travel costs.

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Table A.1 — Continued from previous page

Category	Variable or measure	Data source(s)	Coverage and construction
Travel distance	Bilateral road distance between municipalities	Newly digitized 1973 road network based on maps from the Geospatial Information Authority of Japan. 1930 and 2005 road networks are also digitized for robustness.	Distances are measured along the road network between municipal centroids and are used in the gravity-equation estimation. Bilateral distances change little when the 1973 network is replaced by the 1930 or 2005 network.
Water infrastructure	Share of settlements with installed water facilities	*Ryukyus Statistical Yearbook*, 8th ed. (1963), Table 323	Measured as of June 30, 1963 and constructed as the number of settlements with installed water facilities divided by the total number of settlements. Predecessor municipalities are aggregated to the fixed 1970 municipal units.
Road infrastructure	Municipal-road improvement rate, paving rate, road density outside base land, and the road quality index	Okinawa Prefectural Archives, *Doro ni kansuru shorui* [Documents Concerning Roads] series; *Kensetsu unyu yoran 1965* for aggregate cross-checks	Municipal-road ledgers refer mainly to 1963–64 and exclude military and Ryukyu Government roads. The first two measures divide improved or paved length by total municipal-road length; density divides total road length by 1954 municipal area net of March 1956 base land. The index is the unweighted sum of the full-sample z-scores of these three measures.
Early market access	Indicator for a 1949 road corridor designated to host markets authorized to transact with US occupation personnel	US Military Government Directives Nos. 38 (1948), 19 (1949), and 21 (1949); US Navy, *Okinawa Shima Road Map* (1945); authors' road-to-municipality crosswalk	The indicator equals one when the reconstructed centerline of a designated road crosses the interior of a fixed 1970 municipality (or for the separately designated Naha establishment). It represents access to authorized markets, not a statutory land-use zone.

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Table A.1 — Continued from previous page

Category	Variable or measure	Data source(s)	Coverage and construction
Terrain controls	Municipality-level mean elevation and mean slope	Ministry of Land, Infrastructure, Transport and Tourism, Digital National Land Information: administrative-area data (N03, 1950 and 1975) and elevation and slope fourth-mesh data (G04-c, 2011 release)	The approximately 500-meter grid cells are intersected with the reconstructed fixed 1970 municipal polygons and aggregated using intersection-area weights. In the infrastructure and fundamentals regressions, log(1 + mean elevation) and mean slope are standardized across the 35 municipalities.

B Supplementary results on stylized facts

B.1 Overview of supplementary results on stylized facts

Levels in 1970. Figure B.1 reproduces Facts 1 and 2 using 1970 levels instead of changes from 1938.

Endogeneity of base location. Table B.1 shows that the patterns persist after controlling for terrain and instrumenting base exposure with early occupation near the US landing point.

Prewar trends. Figure B.3 examines changes from 1934 to 1938. The population-density coefficient is positive ($p = 0.057$), with a cumulative point estimate about 3% of the post-war coefficient, but the magnitude is much smaller than the observed association. Sectoral employment-share coefficients are not statistically significant.

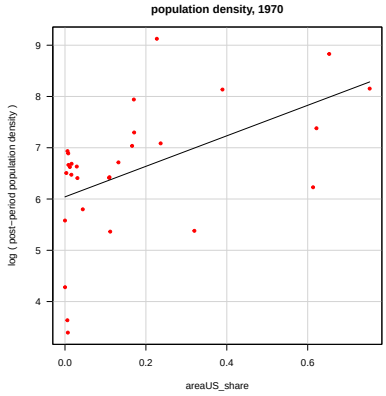
Base returns. Using the 1956 base-area share as an instrument yields the same qualitative conclusions (Figure B.4).

Alternative exposure measures. Replacing base-area shares with the densities of Okinawan base workers or US military personnel preserves the positive pre-reversion associations with population density and services (Figures B.5 and B.6).

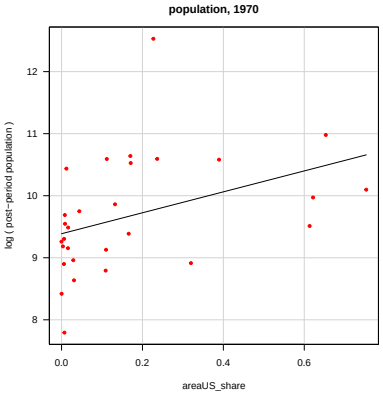
Patterns over time. Figure B.7 shows similar patterns across years, with smaller coefficients after reversion in 1972 as the bases' GNI share declined.

Service subsectors. Table B.2 shows that wholesale and retail and intangible services have the strongest positive associations with base exposure.

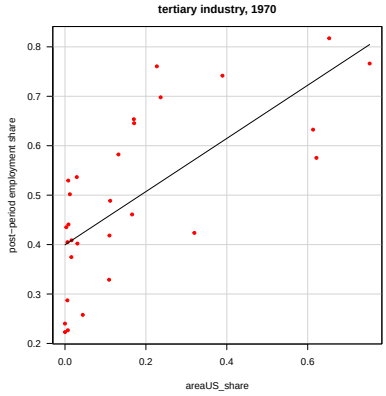
B.2 Stylized Facts for the 1970 level variables



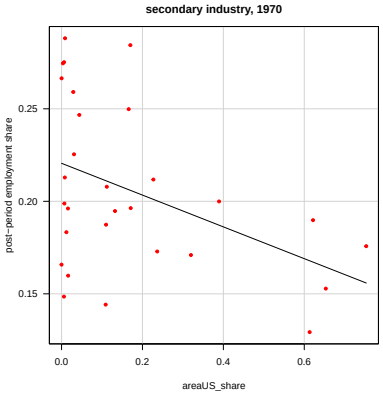
(a) Population density



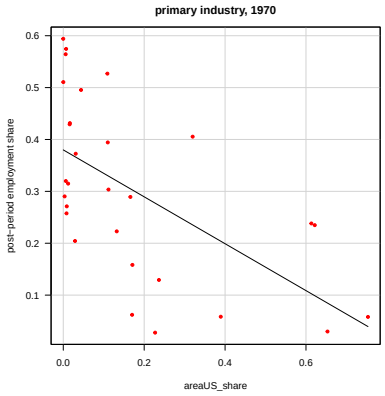
(b) Population level



(c) Service share



(d) Manufacturing share



(e) Agriculture share

Figure B.1: Relationship between US bases, Urbanization, and Industry Structure (1970, without differencing)

Note: Each point represents a municipality. The horizontal axis is the 1970 US-base area share. The outcomes are log population density, log population, and the employment shares of services, manufacturing, and agriculture. Population density uses municipal land area outside US bases. Lines are OLS fits.

B.3 Endogeneity of US base location

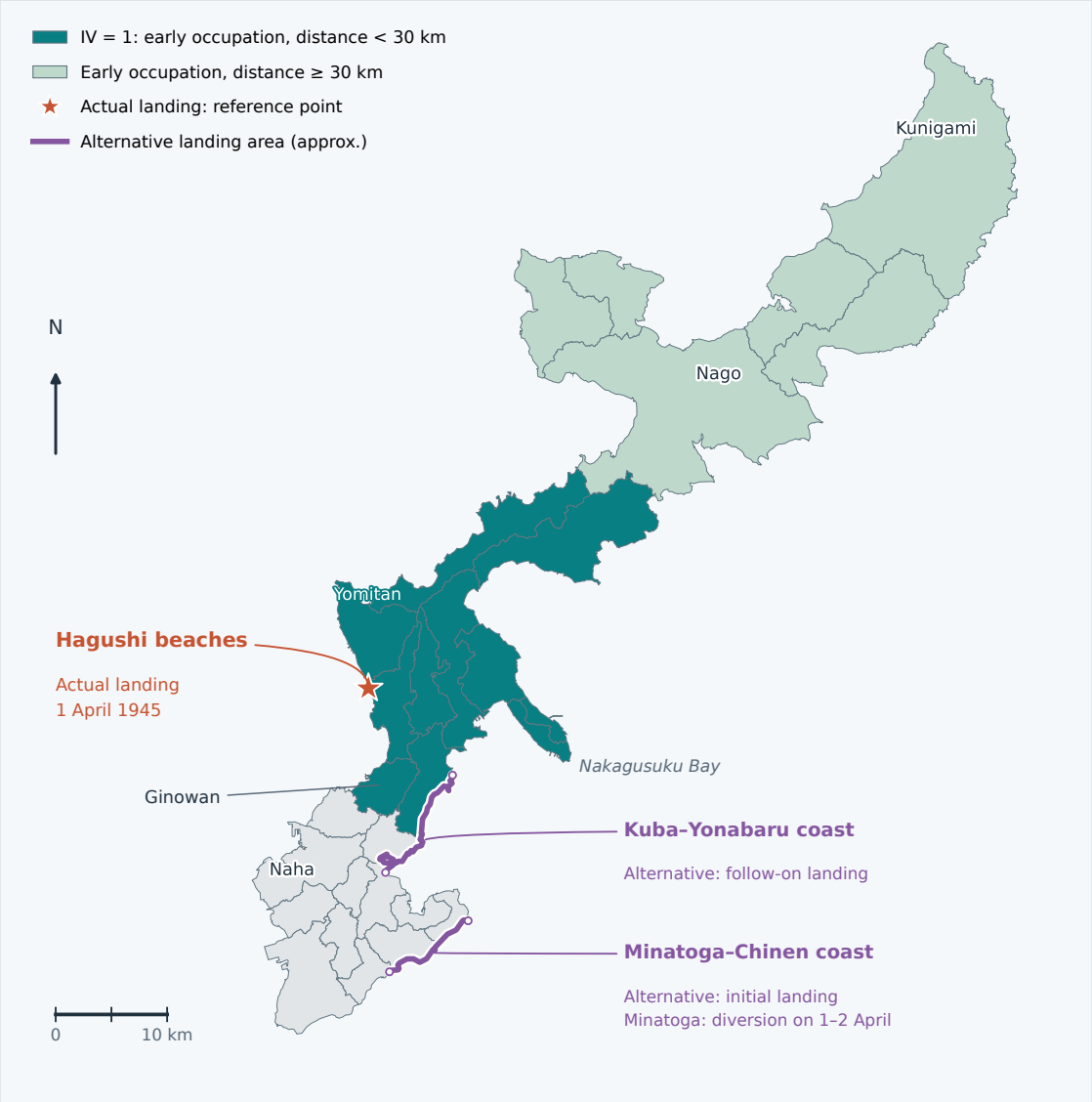


Figure B.2: US landing areas and the early-occupation instrument

Note: The star marks the US landing reference point; purple lines show alternative landing sectors. Dark teal denotes early-occupation municipalities with a mean distance below 30 km from the landing point, and light green denotes the remaining early-occupation municipalities. Boundaries are harmonized between 1938 and 1970; only the main island is shown. Sources: project GIS data and the historical accounts cited below. Place names are from the CODH historical place-name dataset (CC BY 4.0) and the Okinawa Prefectural Archives.

Figure B.2 shows that the Hagushi beaches, where US forces landed on April 1, 1945, were not the only landing areas considered. If the west-coast landing proved impracticable, the alternative plan called for Marines to land between Minatoga and Chinen Point, followed by Army divisions between Kuba and Yonabaru (Appleman, Burns, Gugeler and Stevens, 1948, pp. 32–34). These were successive components of the same alternative plan. In assessing this alternative, Appleman et al. (1948) write (p. 34):

> "Although the alternate plan met most of the requirements for a successful landing operation, it was distinctly a second choice because it would allow the enemy reserves to offer maximum opposition to the second landings and would require a prolonged assault against all the enemy forces on the island to complete the first phase of the mission."

The alternative was less attractive because of anticipated Japanese resistance. The Marine Corps history also describes the east-coast alternative and the weather concerns that prompted it (Nichols and Shaw, 1955, pp. 23–25). These accounts emphasize military and logistical considerations, including airfield access, unloading capacity, and the expected enemy response.

The colored areas in Figure B.2 were occupied within about a week of the landing, whereas resistance delayed occupation of southern Okinawa for several months. The US began building bases in the early-occupation area during the battle, partly to support air raids on mainland Japan. Many remained after the war, making early occupation a predictor of subsequent base exposure.

To implement this idea, we construct an instrumental variable as follows. We combine the early-occupation classification shown in Figure B.2 with proximity to the actual landing point:

$$Z_i = \text{EarlyOccupation}_i \, \mathbf{1}\{\bar{d}_i < 30 \text{ km}\}. \tag{B.1}$$

Here, $\bar{d}_i$ is the distance from the landing reference point to the municipal representative point, averaged across constituent 1970 municipalities when harmonizing boundaries. EarlyOccupation_i equals one if municipality i was occupied during the early stage of the battle. The instrument combines proximity to the landing point with occupation timing. We control for municipal mean altitude and slope in both stages.

	OLS			IV (2SLS)		
	(1) Population density	(2) Service share	(3) Manufacturing share	(4) Population density	(5) Service share	(6) Manufacturing share
Base area share (1970)	2.4584***	0.4050***	-0.0500	2.3998***	0.6598***	0.0204
	(0.4322)	(0.0760)	(0.0429)	(0.4483)	(0.1849)	(0.0872)
Mean altitude (m)	0.0034	0.0004	-0.0008	0.0036	-0.0006	-0.0011*
	(0.0041)	(0.0008)	(0.0005)	(0.0039)	(0.0009)	(0.0006)
Mean slope	-0.1316**	-0.0249*	0.0096	-0.1356**	-0.0076	0.0144
	(0.0550)	(0.0126)	(0.0067)	(0.0540)	(0.0160)	(0.0092)
Constant	0.7066***	0.4658***	0.1838***	0.7270***	0.3771***	0.1592***
	(0.1640)	(0.0596)	(0.0356)	(0.1935)	(0.0834)	(0.0482)
Observations	29	29	29	29	29	29
First-stage KP F				17.27	17.27	17.27

Notes: Outcomes are 1938–1970 changes in log population density and the service (broad tertiary) and manufacturing (broad secondary) employment shares. All columns use 29 harmonized municipalities and control for mean altitude and slope. Columns (4)–(6) instrument the 1970 base-area share with early occupation interacted with mean distance below 30 km from the landing point; both stages include the terrain controls. KP F is the Kleibergen–Paap first-stage F statistic. Robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table B.1: Base exposure, population density, and sectoral employment: OLS and IV

Table B.1 reports the results for population density growth and changes in the service and manufacturing employment shares. Columns (1)–(3) report the OLS results, while columns (4)–(6) report two-stage least squares estimates using the variable (B.1) as an instrument. The OLS estimates are 2.4584 for population density growth and 0.4050 for the change in the service employment share. The corresponding IV estimates are 2.3998 and 0.6598, implying that the population density estimate is almost unchanged under IV, while the service-share estimate is larger. For the manufacturing employment share, the OLS and IV estimates are −0.0500 and 0.0204, respectively, and neither is statistically significant. The first-stage Kleibergen–Paap F statistic is 17.27 in all three IV columns. These results indicate that the positive association of US bases with urbanization and service-sector expansion survives, while neither specification provides evidence of an increase in the manufacturing employment share.

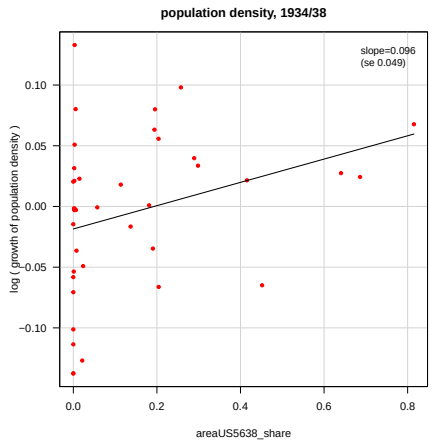
B.4 Testing the pre-trends

Figure B.3 relates changes from 1934 to 1938 to the 1956 US-base area share. The coefficient for log population-density growth is 0.096 (standard error 0.049, $p = 0.057$). Because municipal area is constant across the prewar years, log population and population-density changes are identical.

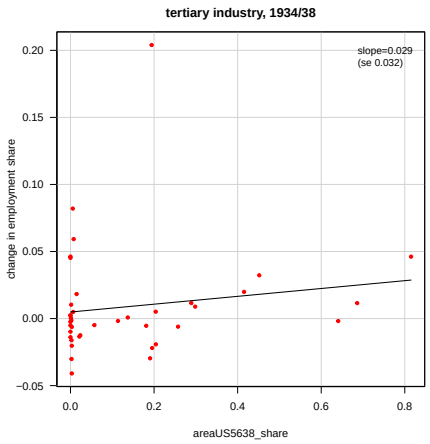
The cumulative prewar coefficient is about 3% of the postwar population-density coefficient of 3.045 in Figure 4a. A 10-percentage-point difference in base share is associated with an approximately 1% higher ratio of end-period to initial density in 1934–1938, compared with 36% in 1938–1970. Postwar density changes reflect both population change and the reduction in civilian land caused by base occupation.[B.1]

The prewar coefficients for service, manufacturing, and agricultural employment shares are 0.029 (standard error 0.032), 0.004 (0.008), and −0.034 (0.031), respectively. None is statistically significant at the 10% level.

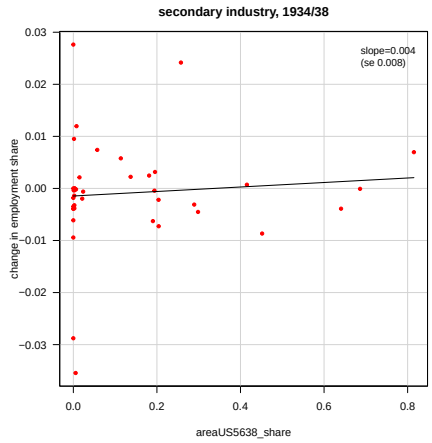
[B.1] The periods span four and 32 years. Annualized log-change coefficients are 0.024 and 0.095, respectively, so the prewar coefficient is about 25% of the postwar coefficient. The prewar regressions use 39 municipalities and the 1956 base-area share; the postwar regressions use 29 aggregated municipalities and the 1970 share.



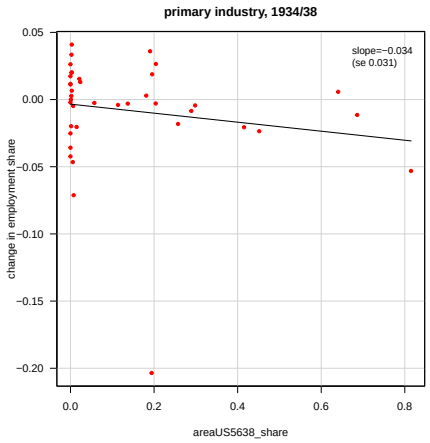
(a) Population density



(b) Service share



(c) Manufacturing share



(d) Agriculture share

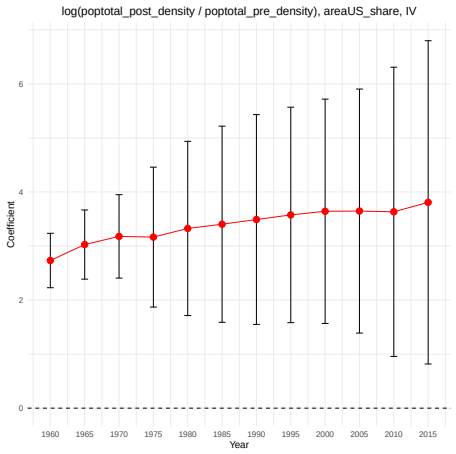
Figure B.3: Prewar trends, 1934–1938

Note: Each point represents one of 39 municipalities. The horizontal axis is the 1956 US-base area share. Outcomes are 1934–1938 changes in log population density and sectoral employment shares; a share change of 0.01 equals one percentage point. Lines are OLS fits, and boxes report slopes and conventional standard errors.

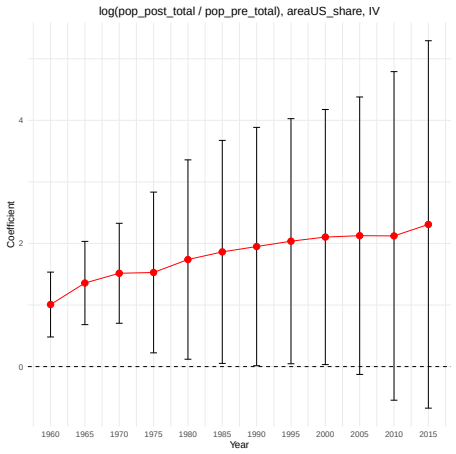
B.5 Addressing endogeneity of US base returns

We instrument each year's base-area share with its 1956 value to address selective base returns. Figure B.4 repeats Figure B.7 using two-stage least squares. The estimates preserve Facts 1 and 2.[B.2]

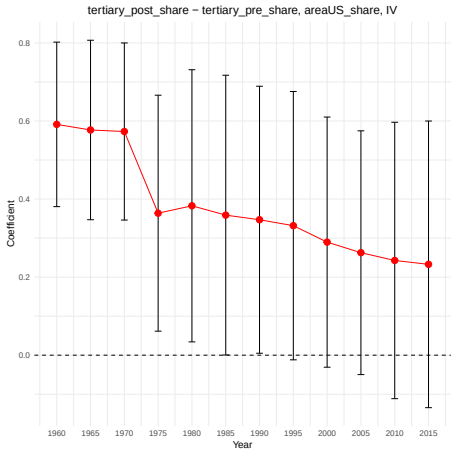
[B.2] First-stage F statistics range from 27.63 in 2015 to 226.16 in 1965.



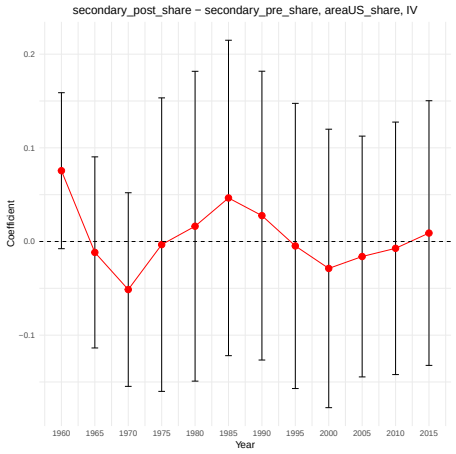
(a) Population density



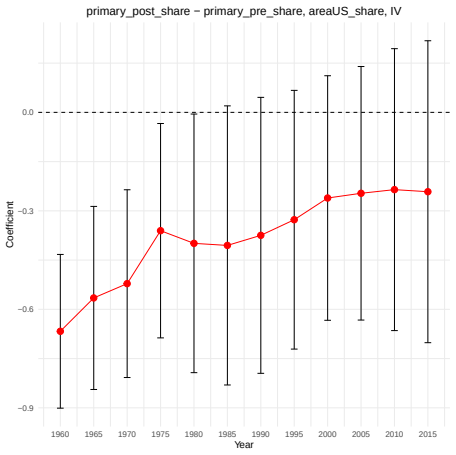
(b) Population level



(c) Service share



(d) Manufacturing share



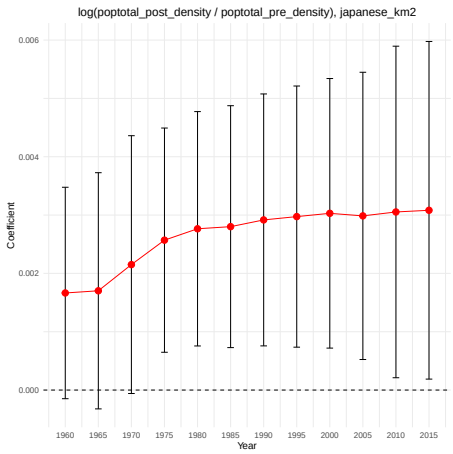
(e) Agriculture share

Figure B.4: Urbanization without industrialization around the US bases (IV estimation)

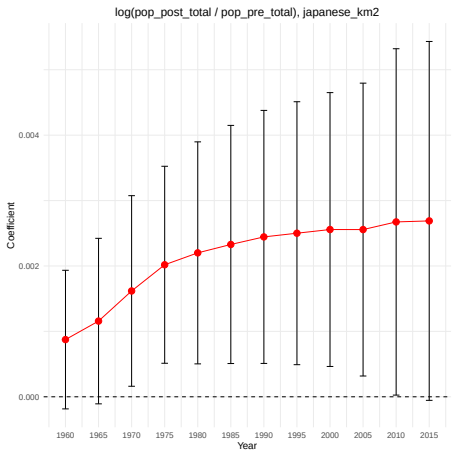
Note: Each panel reports year-specific two-stage least squares coefficients on the contemporaneous US-base area share, instrumented by its 1956 value. Outcomes and confidence intervals follow Figure B.7.

B.6 Using alternative measures of base importance

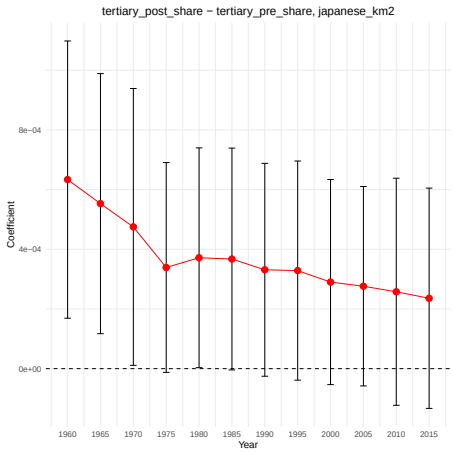
Instead of the share of land occupied by US bases in a municipality, we use alternative measures of base importance: the densities of Okinawan base workers and US military personnel. Komeito (1969) reports the number of Okinawan workers and US military personnel at each base. We aggregate these counts to municipalities and divide by total municipal land area, including base land. Figures B.5 and B.6 use the resulting densities. Both measures are positively associated with population density and service employment shares before reversion in 1972.



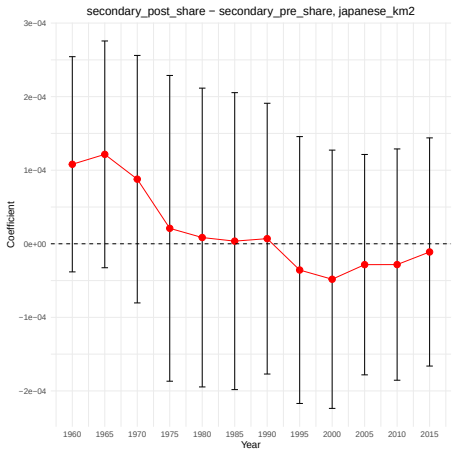
(a) Population density



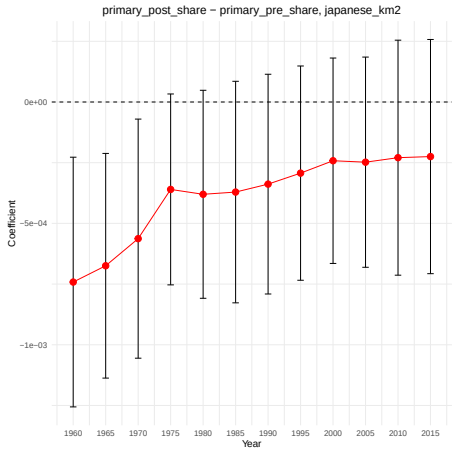
(b) Population level



(c) Service share



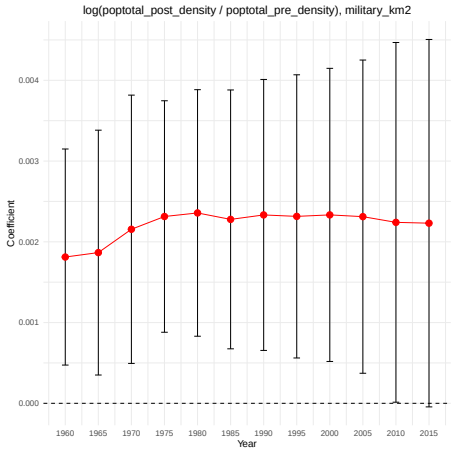
(d) Manufacturing share



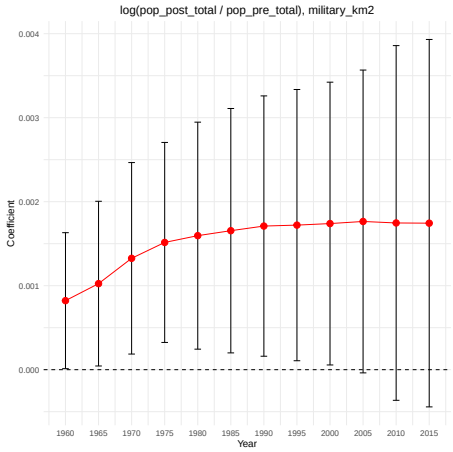
(e) Agriculture share

Figure B.5: Urbanization without industrialization and the US bases (Okinawan base-worker density)

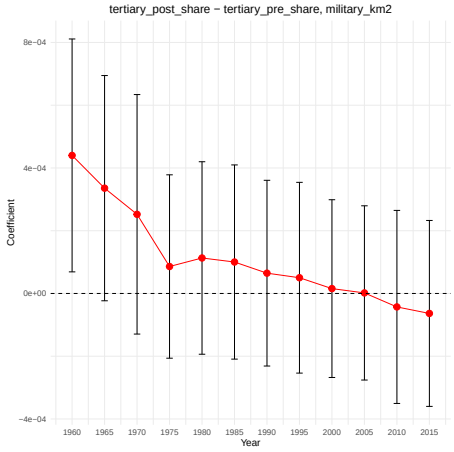
Note: The explanatory variable is the number of Okinawan base workers per square kilometer of total municipal land area, using counts from Komeito (1969). Outcomes and confidence intervals follow Figure B.7.



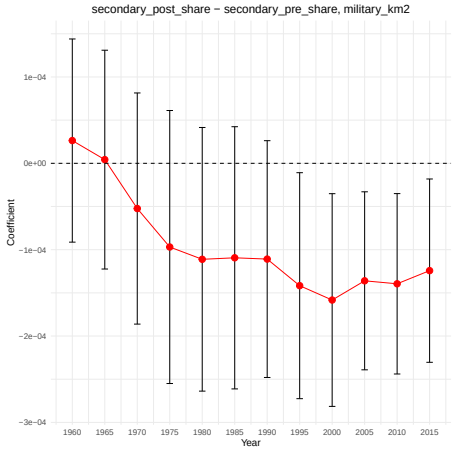
(a) Population density



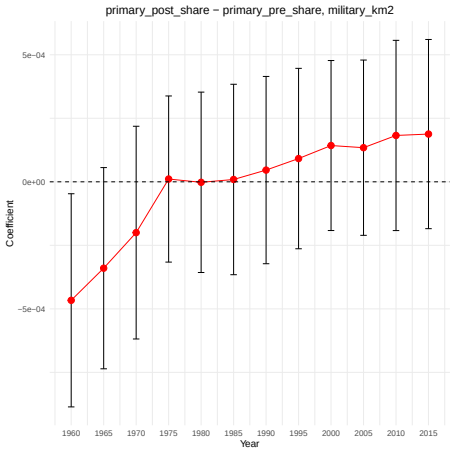
(b) Population level



(c) Service share



(d) Manufacturing share



(e) Agriculture share

Figure B.6: Urbanization without industrialization and the US bases (US military personnel density)

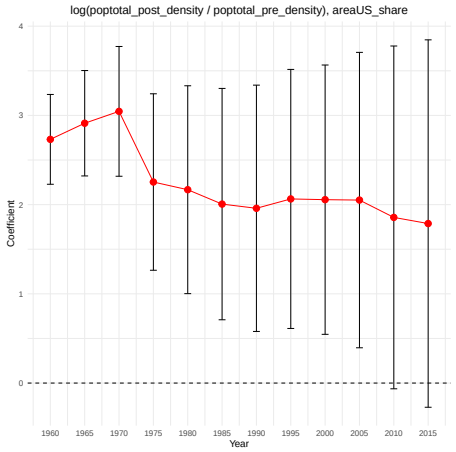
Note: The explanatory variable is the number of US military personnel per square kilometer of total municipal land area, using counts from Komeito (1969). Outcomes and confidence intervals follow Figure B.7.

B.7 The association with US bases in different years

Figure B.7 reports OLS estimates of β_t from

$$y_{it} - y_{i,1938} = \beta_t \text{USBaseShare}_{it} + \text{constant} + \text{error},$$

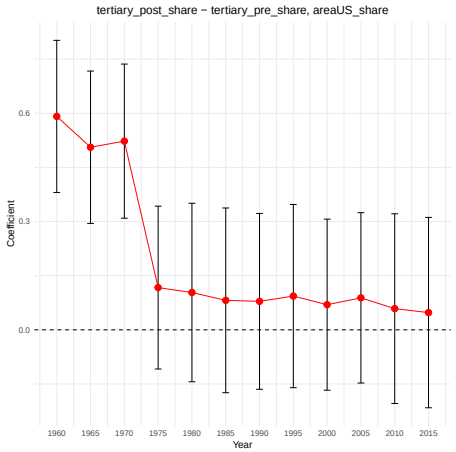
for $t = 1960, 1965, \ldots, 2015$. Here, USBaseShare_{it} is the municipal base-area share and y_{it} is the outcome of interest. Population, population density, and service employment shares are positively associated with base exposure in all years. The coefficients become smaller after reversion in 1972, when base-related income fell as a share of GNI (Figure 3). The service-share pattern mainly reflects growth in municipalities away from the bases. Manufacturing shares show little association with base exposure, while agricultural shares are negatively associated with it.



(a) Population density



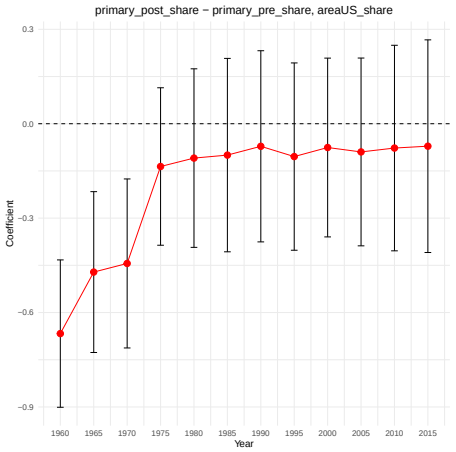
(b) Population level



(c) Service share



(d) Manufacturing share



(e) Agriculture share

Figure B.7: Urbanization without industrialization around the US bases: Long-run Perspective

Note: Each panel reports year-specific OLS coefficients on the contemporaneous US-base area share. Outcomes are changes from 1938 in log population density, log population, and sectoral employment shares. Postwar population density uses land outside US bases; prewar density uses total municipal land area. Black bars are 95% confidence intervals.

B.8 Disaggregating service-sector employment

We examine 1970 employment shares in wholesale and retail, finance and insurance, real estate, transportation and communications, utilities, intangible services, public administration, and unclassified jobs. Since comparable prewar categories are unavailable, we estimate a cross-sectional regression for each sector j:

$$y_{ij1970} = \beta_j \text{USBaseShare}_{i1970} + \text{constant} + \text{error term}. \tag{B.2}$$

Here, y_{ij1970} is employment in sector j divided by total employment across all sectors in municipality i in 1970, and β_j is the sector-specific coefficient.

Table B.2 shows the strongest positive associations for wholesale and retail and intangible services, consistent with demand for consumer services around bases.[B.3] Finance and insurance, real estate, and utilities also have positive, statistically significant coefficients. Transportation and communications and public administration do not.

	Employment share in total municipal employment							
	Wholesale and retail	Finance and insurance	Real estate	Transportation and communications	Utilities	Intangible service	Public administration	Unclassified
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
US base area share	0.214***	0.009*	0.004***	−0.035	0.013***	0.339***	−0.007	0.0001
	(0.047)	(0.005)	(0.001)	(0.023)	(0.003)	(0.054)	(0.008)	(0.0001)
Average employment share	0.162	0.009	0.002	0.056	0.006	0.220	0.038	0.000
Observations	29	29	29	29	29	29	29	29
R^2	0.434	0.111	0.318	0.082	0.442	0.593	0.027	0.061

*p<0.1; **p<0.05; ***p<0.01

Table B.2: Service employment share by sector and the US base area share (1970)

Notes: The table reports OLS estimates of equation (B.2) for 29 municipalities harmonized between 1938 and 1970. Outcomes are sectoral employment shares in 1970, measured relative to total municipal employment. Conventional standard errors are in parentheses.

[B.3] Fan, Peters and Zilibotti (2023) distinguish consumer and business services in their analysis of Indian growth.

C Calibration details

C.1 Supplementary calibration details

Demand weights. A more general version of the price index in equation (2) is

$$\mathbb{P}_i = \left[\beta_T (P_i^T)^{1-\kappa} + \beta_N (\mathbb{P}_i^N)^{1-\kappa}\right]^{\frac{1}{1-\kappa}},$$

where $\beta_T, \beta_N > 0$ are preference weights for tradable goods and non-tradable services. These weights are not separately identified from sectoral productivity in our calibration. Following Faber and Gaubert (2019), we set $\beta_T = \beta_N = 1$; the calibrated productivity terms therefore also absorb the demand weights.

Wage scales. Let $L_s \equiv \sum_i L_{is}$ and $\lambda_s \equiv \sum_i \sum_j \lambda_{ijs}$. Given the private-sector amenity normalizations in Step 3 of Section 5, we normalize the geometric mean of ω_{iA} to one and match the manufacturing and service wage scales to observed employment shares. Using equation (17), the restriction $\lambda_s/\lambda_A = L_s/L_A$ becomes

$$\frac{L_s}{L_A} = \frac{\sum_i \sum_j \widetilde{A}_{is}\omega_{js} d_{ij}^{-\vartheta}}{\sum_i \sum_j \widetilde{A}_{iA}\omega_{jA} d_{ij}^{-\vartheta}}, \qquad s \in \{M, N\}. \tag{C.1}$$

Appendix C.3 applies the analogous normalization to the US-base sector using its primed variables.

Service-price recovery. For a given κ, the service-price vector $\{P_i^N\}_{i=1}^I$ determines all the price indexes $\mathbb{P}_k^N$ through equation (3). Under the benchmark normalization $P^T = 1$, equations (2) and (5) also imply

$$\chi_k^N = \frac{\alpha \left[\sum_l \xi_{kl} (P_l^N)^{1-\sigma}\right]^{(1-\kappa)/(1-\sigma)}}{1 + \left[\sum_l \xi_{kl} (P_l^N)^{1-\sigma}\right]^{(1-\kappa)/(1-\sigma)}}.$$

Substituting these functions into the service-market clearing condition (30) leaves the service-price vector to be recovered, conditional on the other calibrated inputs.

Housing expenditure share. The housing expenditure share $1 - \alpha = 0.25$ used in Step 1 of Section 5 is also consistent with the 1999 National Survey of Family Income and Expenditure for Okinawa Prefecture (in Japanese; accessed May 16, 2025).

C.2 Gravity equation estimation

We estimate commuting and shopping gravity equations using the 1977 Okinawa person trip survey and intermunicipal distances from the digitized 1973 road network. For within-municipality trips, we use the expected distance between two points drawn independently and uniformly from a disk with municipality i's area, $Area_i$: $d_{ii} = \frac{128}{45\pi}\sqrt{\frac{Area_i}{\pi}}$. Commuting costs take the form $\tau_{ij} = d_{ij}^{\zeta}$.

Commuting gravity equation. We classify workers into agriculture, manufacturing, and non-tradable services using their reported industry and pool the three municipality-to-municipality commuting flow matrices. We estimate the multiplicative counterpart of equation (32), with origin-by-industry and destination-by-industry fixed effects. Table C.1 reports distance coefficients of -1.975 under PPML and -1.714 under log OLS, both significant at the 1 percent level. We set $\vartheta = 1.85$, approximately the midpoint of their absolute values. Given $\theta = 2.19$, the commuting cost elasticity is $\zeta = \vartheta/\theta \simeq 0.845$.

Shopping gravity equation. Using home-originating shopping and leisure trips from the 1977 person trip survey, we estimate

$$\ln \xi_{ij}^{\text{trip}} = -\psi_S \ln d_{ij} + \Psi_i^R + \Psi_j^D, \tag{C.2}$$

where Ψ_i^R and Ψ_j^D are origin and destination fixed effects. We also estimate the specification without these fixed effects.

Table C.2 reports PPML estimates using destination shares, including zeros, and log-OLS estimates using positive shares. We set $\psi_S = 2.3$ from the PPML estimate without fixed effects in Column 1 (-2.2978) and construct the model's shopping weights using equation (33).

	(1)	(2)
Log distance ($-\zeta\theta$)	-1.9750***	-1.7138***
	(0.0329)	(0.0422)
Method	PPML	OLS
Origin × industry fixed effects	Yes	Yes
Destination × industry fixed effects	Yes	Yes
N	1,728	672
R^2		0.850

Heteroskedasticity-robust standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C.1: Pooled industry-conditioned commuting gravity estimates

Notes: The table reports gravity estimates for commuting flows from the 1977 Okinawa person trip survey. Agriculture, manufacturing, and non-tradable services are pooled with a common distance coefficient and origin-by-industry and destination-by-industry fixed effects. PPML includes all $24 \times 24 \times 3$ origin–destination–industry cells, including zeros; OLS uses log flows for positive cells. Within-municipality distances are imputed as described in the text.

	(1)	(2)	(3)	(4)
Log distance	-2.2978***	-1.9928***	-3.6199***	-2.4406***
	(0.1506)	(0.1400)	(0.1475)	(0.1369)
Method	PPML	OLS	PPML	OLS
Origin fixed effects	No	No	Yes	Yes
Destination fixed effects	No	No	Yes	Yes
N	576	146	576	146
R^2		0.495		0.885

Heteroskedasticity-robust standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table C.2: Shopping gravity estimates with and without origin and destination fixed effects

Notes: The sample contains one home-originating shopping or leisure trip per respondent from the 1977 Okinawa person trip survey. Destination shares use survey expansion weights and are normalized within each origin. PPML includes all 24×24 origin–destination pairs, including zero shares; OLS uses log shares for positive pairs. Columns (1)–(2) omit origin and destination fixed effects; columns (3)–(4) include both. Distances use the digitized 1973 road network, with within-municipality distances imputed as described in the text.

C.3 Calibration using GNI component shares

We use the three base-related components reported separately in the Okinawan GNI statistics: labor income earned by Okinawan workers employed at US bases, land rents paid for US-base land, and US personnel's purchases of Okinawan goods and services. Let their observed shares of GNI be denoted by s_E, s_R, and s_C, respectively.

GNI in the equilibrium. Model-implied GNI is the total income of Okinawan workers and landowners:

$$\begin{aligned} Y &= \sum_i \sum_s w_{is} L_{is} + \sum_i \Omega_i \\ &= \underbrace{\sum_i \sum_{s \in \{A,M,N\}} \frac{(1-\mu\gamma^s) w_{is} L_{is}}{1-\gamma^s} + (1-\alpha)(1-\mu) \sum_i \sum_{s \in \{A,M,N,B\}} w_{is} L_{is}}_{\equiv Y_P} \\ &\quad + \underbrace{\sum_i w_{iB} L_{iB}}_{\equiv Y_E} + \underbrace{\sum_i m_i \mathbb{Q}_i^{US} S_i}_{\equiv Y_R} . \end{aligned}$$

Here, Y_P combines private-sector labor income and private-land income from production and housing; Y_E is labor income from US-base employment; and Y_R is rent paid for US-base land.

Matching the GNI share of the US-base labor income. Step 3 in Section 5 recovers private-sector wages and composite amenities. For the US-base sector, define $\omega'_{iB} \equiv a_B \omega_{iB}$ and $\widetilde{A}'_{iB} \equiv \widetilde{A}_{iB}/a_B$. We recover these variables from equations (34) and (35), normalize the geometric mean of $\widetilde{A}'_{iB}$ to one, and set the scale of ω'_{iB} to match the base-sector employment share. Equation (C.1) applies with $s = B$ and the primed variables because $\widetilde{A}_{iB}\omega_{iB} = \widetilde{A}'_{iB}\omega'_{iB}$.

We choose a_B so that $Y_E/Y_P = s_E/(1 - s_E - s_R)$. Given private-sector wages and ω'_{iB},

$$\begin{aligned} Y_P = &\sum_i \sum_{s \in \{A,M,N\}} \frac{(1-\mu\gamma^s) w_{is} L_{is}}{1-\gamma^s} + (1-\alpha)(1-\mu) \sum_i \sum_{s \in \{A,M,N\}} w_{is} L_{is} \\ &+ (1-\alpha)(1-\mu) a_B^{-\frac{1}{\theta}} \sum_i (\omega'_{iB})^{\frac{1}{\theta}} L_{iB} \end{aligned}$$

and

$$Y_E = a_B^{-\frac{1}{\theta}} \sum_i (\omega'_{iB})^{\frac{1}{\theta}} L_{iB}.$$

Solving $Y_E/Y_P = s_E/(1 - s_E - s_R)$ determines a_B. We then recover $w_{iB} = (\omega'_{iB}/a_B)^{1/\theta}$ and $\widetilde{A}_{iB} = a_B \widetilde{A}'_{iB}$.

Matching the GNI share of the land rents for the US bases. We set municipal US-base rent rates proportional to private-sector land rents and choose their scale to satisfy $Y_R/Y_E = s_R/s_E$:

$$\mathbb{Q}_i^{US} = \frac{s_R}{s_E} \frac{Y_E}{\sum_k m_k \mathbb{Q}_k S_k} \mathbb{Q}_i.$$

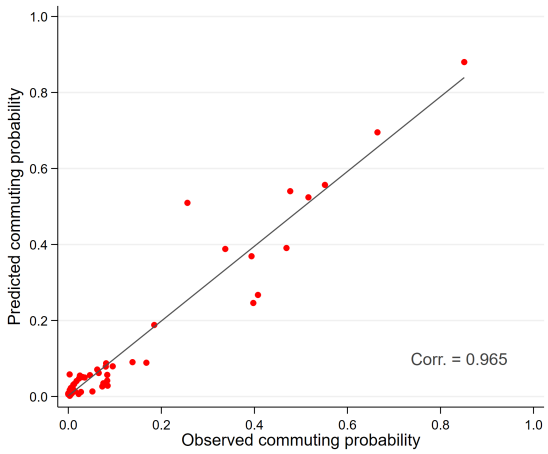
Thus, aggregate US-base land rents equal $(s_R/s_E)Y_E$.

Matching the GNI share of consumption by the US military personnel. We allocate the observed total number of US military personnel, L^{US}, across municipalities in proportion to US-base land area:

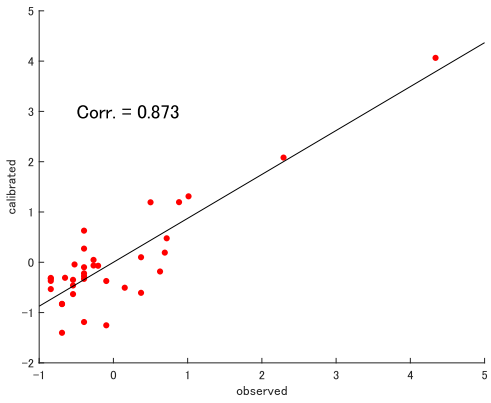
$$L_i^{US} = L^{US} \frac{m_i S_i}{\sum_k m_k S_k}.$$

Per-capita spending is $e_{US} = s_C Y/L^{US}$, which matches US personnel's purchases to their observed share of GNI. We then obtain consumption as $\Phi_i = m_i \mathbb{Q}_i^{US} S_i + e_{US} L_i^{US}$.

C.4 Model fit



(a) Bilateral commuting probability



(b) House prices and floor-space rents

Figure C.1: Comparison of the observed and calibrated values

Note: Panel C.1a compares 1967 commuting probabilities with model predictions for 1970. Each point represents one of 56 origin–destination pairs across eight regions; probabilities are normalized within origin regions. Panel C.1b compares municipal house prices with model-implied floor-space rents. Each point represents a municipality, and both variables are standardized to mean zero and standard deviation one. Observed values are on the horizontal axes and model predictions on the vertical axes. Lines are OLS fits; correlations are 0.958 and 0.873, respectively. Sources: the 1967 Person Trip Survey (Okinawa Prefectural Library) and the 1970 *shichoson koufuzei seido kaisetsu*.

D Counterfactual details

D.1 Computational procedure

At a given aggregate workforce L, we iterate between wages and sector-specific residence counts. The benchmark sets $\Phi_i = L_{iB} = 0$ and holds occupied base land fixed. We initialize $\{R_{is}\}$ with $\sum_i \sum_s R_{is} = L$ and $R_{iB} = 0$.

Inner loop. Holding $\{R_{is}\}$ fixed, we proceed as follows:

1. Initialize $\{w_{iA}, w_{iM}, w_{iN}\}$ and set $w_{iB} = 0$.
2. Compute private-sector employment from (21).
3. Solve (28) for Q_i, using $H_i = (1 - m_i) S_i Q_i^{\phi}$ under the normalization $P^T = 1$. Obtain H_{iN} from (27) and recover service prices from (25): $P_i^N = (\widetilde{b}_i^N)^{-1} w_{iN}^{1-\gamma^N} Q_i^{\gamma^N} L_{iN}^{\frac{1-\gamma^N}{1-\sigma}} H_{iN}^{\frac{\gamma^N}{1-\sigma}}$.
4. Update agricultural and manufacturing wages from (23) and (24), and service wages from (30).
5. Repeat steps 2–4 until wages converge.

Main loop. Update $R_{is} = L\lambda_{is}^R$ using (18) and (20), then repeat the inner loop. Iterate until the residence distribution converges.

For experiments retaining base employment, each inner-loop iteration also updates base wages. Combining (17) and (21) gives

$$w_{iB} = \left[\frac{L_{iB}}{\sum_k \frac{A_{kB} R_k^P \tau_{ki}^{-\theta}}{\sum_{s \in \{A,M,N\}} A_{ks} \sum_l w_{ls}^{\theta} \tau_{kl}^{-\theta}}} \right]^{1/\theta}, \qquad R_k^P = \sum_{s \in \{A,M,N\}} R_{ks}.$$

Base wages are zero where $L_{iB} = 0$. The open-population model adds the outer loop in Appendix D.7.

D.2 Local price and wage adjustment in the benchmark counterfactual

Figure D.1 maps benchmark changes in wages, floor-space rents, service prices, and varieties. Non-tradable wages fall in every municipality, and varieties fall in 32 of 35. Population inflows sustain service demand in the three municipalities where varieties increase. Manufacturing wages rise everywhere as employment expansion raises productivity through agglomeration. Lower floor-space rents reinforce these gains around the bases; elsewhere, productivity gains outweigh higher rents. With fixed agricultural prices and productivity, agricultural wages rise where rents fall and decline where rents rise, as implied by equation (23). Floor-space rents fall in Chatan (-12.19%), Koza (-10.77%), and Kin (-10.96%), but rise in 19 municipalities, consistent with the residential reallocation in Figure 5. Lower marginal costs outweigh the effect of fewer varieties on the CES price index, so service prices fall in every municipality.

Figure D.2 reports exposure gradients of -0.610 for non-tradable varieties, -0.167 for floor-space rents, and -0.114 for service prices. The wage gradients are -0.317 for non-tradables, 0.060 for manufacturing, and 0.075 for agriculture. These responses are stronger in municipalities with greater base exposure.

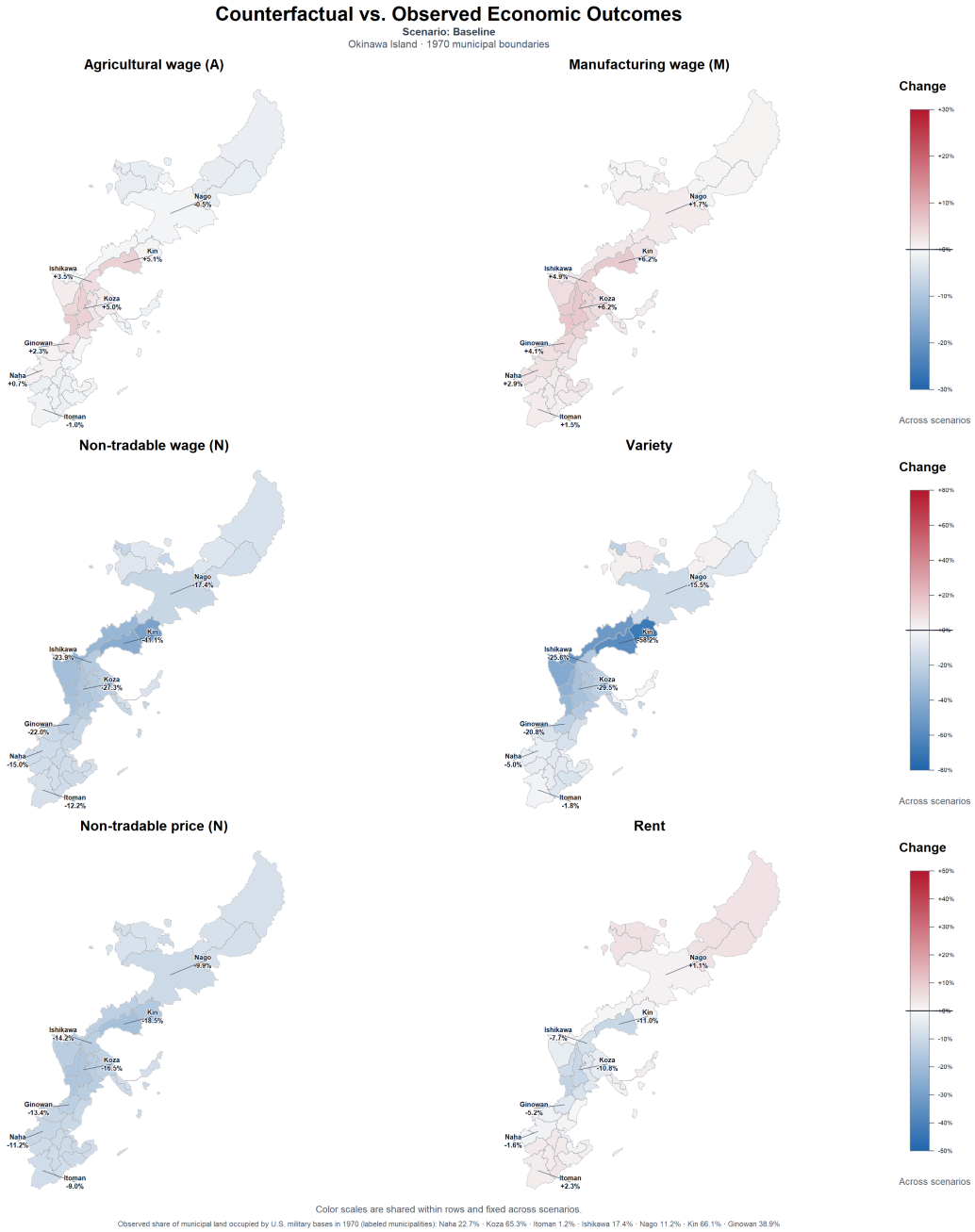
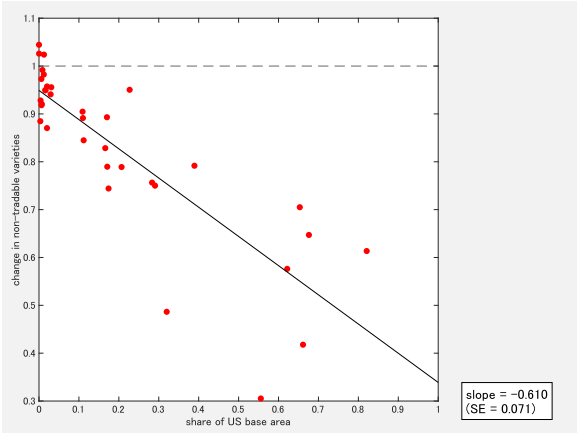
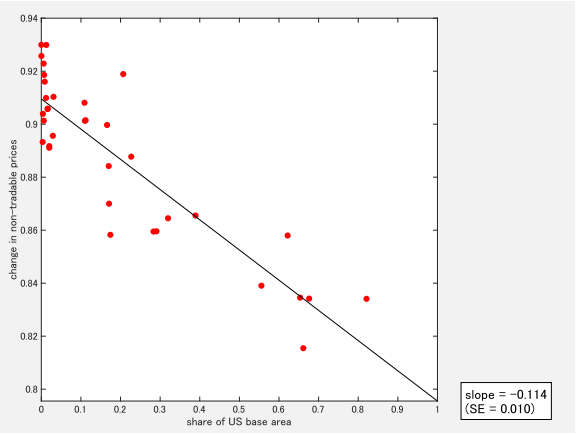


Figure D.1: Municipal wage, rent, price, and variety responses in the benchmark counterfactual

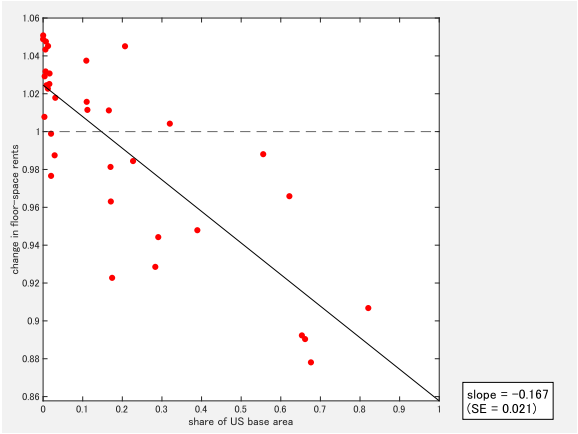
Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. The benchmark sets $\Phi_i = L_{iB} = 0$ and leaves base land unavailable for private use. Red denotes an increase and blue a decrease.



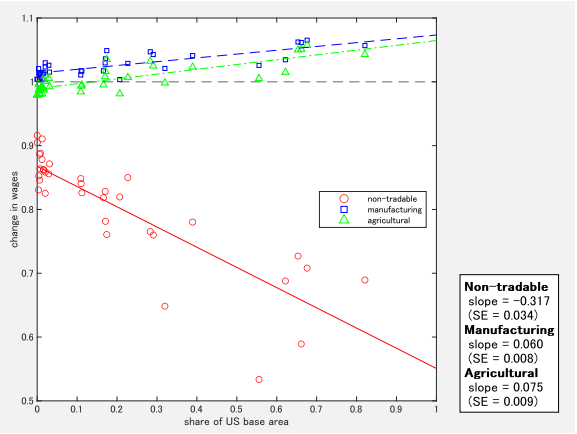
(a) Non-tradable varieties



(b) Non-tradable price index



(c) Floor-space rent



(d) Sectoral wages

Figure D.2: Exposure gradients in prices, rents, varieties, and wages in the benchmark counterfactual

Note: Each panel plots a counterfactual-to-observed ratio against the 1970 base-area share. In panel D.2d, red circles, blue squares, and green triangles denote non-tradable, manufacturing, and agricultural wages. Lines are linear fits, and the boxes report slopes and standard errors. The benchmark sets $\Phi_i = L_{iB} = 0$ while leaving base acreage unavailable for private use.

D.3 Spatial results for the consumption and employment channels

This subsection reports the maps and fitted gradients underlying the channel comparison in Section 6.1.1. Figures D.3 and D.4 show population responses when $\Phi_i = 0$ and when local base employment is removed, respectively.

Figures D.6 and D.7 show the corresponding local prices, wages, rents, and varieties. Their common color scales facilitate direct comparison of the two mechanisms.

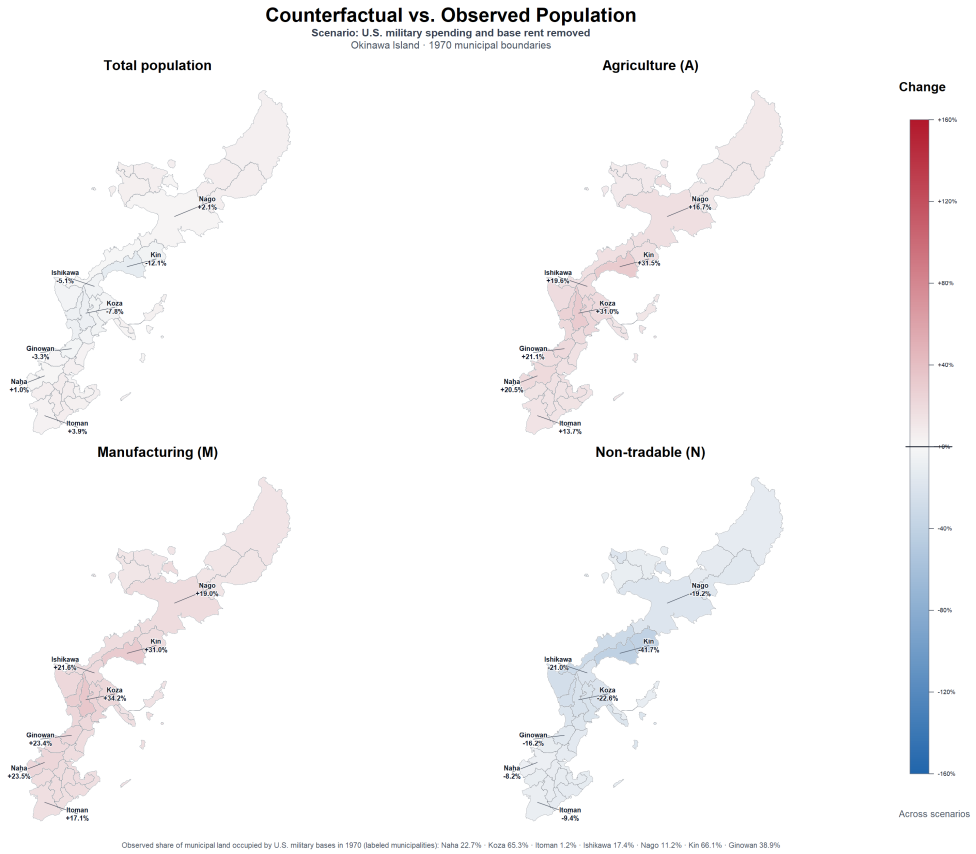


Figure D.3: Municipal population responses when base-sourced expenditure and rent income are removed

Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for total residents and residents working in agriculture (A), manufacturing (M), and civilian non-tradable services (N). The total includes base workers. The counterfactual sets $\Phi_i = 0$, removing US personnel's purchases and base-rent income while retaining base employment and occupied base land. Red denotes an increase and blue a decrease.

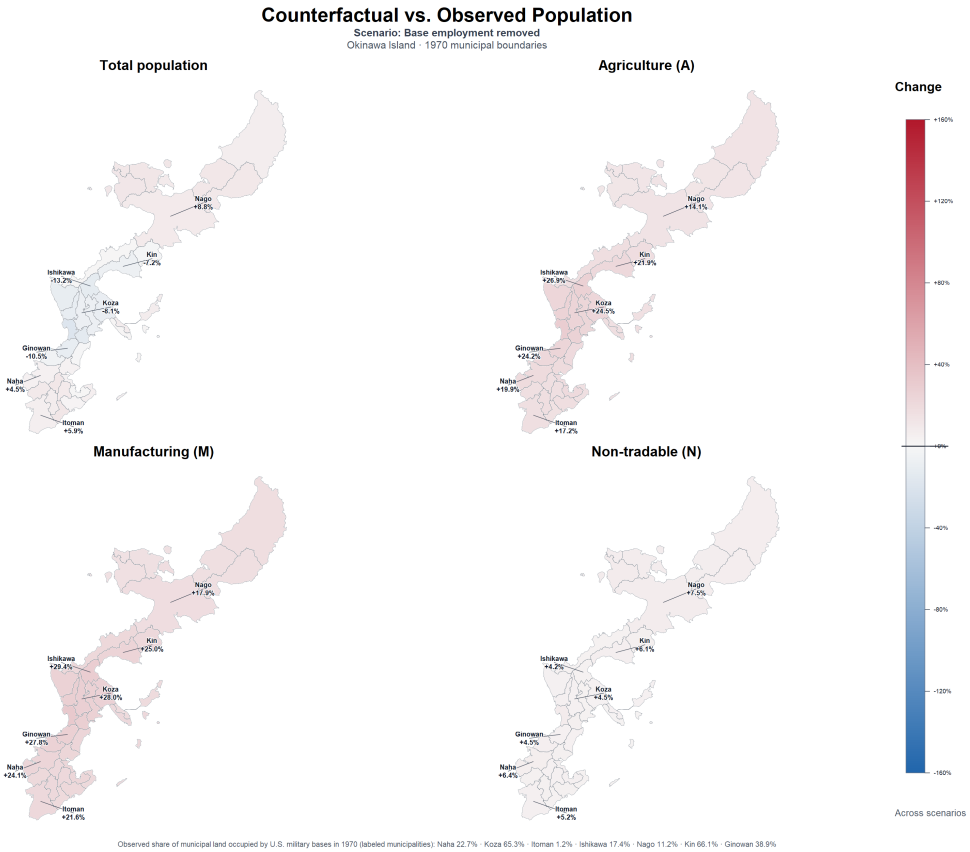
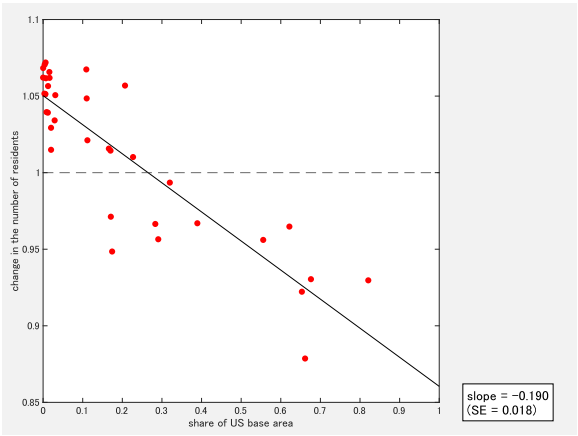
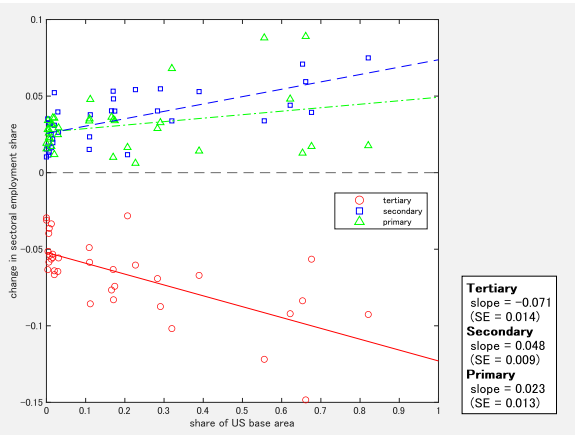


Figure D.4: Municipal population responses when local base employment is removed

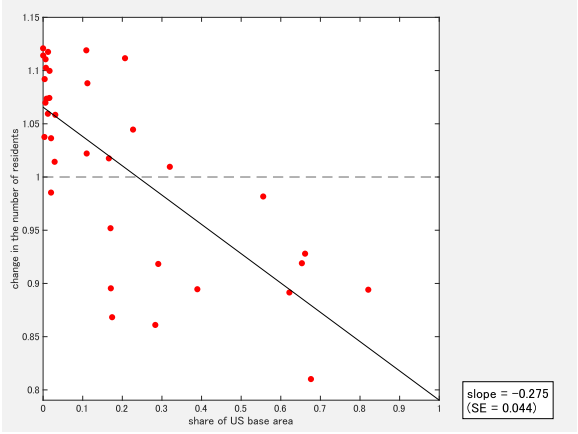
Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total residents and residents working in agriculture (A), manufacturing (M), and civilian non-tradable services (N). The observed total includes base workers. The counterfactual sets $L_{iB} = 0$, retaining US personnel's purchases, base-rent income, and occupied base land. Red denotes an increase and blue a decrease.



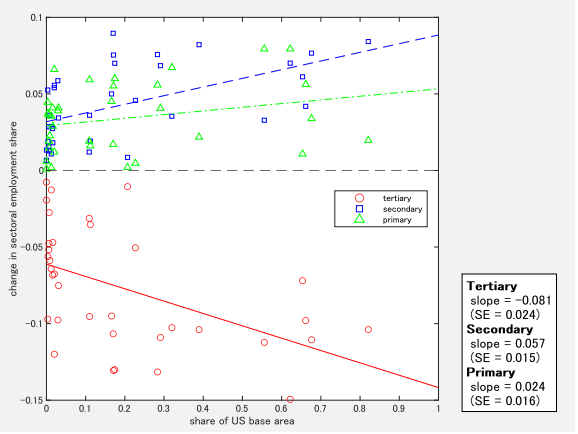
(a) Purchases and rents removed: residents



(b) Purchases and rents removed: sector shares



(c) Base employment removed: residents



(d) Base employment removed: sector shares

Figure D.5: Exposure gradients for the consumption and employment channels

Note: Each point is a municipality; the horizontal axis is the 1970 base-area share. Panels D.5a and D.5b set $\Phi_i = 0$; panels D.5c and D.5d set $L_{iB} = 0$. The other channel and occupied base land remain fixed. Resident panels plot counterfactual-to-observed ratios; share panels plot counterfactual-minus-observed sector-share changes among residents. Red circles denote tertiary (civilian non-tradables plus base employment), blue squares manufacturing, and green triangles agriculture. Lines are linear fits; boxes report slopes and standard errors.

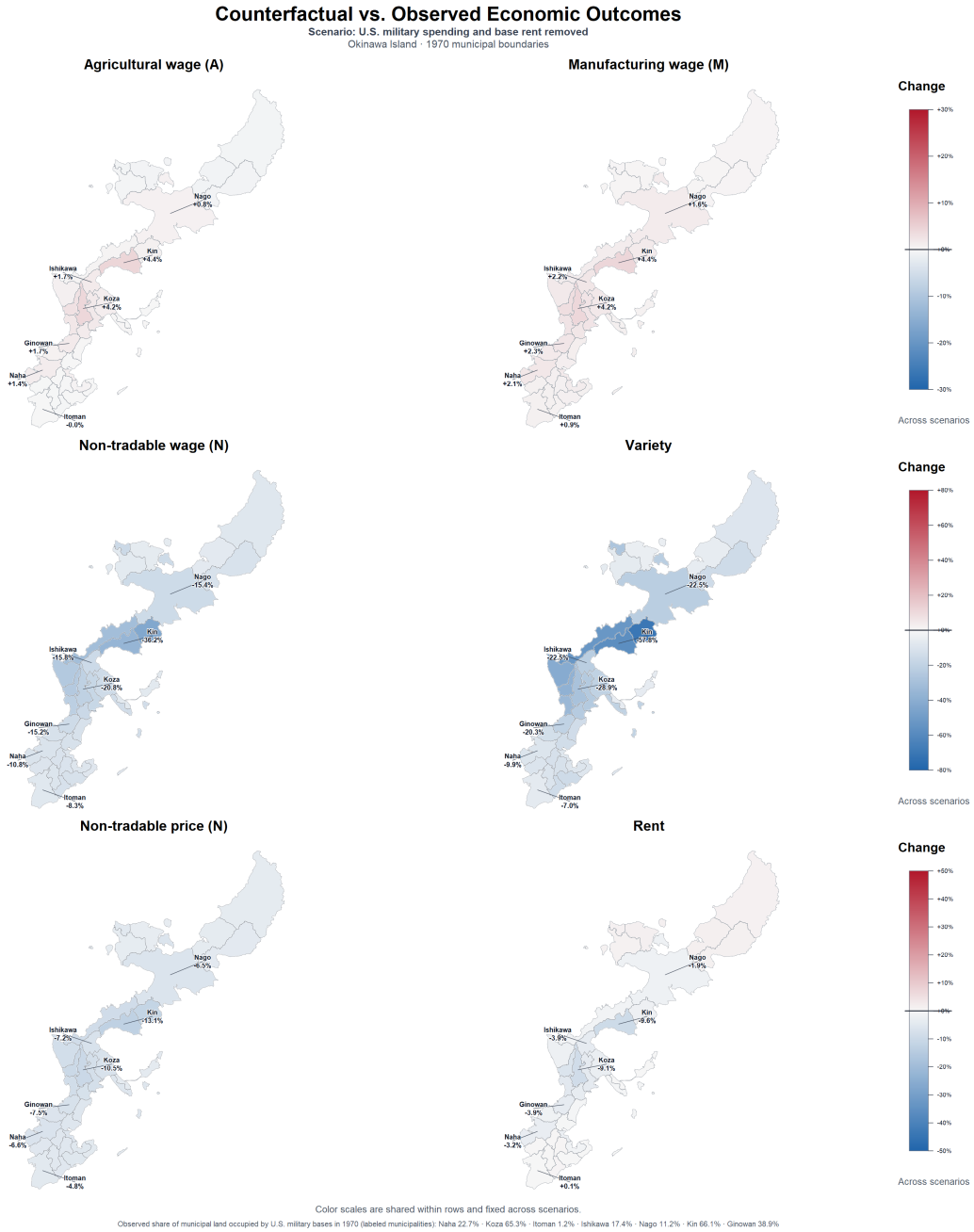


Figure D.6: Municipal economic outcomes when base-sourced expenditure and rent income are removed

Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. The counterfactual sets $\Phi_i = 0$, removing US personnel's purchases and base-rent income while retaining base employment and occupied base land. Red denotes an increase and blue a decrease.

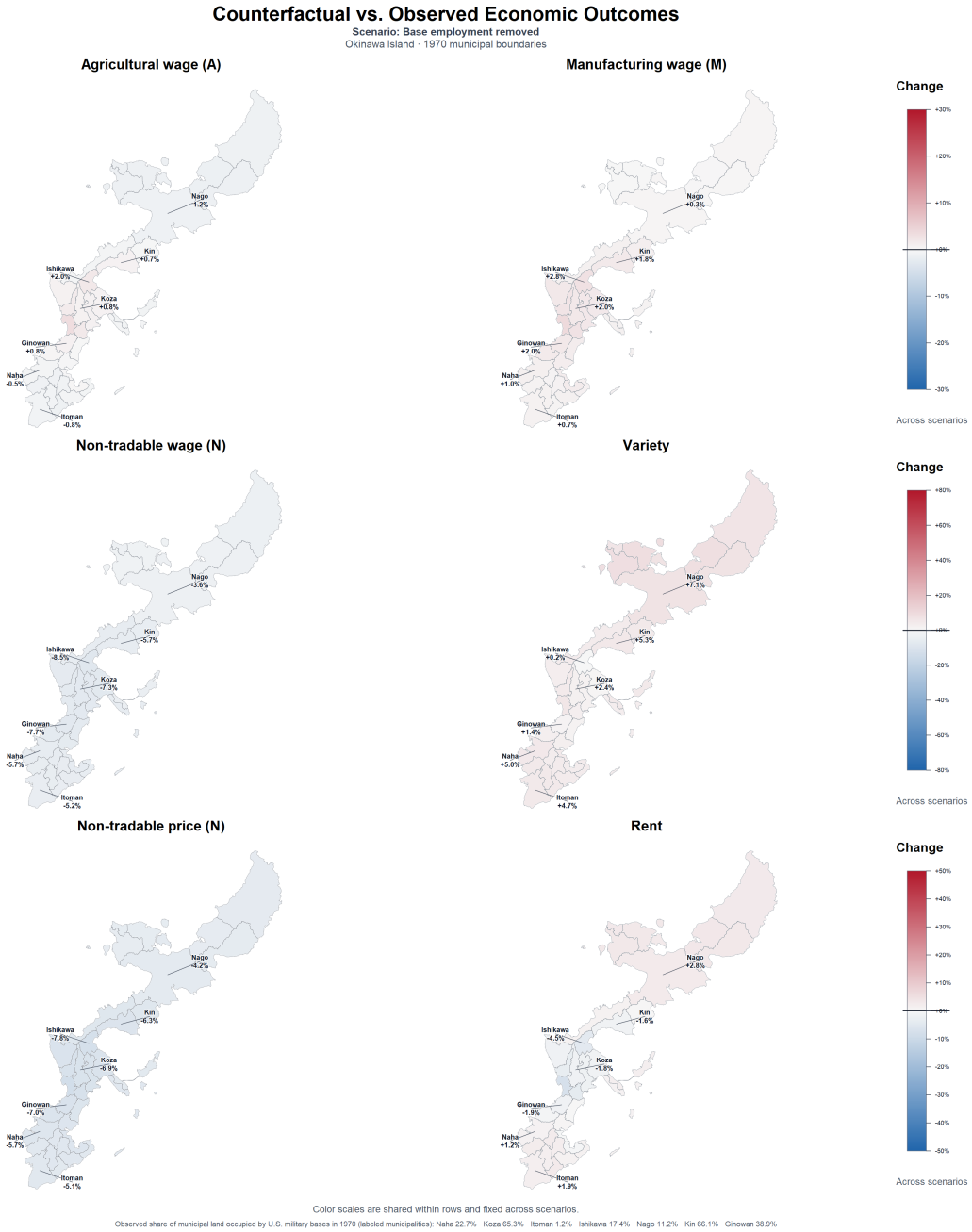


Figure D.7: Municipal economic outcomes when local base employment is removed

Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. The counterfactual sets $L_{iB} = 0$, retaining US personnel's purchases, base-rent income, and occupied base land. Red denotes an increase and blue a decrease.

D.4 Fixed municipal residence

The fixed-residence counterfactual in Section 6.1.2 imposes $R_i^{cf} = R_i$ while allowing sector and workplace choices, commuting, and local prices to adjust. Figure D.8 shows resident-sector reallocation, Figure D.9 reports exposure gradients, and Figure D.10 shows wages, rents, prices, and varieties. Sectoral and workplace reallocation remains strongly related to base exposure, while the rent and variety exposure gradients are flatter than in the benchmark.

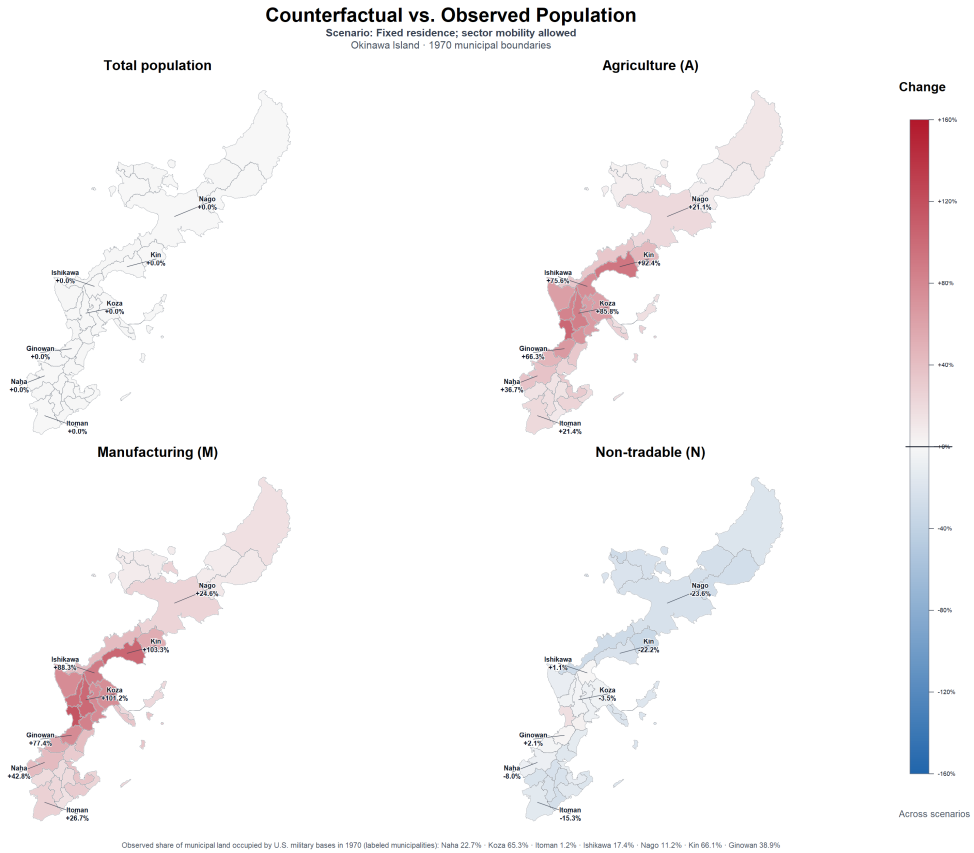
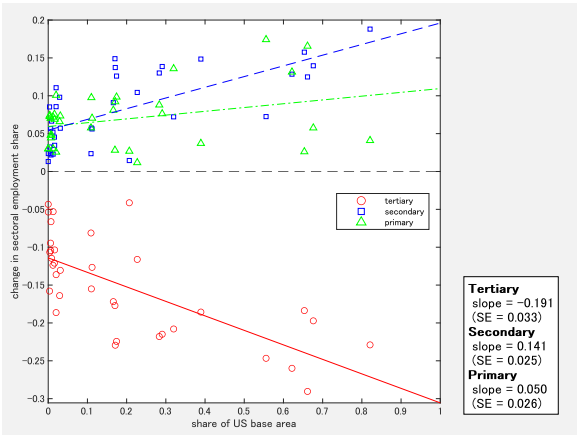
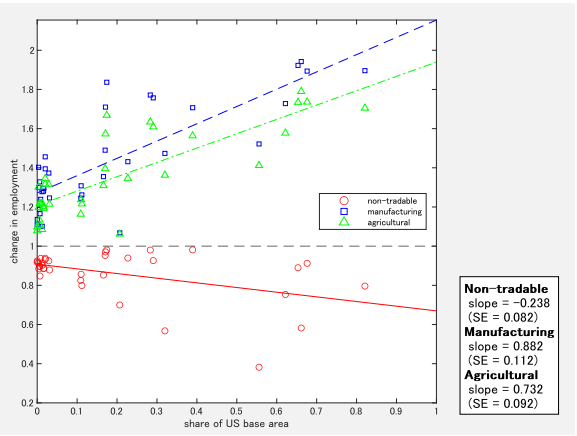


Figure D.8: Municipal resident-sector reallocation with municipal residence fixed

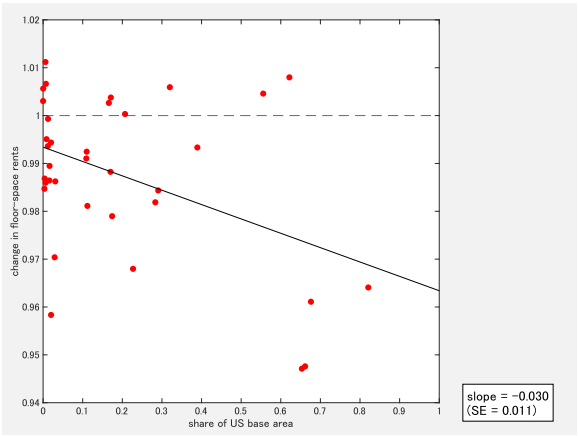
Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for total residents and residents working in agriculture (A), manufacturing (M), and civilian non-tradable services (N). The observed total includes base workers. The counterfactual fixes $R_i^{cf} = R_i$, sets $\Phi_i = L_{iB} = 0$, and leaves base land unavailable for private use. Red denotes an increase and blue a decrease. Municipal boundaries are those of 1970; labels report changes for selected municipalities.



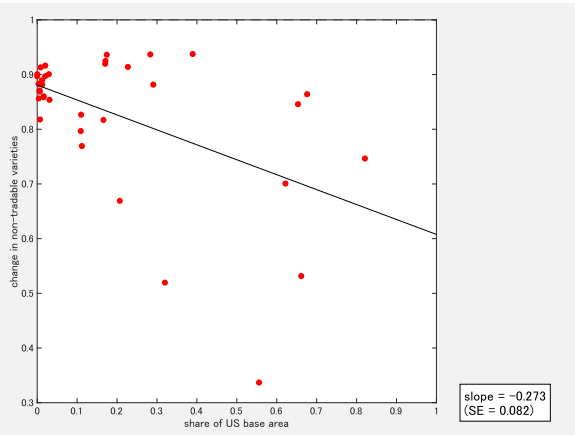
(a) Broad sector-of-employment shares



(b) Civilian workplace employment



(c) Floor-space rent



(d) Non-tradable varieties

Figure D.9: Exposure gradients with municipal residence fixed

Note: Each point is a municipality, and the horizontal axis is the 1970 base-area share. Panel D.9a plots counterfactual-minus-observed changes in resident sector-of-employment shares; the other panels plot counterfactual-to-observed ratios. In the sectoral panels, red circles, blue squares, and green triangles denote non-tradables (tertiary in panel D.9a), manufacturing, and agriculture. Tertiary combines civilian non-tradables and base employment. Lines are linear fits, and boxes report slopes and standard errors. The counterfactual fixes $R_i^{cf} = R_i$, sets $\Phi_i = L_{iB} = 0$, and leaves base land unavailable for private use.

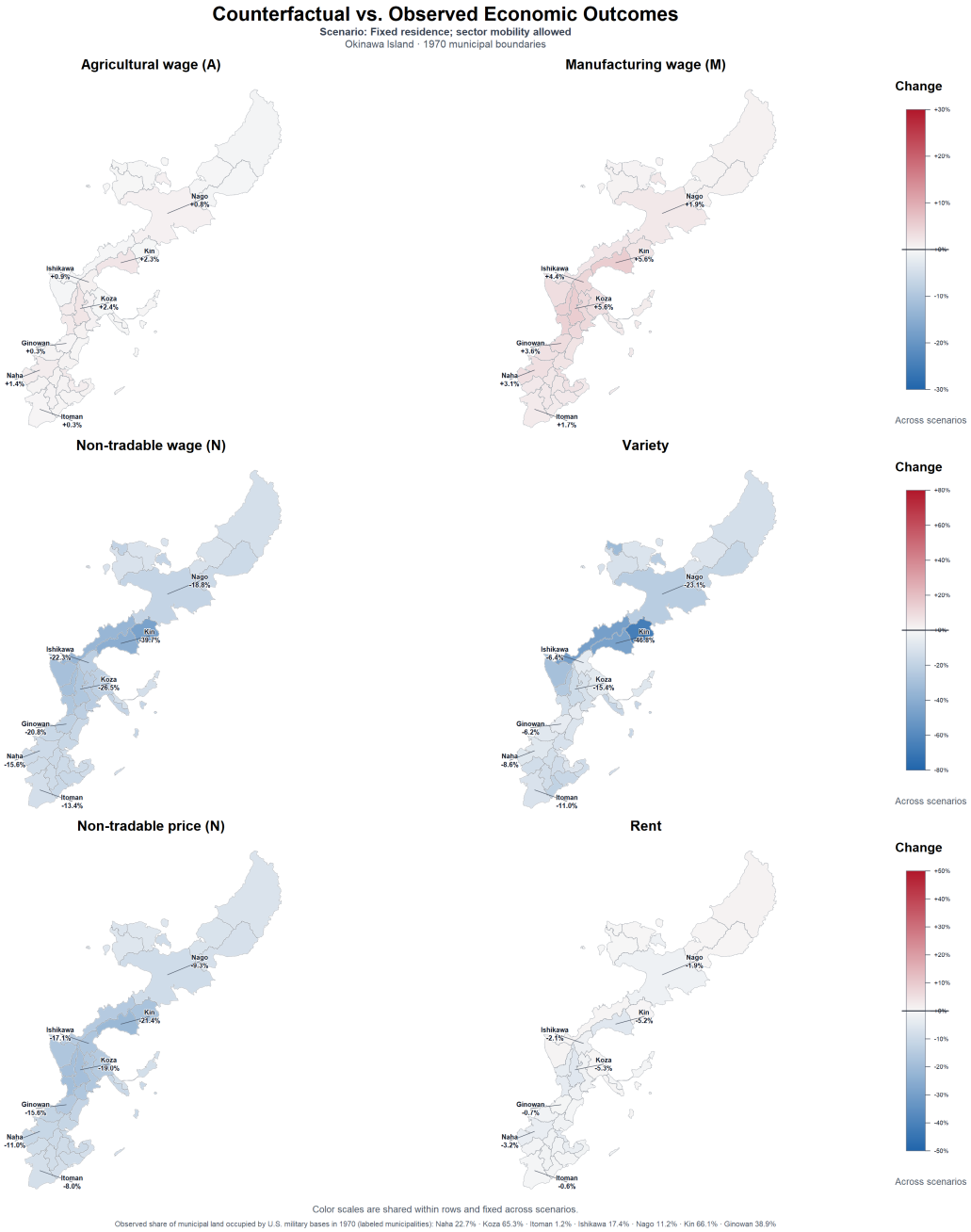
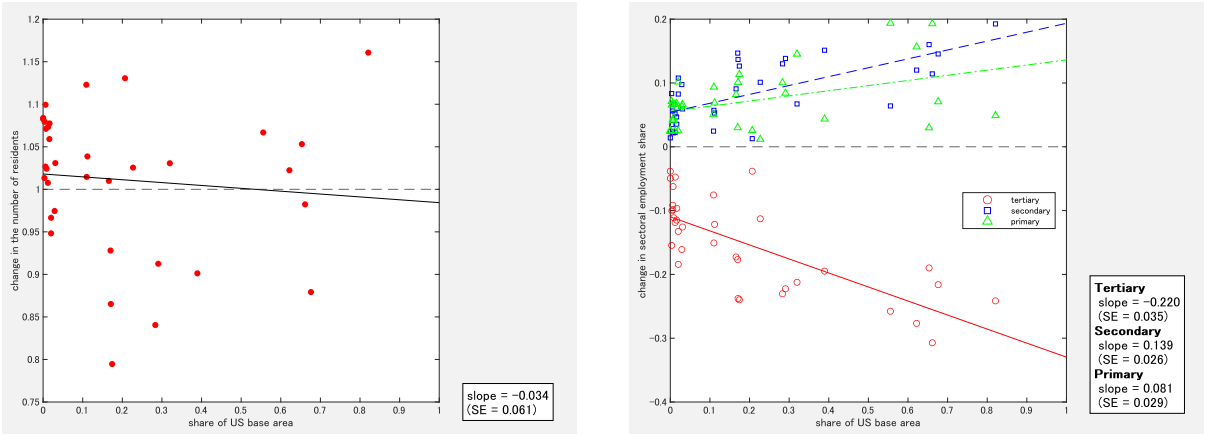


Figure D.10: Municipal wage, rent, price, and variety responses with municipal residence fixed

Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. The counterfactual fixes $R_i^{cf} = R_i$, sets $\Phi_i = L_{iB} = 0$, and leaves base land unavailable for private use. Red denotes an increase and blue a decrease.

D.5 Returning base land to private use



(a) Resident population

(b) Broad sector-of-employment shares

Figure D.11: Spatial effects when US-base land is returned to private use

Note: The horizontal axis is the 1970 base-area share. The left panel plots the counterfactual-to-observed resident ratio. The right panel plots counterfactual-minus-observed changes in primary, secondary, and tertiary sector-of-employment shares among residents; tertiary combines civilian non-tradable services and base employment. The counterfactual sets $\Phi_i = L_{iB} = m_i = 0$, so base land is available for private use. Symbols, fitted lines, slopes, and standard errors follow Figure 6.

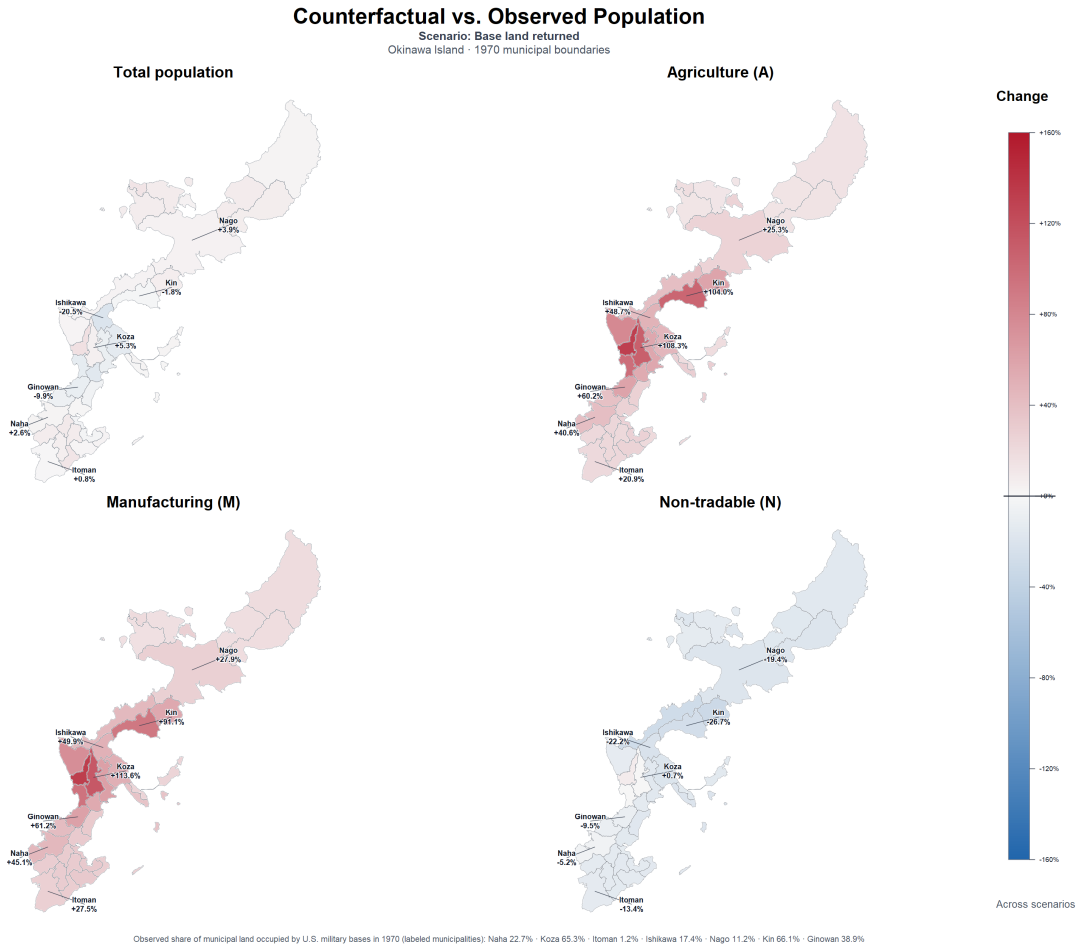
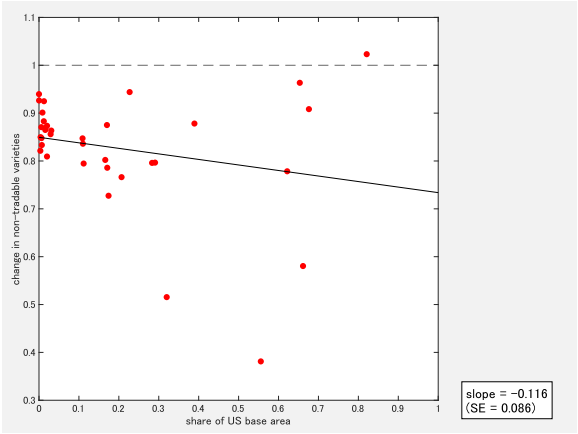
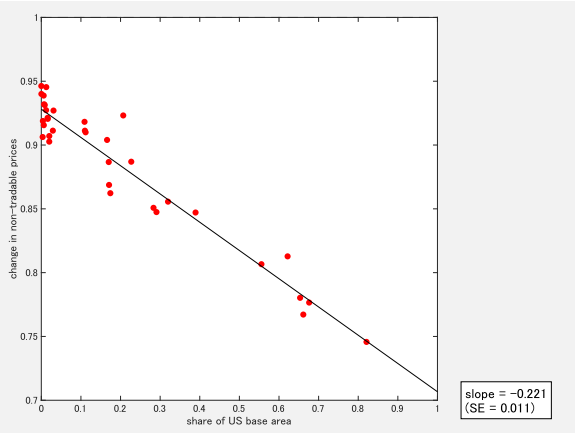


Figure D.12: Municipal population changes when US-base land is returned to private use

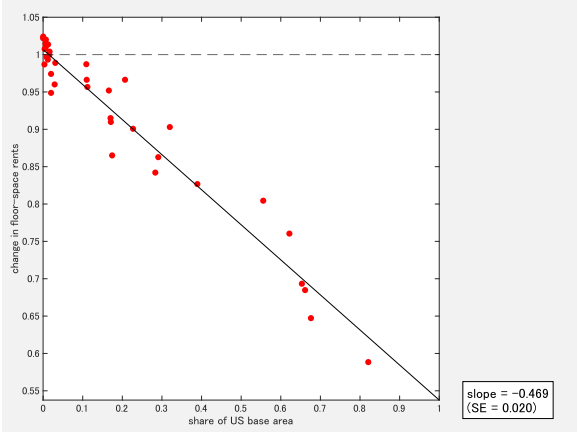
Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total residents and for residents working in agriculture (A), manufacturing (M), and non-tradable services (N). Red denotes an increase and blue a decrease. The counterfactual sets $\Phi_i = L_{iB} = m_i = 0$. Municipal boundaries are those of 1970; labels report changes for selected municipalities.



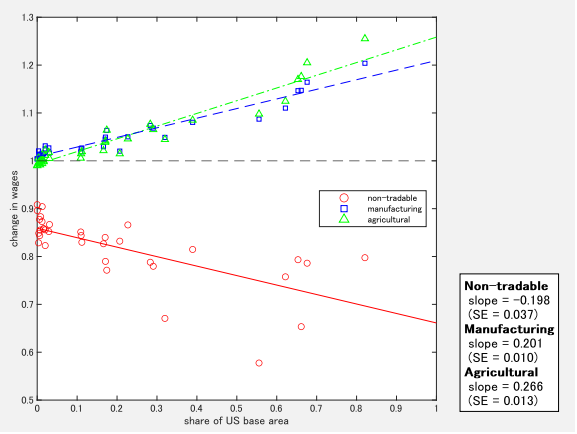
(a) Non-tradable varieties



(b) Non-tradable price index



(c) Floor-space rent



(d) Sectoral wages

Figure D.13: Prices, rents, varieties, and wages when US-base land is returned

Note: Each panel plots a counterfactual-to-observed ratio against the 1970 base-area share. In the wage panel, red circles, blue squares, and green triangles denote non-tradable, manufacturing, and agricultural wages, respectively. Lines are linear fits, and the boxes report slopes and standard errors. The counterfactual sets $\Phi_i = L_{iB} = m_i = 0$.

D.6 Local and aggregate elasticity calculations

Experiments and elasticities. Section 6.2 considers separate 1% increases in base-sourced consumption, Φ_j, and Okinawan base employment, L_{jB}, in one municipality j from the calibrated 1970 equilibrium. All other exogenous inputs, including fundamental amenities and productivity, occupied base land and Okinawa's total workforce, remain fixed.

For outcome Z_i in response municipality i and input x_j in source municipality j, define

$$E_{ij}^{Z,x} \equiv \frac{\partial \ln Z_i}{\partial \ln x_j}, \qquad x_j \in \{\Phi_j, L_{jB}\}. \tag{D.1}$$

We compute these elasticities by symmetric log differences, changing x_j to $1.01x_j$ and $x_j/1.01$ and dividing the difference in log outcomes by $2\ln(1.01)$. The diagonal entry is the local response; off-diagonal entries are spillovers. For a 1% input increase, $E_{ij}^{Z,x}$ gives the first-order percentage change in Z_i. Both panels of Table 3 use the same 29 source municipalities with positive consumption and base employment, denoted by $\mathcal{J}$, and all 35 response municipalities. For comparability across panels, we exclude five municipalities with zero base employment and one with zero base-sourced consumption, since the corresponding log elasticities are undefined at zero. This leads to 29 (out of 35) source municipalities.

Aggregation. From the matrix of elasticity of the response in municipality i with respect to a shock in municipality j, we compute the aggregate response in level as follows. Let s_i^Z denote municipality i's aggregation weight for outcome Z, with $\sum_i s_i^Z = 1$; superscript 0 denotes the initial equilibrium. Employment and resident-worker responses use shares of the corresponding initial worker counts as the weights. Non-base employment is $L_i^P = \sum_{s\in\{A,M,N\}} L_{is}$, and total workplace employment is $L_i = L_i^P + L_{iB}$. For sectoral wages, floor-space rents, and service prices, the weights are

$$s_i^{w_s} = \frac{w_{is}^0 L_{is}^0}{\sum_k w_{ks}^0 L_{ks}^0}, \qquad s_i^Q = \frac{Q_i^0 H_i^0}{\sum_k Q_k^0 H_k^0}, \qquad s_i^{P^N} = \frac{w_{iN}^0 L_{iN}^0/(1-\gamma^N)}{\sum_k w_{kN}^0 L_{kN}^0/(1-\gamma^N)}.$$

The island-level service-price index is defined as $\prod_i (P_i^N)^{s_i^{P^N}}$, weighted by initial service-revenue shares across production locations.

For a shock in j, the aggregate response is $\sum_i s_i^Z E_{ij}^{Z,x}$, comprising the local contribution

$s_j^Z E_{jj}^{Z,x}$ and spillovers $\sum_{i \neq j} s_i^Z E_{ij}^{Z,x}$. Table 3's three columns are

$$\text{Local response} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} E_{jj}^{Z,x},$$

$$\text{Okinawa-wide response} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \sum_{i} s_i^Z E_{ij}^{Z,x},$$

$$\text{Local share of aggregate impact} = 100 \frac{\sum_{j \in \mathcal{J}} s_j^Z E_{jj}^{Z,x}}{\sum_{j \in \mathcal{J}} \sum_i s_i^Z E_{ij}^{Z,x}}.$$

The third column is the local share of the net aggregate change; negative values indicate offsetting local responses. NA denotes a zero aggregate response: resident workers are fixed in both experiments, as is non-base employment under consumption shocks.

Worker-count effects. Here we express the employment response as the absolute number of jobs in response to a one-job increase in the US bases. For base hiring, divide the count change by the additional $0.01 L_{jB}^0$ base jobs. The local effect per added base job is

$$m_j^Z = \frac{Z_j^0 E_{jj}^{Z,L_B}}{L_{jB}^0}, \qquad \bar{m}^Z = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} m_j^Z.$$

The mean non-base response is -0.044, giving a total local employment gain of $1 - 0.044 = 0.956$ jobs per added base job. Net inward commuting equals workplace employment minus resident workers, $L_i - R_i$. The increase of 0.956 local jobs therefore comprises 0.473 additional resident workers and an increase of $0.956 - 0.473 = 0.483$ in net inward commuting.

D.7 Computing the open-population equilibrium

Let θ_{out} govern workers' choice between Okinawa and the outside economy. We use $\theta_{\text{out}} \in \{0.5, 1.1, 2.19\}$ in the open-population exercises, while retaining $\theta = 2.19$ for residence, workplace, and sector choice within Okinawa. With Fréchet preferences at the outer margin, Okinawa's share of Japan's workforce is

$$\Lambda = \frac{\mathbb{W}^{\theta_{\text{out}}}}{\mathbb{W}^{\theta_{\text{out}}} + \overline{u}^{\theta_{\text{out}}}}, \tag{D.2}$$

where $\mathbb{W}$ is expected welfare in (31) and $\overline{u}$ is welfare outside Okinawa. The benchmark fixes Okinawa's workforce and omits this migration margin.

Holding Japan's total workforce and $\overline{u}$ fixed, let $\widehat{\mathbb{W}} = \mathbb{W}/\mathbb{W}^0$ denote welfare relative to the initial equilibrium. Then

$$\widehat{\Lambda} \equiv \frac{\Lambda}{\Lambda^0} = \frac{\widehat{\mathbb{W}}^{\theta_{\text{out}}}}{\Lambda^0 \widehat{\mathbb{W}}^{\theta_{\text{out}}} + 1 - \Lambda^0} = \frac{L}{L^0}. \tag{D.3}$$

We set $\Lambda^0 = 0.0077$, using the Okinawa main island's share of Japan's 1970 population as a proxy for its worker share.

We use the following "outer loop" to solve the model with endogeneous total population:

Outer loop.

1. Solve the inner and main loops in Appendix D.1 at a trial aggregate workforce L and compute $\mathbb{W}$.
2. Use (D.3) to obtain the target workforce $L^0\widehat{\Lambda}$, with L^0, Λ^0, and $\mathbb{W}^0$ fixed at their initial values.
3. Rescale all R_{is} by $L^0\widehat{\Lambda}/L$ and repeat until $L = L^0\widehat{\Lambda}$.

D.8 Alternative external-market assumptions: specification and results

This subsection provides the specifications and detailed results for the external-validity exercises in Section 6.3. Table D.1 reports aggregate outcomes across all parameter values; the figures below report spatial outcomes for the middle cases.

Table D.1: Okinawa-wide effects of counterfactual scenarios under extended models

	Aggregate effects (%)			Employment-share changes (pp)			
Scenario	Workers' Welfare	GNI	Worker population	Agric.	Manuf.	Non-trad.	U.S. base
Benchmark							
Benchmark counterfactual	−7.25	−11.99	0.00	+5.58	+10.26	−4.92	−10.93
Larger geographical mobility in and out of Okinawa							
Low migration elasticity ($\theta_{\text{out}} = 0.5$)	−7.26	−15.21	−3.67	+5.74	+10.19	−5.01	−10.93
Middle migration elasticity ($\theta_{\text{out}} = 1.1$)	−7.27	−18.93	−7.91	+5.93	+10.11	−5.12	−10.93
High migration elasticity ($\theta_{\text{out}} = 2.19$)	−7.28	−25.29	−15.16	+6.28	+9.96	−5.32	−10.93
Endogenous manufacturing goods prices							
High import elasticity ($\varepsilon_M = 20$)	−6.68	−16.19	0.00	+7.89	+8.39	−5.35	−10.93
Middle import elasticity ($\varepsilon_M = 10$)	−6.25	−18.64	0.00	+9.41	+7.13	−5.62	−10.93
Low import elasticity ($\varepsilon_M = 5$)	−5.63	−21.53	0.00	+11.43	+5.45	−5.95	−10.93
Fixed import quantity ($\varepsilon_M = 0$)	−2.10	−30.04	0.00	+19.74	−1.77	−7.04	−10.93

Notes: Welfare, GNI, and worker-population entries are percentage changes from the initial equilibrium, $100(\text{counterfactual}/\text{initial} - 1)$. Worker population is $\sum_i R_i$ across the 35 municipalities and adjusts in the external-migration scenarios. Employment entries are changes in Okinawa-wide sectoral shares in percentage points (pp). Initial shares are 16.55%, 21.44%, 51.08%, and 10.93% for agriculture, manufacturing, non-tradable services, and US-base employment, respectively. All scenarios set $\Phi_i = L_{iB} = 0$ while keeping base land unavailable for private use.

Population adjustment through external migration. The open-population specification is described in Appendix D.7. We use $\theta_{\text{out}} = 1.1$ as the middle case, informed by Ortega and Peri (2013), who estimate income elasticities of international migration flows of 0.76 overall and 1.90 within Europe. The low case sets $\theta_{\text{out}} = 0.5$; the high case uses 2.19, equal to the within-Okinawa migration elasticity based on Hayakawa, Koster, Tabuchi and Thisse (2021).

As θ_{out} rises from 0.5 to 1.1 and 2.19, worker population falls by 3.67%, 7.91%, and 15.16%, and GNI falls by 15.21%, 18.93%, and 25.29% (Table D.1). Welfare losses remain between 7.26% and 7.28%, close to the benchmark loss of 7.25%, and sectoral employment-share changes remain similar. External migration mainly amplifies the contraction in population and total GNI.

Endogenous manufacturing prices and finite external trade elasticity. We next endogenize Okinawa's manufacturing-goods price through an external import-supply schedule while retaining agriculture as the numeraire. Let $\mathcal{I}^M$ denote Okinawa's net manufacturing import value and let a subscript 0 denote the observed equilibrium. We assume that the manufacturing price P^M is determined by

$$\frac{P^M}{P_0^M} = \left(\frac{\mathcal{I}^M}{\mathcal{I}_0^M}\right)^{\frac{1}{1+\varepsilon_M}}, \tag{D.4}$$

where ε_M is the price elasticity of manufacturing import supply.[D.1] We specify the homothetic aggregator of tradable prices as $\Upsilon(P^A, P^M) = \left(P^A\right)^{1-\beta}\left(P^M\right)^{\beta}$ and set the manufacturing share in the tradable composite to $\beta = 0.8$ to match the observed trade balance of the agricultural goods in the calibrated economy.[D.2]

Net manufacturing import value equals expenditure on manufacturing goods less local manufacturing output:

$$\mathcal{I}^M = \sum_i \left[\frac{\chi_i^M}{\alpha} C_i + \beta\mu Q_i H_i - \frac{w_{iM} L_{iM}}{1-\gamma^M}\right]. \tag{D.5}$$

Here, $\chi_i^M = \alpha\beta(P^T/\mathbb{P}_i)^{1-\kappa}$, and $C_i = \alpha W_i + (1-\mu)Q_i H_i + \Phi_i$ is expenditure on goods and services excluding housing. Residents' wage income W_i includes commuting earnings. The three terms are manufacturing consumption, manufacturing inputs used in floor-space construction, and local manufacturing output.

We use $\varepsilon_M = 10$ as the middle case, guided by Nicita, Olarreaga and Silva (2018), and 5 and 20 for sensitivity analysis. At $\varepsilon_M = 0$, the physical import quantity is fixed; as $\varepsilon_M \to \infty$, $P^M = P_0^M$, recovering the benchmark.

Table D.1 shows that at $\varepsilon_M = 10$, welfare falls by 6.25% and GNI by 18.64%, compared with 7.25% and 11.99% in the benchmark. Agricultural and manufacturing employment shares rise by 9.41 and 7.13 percentage points, compared with 5.58 and 10.26 points. Lower import-supply elasticity produces a larger manufacturing-price decline, redirects more workers toward agriculture, and increases the GNI loss. Cheaper manufactured goods reduce consumption and construction costs, cushioning the welfare loss. With fixed import quantity, manufacturing's employment share falls by 1.77 points.

Spatial results. For external migration with $\theta_{\text{out}} = 1.1$, Figures D.14 and D.15 map residents and workplace employment, and Figure D.16 maps wages, rents, service prices, and varieties. Figure D.17 plots the corresponding municipal outcomes against base-area shares. The spatial patterns remain close to the benchmark: more base-exposed municipalities experience larger population losses and stronger shifts from tertiary employment toward manufacturing and agri-

[D.1] Equation (D.4) implies that the import quantity of manufacturing goods, denoted by Q^M, is given by $Q^M/Q_0^M = \left(P^M/P_0^M\right)^{\varepsilon_M}$, where Q_0^M represents the import quantity in the observed equilibrium.

[D.2] According to the Okinawa Statistical Yearbook, Okinawa's trade balance for the agricultural goods around 1970 was roughly balanced.

culture. The resident-exposure gradient is −0.445, compared with −0.477 in the benchmark, and the sectoral-share gradients are also nearly unchanged. External migration amplifies the islandwide population decline while preserving these spatial patterns.

For elastic manufacturing import supply with $\varepsilon_M = 10$, Figures D.18 and D.19 map residents and workplace employment, and Figure D.20 maps wages, rents, service prices, and varieties. Figure D.21 reports the corresponding scatterplots against base-area shares. These spatial patterns also remain close to the benchmark: population losses and the shift from tertiary employment toward tradable production are larger in more base-exposed municipalities. The resident-exposure gradient is −0.538, compared with −0.477 in the benchmark. Manufacturing-price adjustment changes aggregate income and the balance between agriculture and manufacturing, but preserves the main spatial pattern of reallocation away from services and base employment.

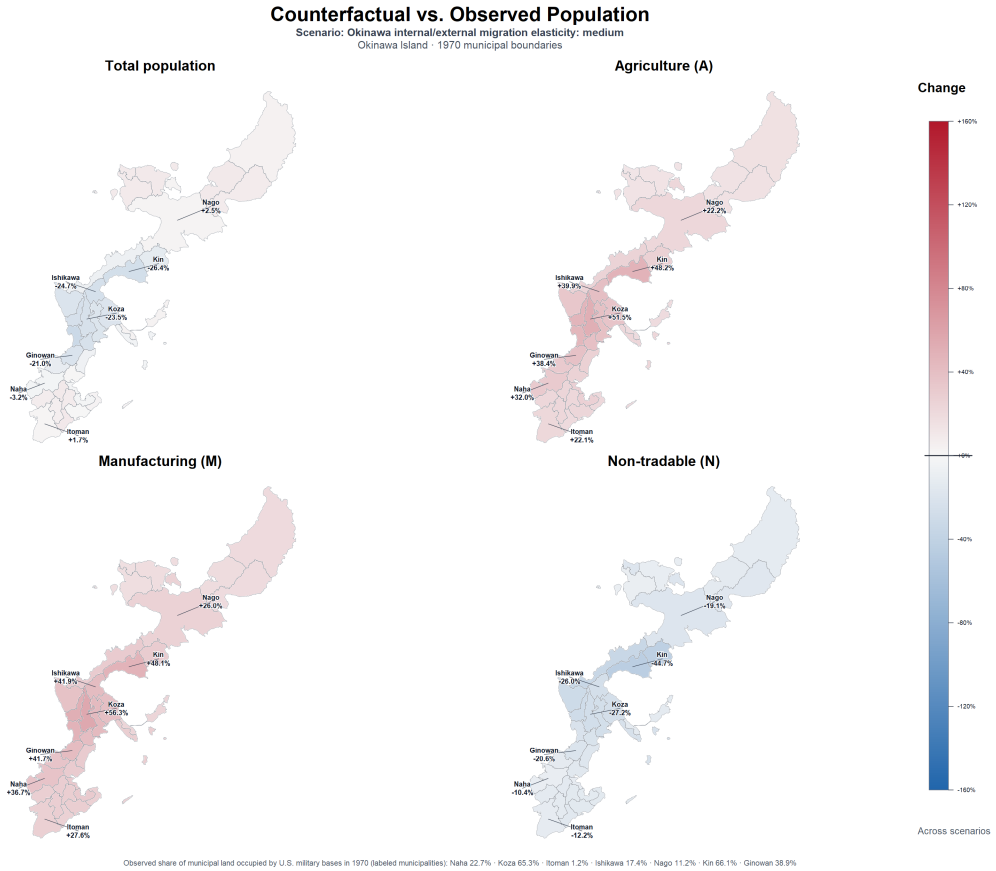


Figure D.14: Municipal worker-resident responses with external migration: middle elasticity

Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for total residents and residents working in agriculture (A), manufacturing (M), and non-tradable services (N). The observed total includes base workers. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\theta_{\text{out}} = 1.1$, with base land unavailable for private use.

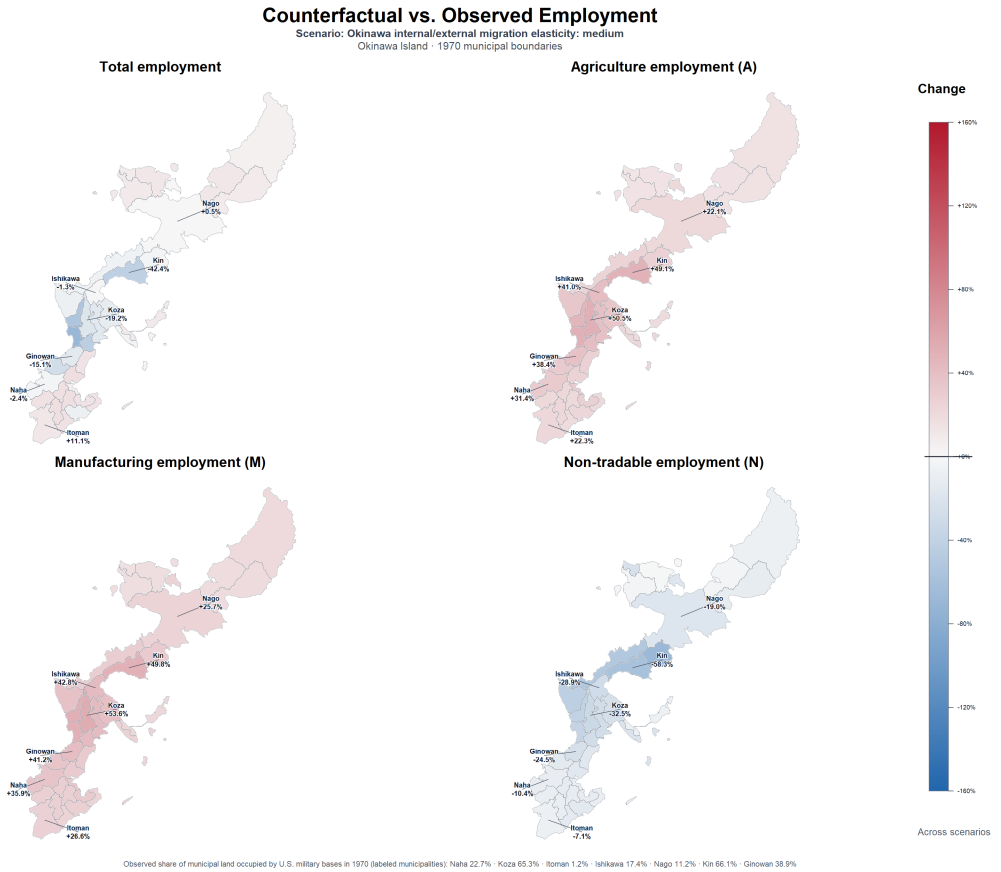


Figure D.15: Municipal workplace-employment responses with external migration: middle elasticity

Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total workplace employment and employment in agriculture (A), manufacturing (M), and non-tradable services (N). The observed total includes base jobs. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\theta_{\text{out}} = 1.1$, with base land unavailable for private use.

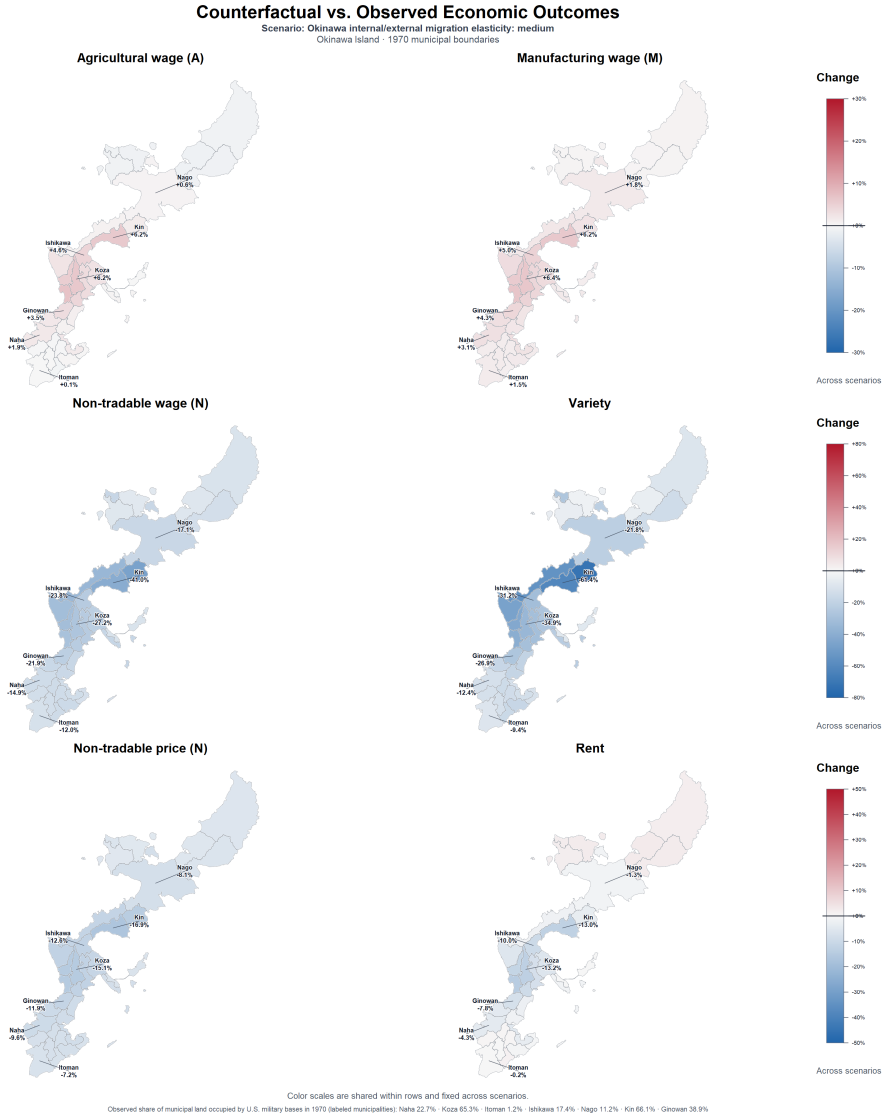
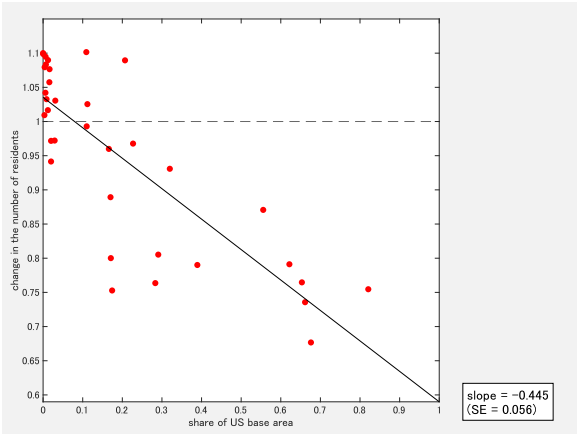
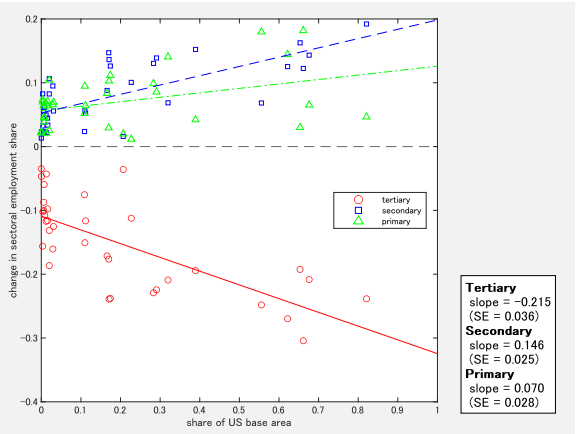


Figure D.16: Municipal wage, rent, price, and variety responses with external migration: middle elasticity

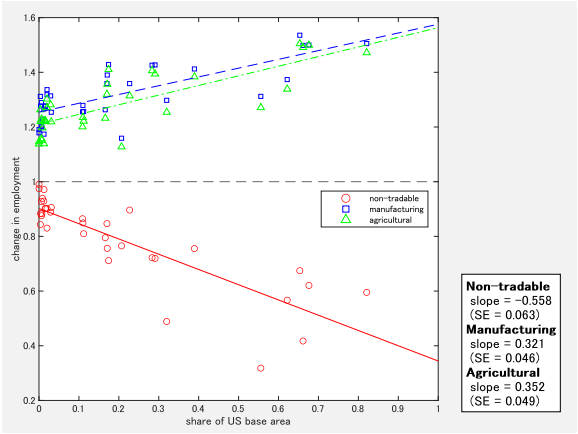
Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\theta_{\text{out}} = 1.1$, with base land unavailable for private use.



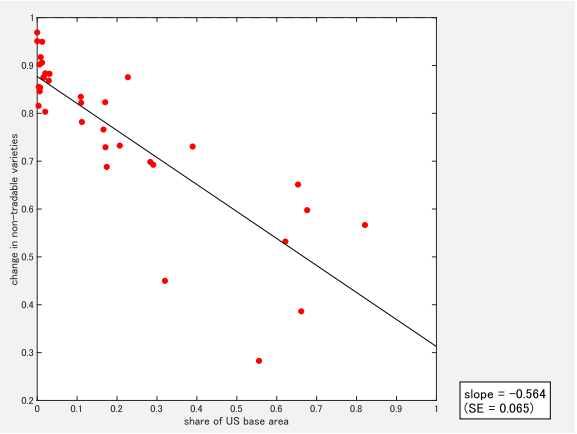
(a) Residents



(b) Broad sector-of-employment shares



(c) Workplace employment



(d) Non-tradable varieties

Figure D.17: Exposure gradients with external migration: middle elasticity

Note: Each point is a municipality, and the horizontal axis is the 1970 base-area share. Panel D.17b plots counterfactual-minus-observed changes in resident sector-of-employment shares; the other panels plot counterfactual-to-observed ratios. In the sectoral panels, red circles denote non-tradable services (including base employment in the tertiary share); blue squares and green triangles denote manufacturing and agriculture. Lines are linear fits, and the boxes report slopes and standard errors. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\theta_{\text{out}} = 1.1$, with base land unavailable for private use.

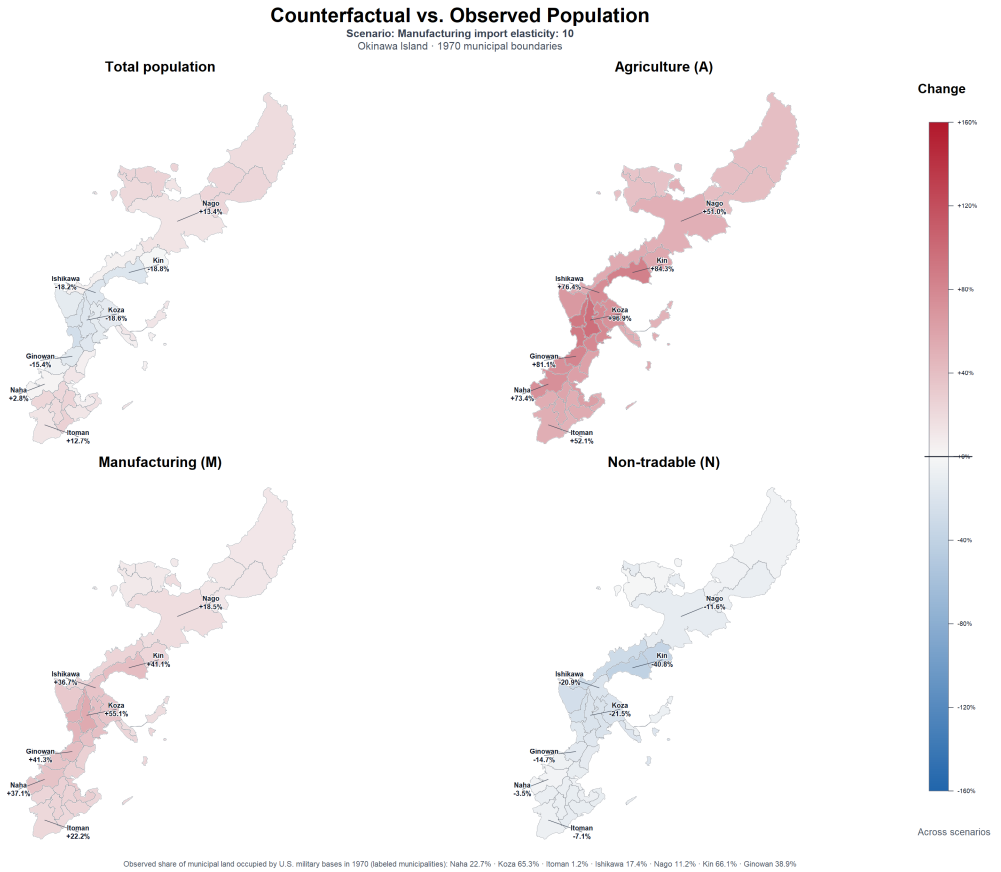


Figure D.18: Municipal worker-resident responses with an endogenous manufacturing price: middle import elasticity

Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total residents and residents working in agriculture (A), manufacturing (M), and non-tradable services (N). The observed total includes base workers. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\varepsilon_M = 10$, with base land unavailable for private use.

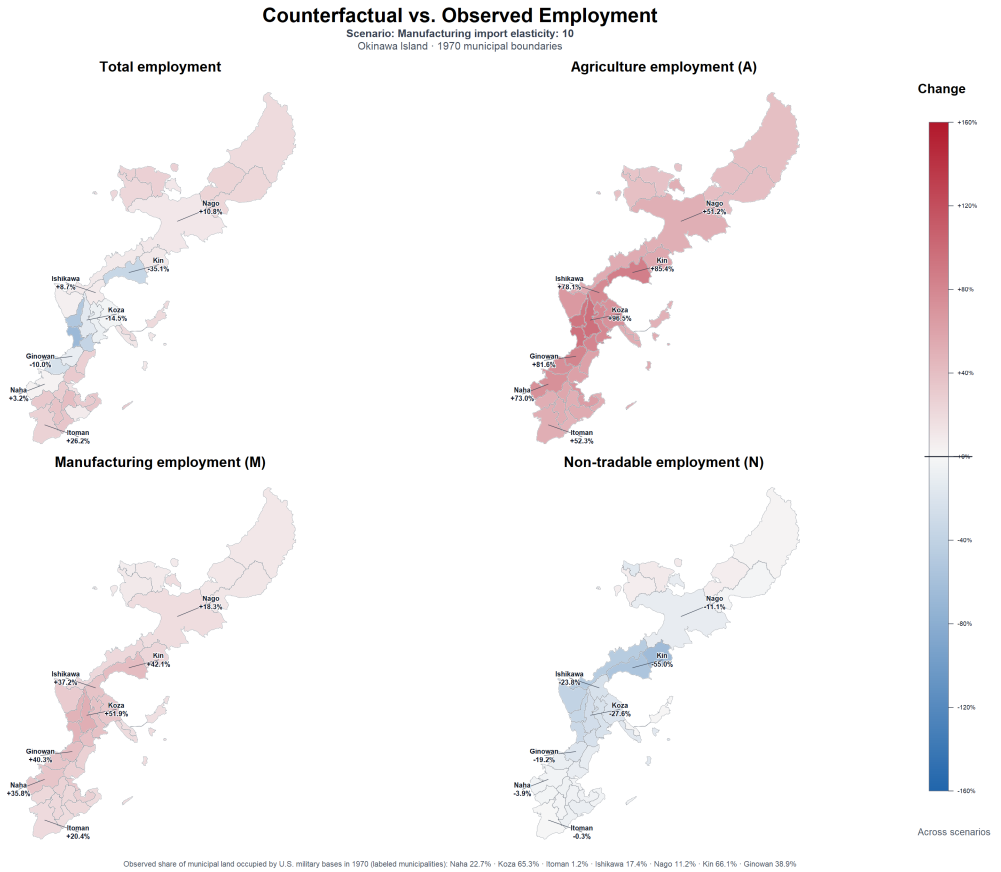


Figure D.19: Municipal workplace-employment responses with an endogenous manufacturing price: middle import elasticity

Note: The maps report $100 \times$ (counterfactual/observed $-$ 1) for total workplace employment and employment in agriculture (A), manufacturing (M), and non-tradable services (N). The observed total includes base jobs. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\varepsilon_M = 10$, with base land unavailable for private use.

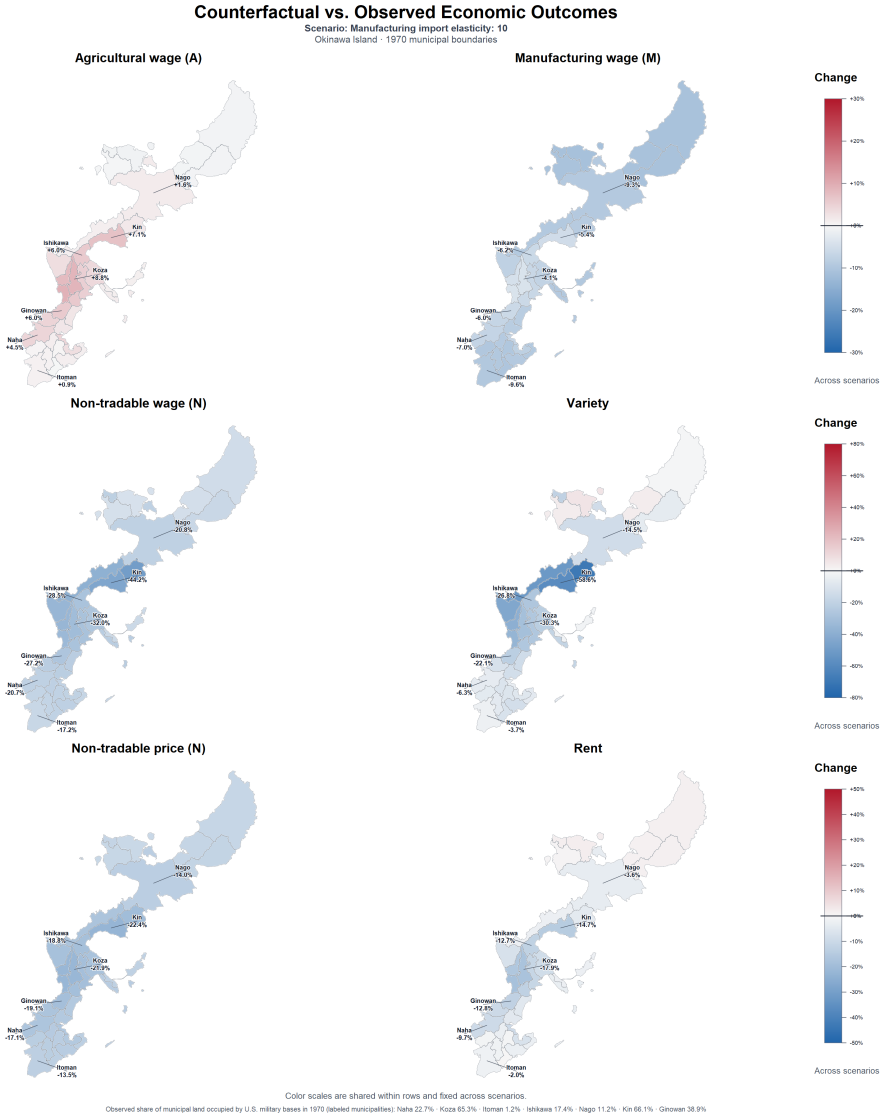
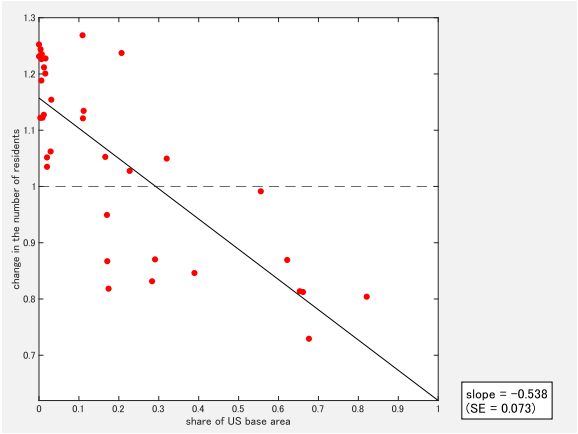
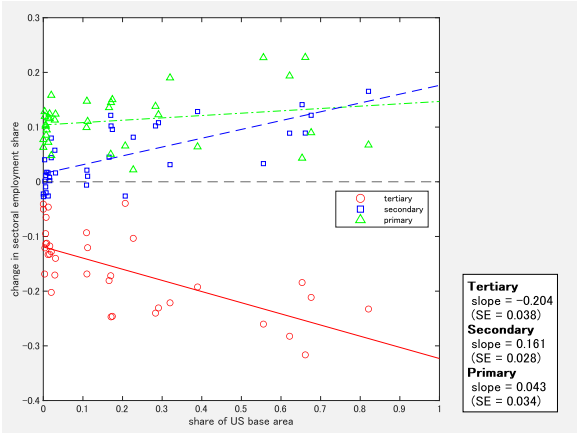


Figure D.20: Municipal wage, rent, price, and variety responses with an endogenous manufacturing price: middle import elasticity

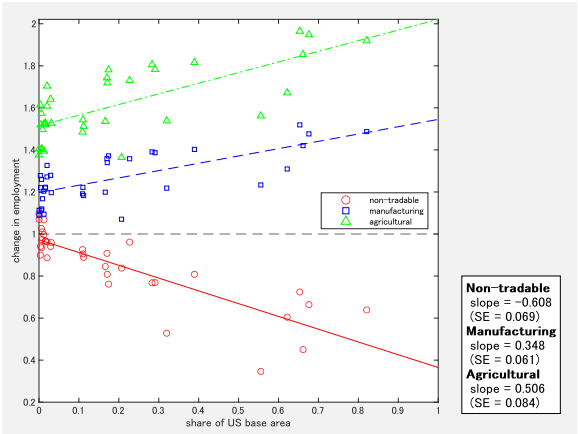
Note: The maps report $100 \times (\text{counterfactual}/\text{observed} - 1)$ for sectoral wages, floor-space rents, the non-tradable price index, and non-tradable varieties. Red denotes an increase and blue a decrease. The maps use 1970 municipal boundaries and common color scales across scenarios. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\varepsilon_M = 10$, with base land unavailable for private use.



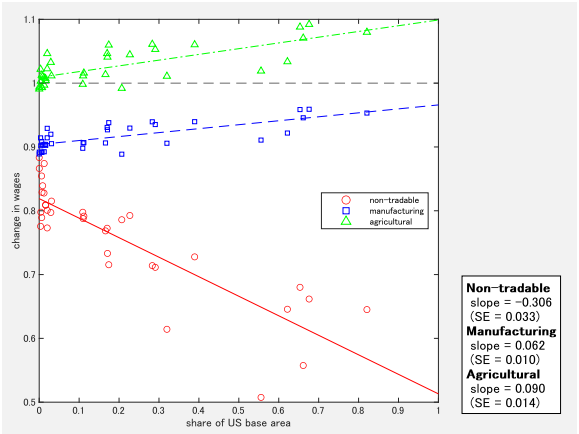
(a) Residents



(b) Broad sector-of-employment shares



(c) Workplace employment



(d) Sectoral wages

Figure D.21: Exposure gradients with an endogenous manufacturing price: middle import elasticity

Note: Each point is a municipality, and the horizontal axis is the 1970 base-area share. Panel D.21b plots counterfactual-minus-observed changes in resident sector-of-employment shares; the other panels plot counterfactual-to-observed ratios. In the sectoral panels, red circles denote non-tradable services (including base employment in the tertiary share); blue squares and green triangles denote manufacturing and agriculture. Lines are linear fits, and the boxes report slopes and standard errors. The counterfactual sets $\Phi_i = L_{iB} = 0$ and $\varepsilon_M = 10$, with base land unavailable for private use.

References for the Appendix

Appleman, Roy E., James M. Burns, Russell A. Gugeler, and John Stevens, *Okinawa: The Last Battle* United States Army in World War II: The War in the Pacific, Washington, DC: Historical Division, Department of the Army, 1948.

Faber, Benjamin and Cecile Gaubert, "Tourism and economic development: Evidence from Mexico's coastline," *American Economic Review*, 2019, *109* (6), 2245–2293.

Fan, Tianyu, Michael Peters, and Fabrizio Zilibotti, "Growing like India — The unequal effects of service-led growth," *Econometrica*, 2023, *91* (4), 1457–1494.

Hayakawa, Kazunobu, Hans R.A. Koster, Takatoshi Tabuchi, and Jacques-François Thisse, "High-speed rail and the spatial distribution of economic activity: Evidence from Japan's Shinkansen," 2021. RIETI Discussion Paper Series 21-E-003.

Komeito, *Okinawa beigun kichi no jittai chosa (Survey on the actual conditions of U.S. military bases in Okinawa)* 1969. (In Japanese).

Nichols, Charles S. and Henry I. Shaw, *Okinawa: Victory in the Pacific*, Washington, DC: Historical Branch, G-3 Division, Headquarters, U.S. Marine Corps, 1955.

Nicita, Alessandro, Marcelo Olarreaga, and Peri Silva, "Cooperation in WTO's Tariff Waters?," *Journal of Political Economy*, 2018, *126* (3), 1302–1338.

Ortega, Francesc and Giovanni Peri, "The Effect of Income and Immigration Policies on International Migration," *Migration Studies*, 2013, *1* (1), 47–74.